CM-SIP — Study Guide

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CM-SIP — Study Guide

Capital Markets – Specified Investment Products (SIP): Derivatives and Collective Investment Schemes

A concise guide to the CM-SIP examination: derivatives, structured products and collective investment schemes. It condenses the official study guide into what the examination tests, section by section.

- **Exam:** 80 multiple-choice/multiple-response questions in 2 hours 30 minutes, plus 6 case studies containing 18 questions (3 per case study); **70% to pass**.
- **Currency of the material:** the official guide states its regulatory information as current at 30 September 2023. Check the latest MAS, SFA, SGX and product terms before relying on a rule or specification.
- This is a study aid, not a replacement for the prospectus, Product Highlights Sheet (PHS), term sheet, risk disclosure statement or current law. If you will conduct regulated activity, you must also meet the relevant MAS representative-notification and competency requirements.

How to use this guide

1. Learn the product mechanism before memorising the payoff.
2. For every product, ask: what is the underlying, what is the investor's position, what triggers payment/conversion/termination, what can be lost, and who is the counterparty?
3. Practise the formulas in Appendix D and the application cases in Chapter 12.
4. Use `CMSIP_QUESTIONS.md` only after reading the relevant section; its answers explain the reasoning rather than merely naming a letter.

Examination orientation

- The examination tests knowledge, understanding and application of SIPs.
- Format: 80 multiple-choice/multiple-response questions in 2 hours 30 minutes, plus 6 case studies containing 18 questions (3 per case study); the passing mark is 70%.
- Unless a question says otherwise, use a **365-day** convention for Singapore Government Securities and a **360-day** convention for other products.
- A non-programmable financial calculator may be used and a formula sheet is provided.
- Scope: exchange-traded and OTC derivatives, structured products and collective investment schemes. Candidates who will conduct regulated activity must also satisfy the relevant MAS representative-notification and competency requirements.

Master mental model

Every derivative or structured product transfers one or more of these exposures:

Exposure	Typical instrument or feature	Main question
Direction	Futures, forwards, calls, puts, CFDs, warrants	Does the investor profit from a rise or fall?
Volatility	Straddles, strangles, barrier options, volatility ETFs	Does the investor need movement or stability?
Time	Option time value, calendars, rollovers, callable notes	How does time decay, maturity or early terminati
Carry/yield	Interest-rate futures, swaps, structured deposits/notes	What is paid or received while the position is ope
Credit	CDS, CLNs, issuer/SPV obligations	Who may fail to pay?
Liquidity	OTC customisation, lock-ups, exotic derivatives	Can the position be closed at a fair price?
Currency	FX forwards/options/swaps, translated warrants, DCIs	Which currency is invested, settled and exposed?
Leverage	Margin, futures, CFDs, DLCs, warrants, accumulators	Can a small move create a large loss or margin ca

Chapter 1 — Overview of Derivatives

A derivative is a contract whose value is linked to the value or volatility of something else. This chapter builds the market map — the underlying asset classes, the exchange and OTC venues, the regulatory response to 2008, and Singapore's position — that every later chapter assumes.

Learning Objectives

The candidate should be able to:

- Describe common underlying asset classes for derivatives and structured products.
- Describe the different types of derivatives and their uses.
- Describe the characteristics of exchange-traded and over-the-counter derivatives.
- Discuss the development of the derivatives market in Singapore and beyond.

1.1 Introduction

A derivative is a financial instrument whose value is linked to the value or volatility of an underlying reference or asset. That reference may be a particular security, an asset, an interest rate, a combination of securities, a basket, an index, or another derivative. Derivatives trade either on centralised, regulated exchanges or bilaterally between counterparties in the less regulated over-the-counter (OTC) market.

Futures, forwards, swaps and options are the basic types. Beyond them sits a wide range of more complex or **exotic** derivatives, built by combining similar and different types. Barrier options, knock-in and knock-out products, swaptions, structured notes and structured funds are examples. By varying the features — strike price, tenure, currency, conditions for execution — and combining different derivatives, an issuer creates a wider range of payoff outcomes and risk profiles aimed at investors holding different views.

The economic uses are four:

- **Hedging** — insuring against adverse price, rate, currency or credit movement.
- **Speculation** — taking a view on the future direction of the market.
- **Arbitrage** — profiting from perceived price differentials between two or more markets or assets.
- **Repackaging** — combining instruments to meet specific investment needs and to create desired synthetic positions.

Investors also trade and hedge on **volatility** — their expectation of the degree of price movement, whether prices stay range-bound, swing wildly, or move steadily in one direction. As volatility trading on indices has grown, leveraged exchange-traded notes (ETNs) and leveraged exchange-traded funds (ETFs) have been created to amplify the impact of rising or falling percentage volatility. Many leveraged ETFs and ETNs actively use futures and options to replicate or track the direction — or the inverse direction — of the underlying index, which makes such funds and notes effectively derivative instruments.

1.2 Underlying Asset Classes for Derivatives and Their Uses

Reduced to building blocks, most derivatives are referenced from or crafted out of five basic asset classes: **foreign exchange, money market instruments, fixed income, equities and commodities**. Indices, baskets, funds and portfolio mixes are extensions and representations of these blocks.

Underlying assets	Commonly traded securities / indic
Currency	Major currencies (USD, EUR, JPY,
Money market, short-term Treasury instruments and their interest rates	Government T-bills (1, 3, 6, 9, 12 m
Fixed income (debt)	Sovereign and corporate debt instru
Equity	Listed stocks; equity or stock indice
Commodities	Commodities; commodities-linked s

Derivatives and structured products allow investors to:

- **a.** Hedge or insure against unfavourable price movements, or potential falls in the value of assets or a portfolio already held — futures, forwards, options.
- **b.** Adopt a view on the future direction of the underlying without investing directly in it.
- **c.** Augment potential profits beyond what pure exposure to the underlying can deliver — options, warrants.
- **d.** Change the payoff profile (cash-flow sequence, size, tenure, currency, risks) in ways the underlying asset cannot offer — swaps, futures, forwards, options.
- **e.** Profit from a fall in the value of the underlying asset or index by taking short positions — options, futures.
- **f.** Secure the right, without the obligation, to decide when to buy or sell the underlying, using a small outlay — long options and warrants. A short call or put is an obligation, not a right, should the buyer choose to exercise.
- **g.** Arbitrage, or take advantage of price differentials between two or more markets.
- **h.** Change the nature of an asset or liability to meet needs that standardised instruments cannot meet.
- **i.** Create synthetic positions without incurring the cost of buying or selling the underlying.

Two footnotes matter for scope. A **debt warrant** resembles an equity or company warrant but attaches to a bond or other debt security, giving the holder the right to buy more bonds or debt securities from the same issuer at a stated price; only equity and company warrants are covered in Chapter 5. **ETFs and ETNs are not derivatives**, but they actively use derivative instruments to enhance or hedge — leveraged and volatility ETFs, for example, use futures and options to replicate or track an index.

1.2.1 Money Market and Other Short Term Interest Rate Instruments

These are short-term debt instruments issued by governments, financial institutions and corporations. They are highly liquid and less risky. They account for **more than 95% of futures and options traded on global exchanges**, and, according to the Bank for International Settlements (BIS), consistently for more than **three-quarters of OTC derivatives trades between 2015 and 2019**.

Minimum denominations are relatively large and transaction sizes can be substantial. Maturities range from **1 day to 1 year**, often under **90 days**. The market is dominated by interbank lending and trading in Treasury bills, commercial paper, banker's acceptances, negotiable certificates of deposit and repurchase agreements. Institutional investors dominate.

Many instruments are benchmarked against interbank rates — **SIBOR, LIBOR, HIBOR, TIBOR, EURIBOR** — the most widely used historically being **LIBOR** (London Interbank Offered Rate). LIBOR is an average interbank lending rate calculated from submissions of actual borrowing rates contracted between banks, which makes it vulnerable to manipulation where submitting banks collude. Colluding to manipulate interbank rates and indices in order to engineer outcomes in the derivatives market is illegal, and traders and dealers have been prosecuted.

LIBOR was phased out and ceased to be used in **2021**, replaced for dollar-denominated derivatives and loans by the **Secured Overnight Financing Rate (SOFR)**.

1.2.2 Foreign Exchange

Cash investments in local or foreign currency are the lowest common denominator for all assets. FX derivatives form the **second largest** category of derivatives traded globally.

Currency forwards and futures are the most basic form, generating different payoffs depending on expected currency movement over time. Swaps and options change the underlying currency or interest payoff under conditional triggers, such as the exchange rate reaching a particular strike. Trading in currency forwards, futures, swaps and options is largely dominated by institutional investors.

In emerging markets with tightly controlled capital flows, inconvertible currency and low liquidity, hedging local-currency risk can be difficult. This is partly solved with a **non-deliverable forward (NDF)**, created as a proxy for the emerging-market currency. NDFs trade OTC in offshore financial centres and settle in a foreign currency other than the NDF currency itself.

1.2.3 Fixed Income Securities

These are borrowings by issuers — mainly governments, government-linked entities or public institutions, financial institutions and large multinational corporations, spanning investment-grade and junk bonds. The holder provides a loan of principal to the issuer and receives periodic interest, with principal returned at maturity. Differences in size, frequency and timing of cash flow produce a return commensurate with the risk taken and the opportunity cost incurred. Fixed-income derivatives — futures, swaps and options — offer alternative cash flows, payoff profiles and modified risk exposures relative to the underlying instrument.

The industry has innovated continuously to populate the investment frontier, and investment banks in the **1980s** popularised creative repackaging of debt for resale. Banks bought long-term fixed-income securities with higher yield and used them as the underlying asset base against which they repackaged and issued smaller denominations of marketable securities. Coupon payments of long-term Treasury bonds were **stripped** to form the asset base for issuing zero-coupon bonds. Portfolios of real-estate loans were bought and used to issue **mortgage-backed securities (MBS)**. Consumer and student loans became the asset base for **collateralised debt obligations (CDOs)**. Even without derivatives there was creative repackaging creating new payoff profiles; read broadly, various CDOs and stripped zero-coupon bonds are *derived* from other bonds and so are fixed-income derivatives.

By the late **1990s**, credit derivative instruments were actively promoted. Investment banks repackaged and modified the credit risk of fixed-income securities using credit insurance and **credit default swaps (CDS)** issued by insurance companies. They also bought select high-yield securities — junk bonds rated **CCC and below** — mixed them with better-rated securities, and added CDS and credit insurance to create a new cocktail for resale.

The credit quality of the underlying asset, and the credit management process, ultimately determines the risk and value of the repackaged product. The **2008 Global Financial Crisis** illustrated this dramatically: defaults on real-estate mortgage loans and CDOs triggered the collapse of **Fannie Mae and Freddie Mac**, the bailout and sale of several global institutions, forced capital restructuring across the global banking community, and a subsequent massive rise in regulatory control and cost.

1.2.4 Equities

Equities let investors participate in company profitability. When earnings prospects and growth are positive the share price rises over time; when they are negative it declines.

Equities are more volatile than other traditional asset classes. The ability of equity derivatives to take **short positions**, or grant a right to sell, is particularly useful in hedging the downside volatility of stocks.

Equity futures let investors buy (long) or sell (short) equities for future delivery at an agreed price. Long equity call options and warrants give the choice to participate in upside with a smaller outlay, augmenting potential profit while limiting loss to that outlay. Long equity put options allow investors to take short positions.

In equity-linked structures, investors can attempt to enhance returns by **selling put options** to collect premium from willing counterparties when the market is flat. This secures yields much higher than typical deposit instruments, but the higher yield carries the obligation to own underperforming securities should they fall to critical trigger levels. Chapters 7 and 8 examine these instruments.

> **Note.** Shorting put options is not a hedge against downside risk. A short put is a bullish or neutral position that loses as the underlying falls.

1.2.5 Commodities

Commodities are the original asset class that spurred the creation of derivatives. Unlike financial derivatives, a significant portion of commodity derivatives trading comes from regular hedging needs and opportunistic demand from agriculture, mining and manufacturing — real underlying business demand.

Supply and demand determine the long-run direction of each commodity, but short-term trading is influenced by other factors, notably the movement of the **USD**. Supply shocks make commodities volatile — the decisions

of the **Organisation of the Petroleum Exporting Countries (OPEC)** on oil prices being the classic case — which makes the ability to take short positions valuable to a commodity investor.

By convention most commodity futures are denominated in **USD**. When the USD appreciates against other major currencies, commodity prices tend to move in the opposite direction, and vice versa. In extreme shocks to currencies or equity markets, **safe-haven commodities** — gold, silver and platinum — tend to rise. Futures and options on these have become increasingly popular both to hedge and to take advantage of the volatility.

1.2.6 Trading with Indices

Index-linked products give exposure to an asset class, market or security without buying its constituents. Futures, option contracts, ETFs and ETNs are pegged to equity indices, commodity indices and popular currency pairs.

Such indirect exposure lets investors:

- **a.** Adopt cost-effective investment exposure, passively over the long term.
- **b.** Hedge and protect profits, or minimise losses, on an existing outstanding position.
- **c.** Take speculative short-term exposure to a short-term opportunity.
- **d.** Augment exposure through leverage — typically **2–3×** a reference index — using ETFs, ETNs, options, futures and warrants on the index of a particular underlying asset class.
- **e.** Speculate on the potential volatility, or lack of volatility, of a security or an asset-class index.

Volatility itself became tradable. The **Chicago Board Options Exchange Volatility Index (CBOE VIX, "VIX")** measures the market's perceived volatility of the S&P 500 in either direction, derived from the price movement of call and put options. VIX options and futures trade on the CBOE daily. When volatility rises the VIX rises; when market volatility falls the VIX falls. VIX options and futures let investors trade volatility regardless of market direction.

Leveraged ETNs and ETFs on the VIX were introduced to widen access — ETNs such as **XIV, ZIV, VIIX, VIIZ, TVIX, TVIZ**, and ETFs such as **SVXY, UVXY** — allowing investors to buy or short volatility, trading the rise or fall in the percentage volatility of the S&P 500.

These structures are extremely risky. On **5 February 2018** the VIX spiked and more than doubled in a single day. The short-volatility ETF **SVXY** (ProShares) and **XIV** (Credit Suisse) each fell **more than 80% in a single day**, wiping out a few billion dollars overnight across just two instruments.

Trading in both was halted temporarily. There was no recovery for holders of the XIV ETN at the open, because a **termination event** — a price level — had been triggered and the issuing bank closed out the synthetic product to protect itself from further downside, forcing traders caught on the wrong side to realise losses. In subsequent weeks, under pressure to address the highly risky leveraged position its product created, ProShares cut its leveraging of the short and long volatility ETFs **SVXY and UVXY by half**.

1.3 Exchange Traded Derivatives

Exchange-traded derivatives are standardised products, mainly futures and options, traded on centralised, regulated exchanges such as **Singapore Exchange Derivatives Trading (SGX-DT)** or **ICE Futures Singapore (IFSG)** — as opposed to bilateral OTC markets. Each product's characteristics are fixed by standard convention: contract size, underlying asset, delivery or expiration dates, and delivery arrangements.

At expiry every contract is settled, either by physical delivery or by cash settlement. Once a trade is executed between a buyer and seller on the exchange, it is recorded with the exchange's **clearing house**. The contract is ultimately not between the original buyer and seller but between the holders at expiry and the clearing house, which guarantees settlement as originally intended using **pooled initial margins** from both sides of the trade.

Global open interest in exchange-traded derivatives, measured in USD trillions, is published by the BIS in its derivatives statistics; the series runs from 2007 through 2020 for futures and options.

1.4 Over-the-Counter Traded Derivatives

OTC derivatives trade directly between two parties without an exchange. Transactions are seldom terminated or assigned — transferred to a third party — before maturity unless both counterparties agree. Banks, other

large financial institutions, fund managers and corporations are the main participants. Historically the two parties settle bilaterally.

Before the credit crisis that began in **2007**, the OTC derivatives market was largely unregulated. In **2018**, OTC derivatives trading was estimated at **more than five times** the value of derivatives traded on global exchanges.

Currency and interest-rate forwards and swaps make up the bulk of OTC derivatives. OTC currency forwards and swaps are driven largely by trades between international banks; corporates hedging underlying currency positions and some sophisticated investors also participate, but they are the minority.

Credit default swaps are the next largest product group. Until 2007, CDS grew in popularity as institutions used credit insurance to enhance the credit rating and pricing of bonds. This "selling off risk" became a lucrative business: insurers and financial institutions that sold credit insurance generated good revenues as long as there was no major systemic downturn. When mortgage-backed securities began to collapse, CDS — profitable for buyers — became a disaster for sellers, who were obliged to pay out on the insurance they had provided against default.

Characteristic	Exchanges	OTC marke
Contract terms	Standardised	Non-standa
Counterparty structure	Clearing house acts as central counterparty: matches trades, eliminates credit risk	Bilateral ag
Collateral	Margin required	Credit risk
Transparency	Transparent	Less transp
Regulation	Regulated	Largely unr

1.4.1 Regulation of OTC Derivatives Trading

Post-mortem reviews of the 2008 crisis attributed much of the blame to the proliferation of products and to the lack of monitoring and regulation of OTC financial derivatives — especially credit derivatives, sold by investment banks to institutional clients and cascading down to private and retail clients. Instruments that began as tools for risk management became instruments for speculative gain. Warren Buffett's 2002 warning is the most quoted: derivatives are "financial weapons of mass destruction, carrying dangers that, while now latent, are potentially lethal."

In **September 2009** the G20 leaders agreed to push for reform of the global financial system. The derivative commitments were:

- greater transparency in derivatives trading;
- moving more standardised OTC contracts onto exchanges or electronic trading platforms;
- clearing through central counterparties (CCPs);
- reporting to trade repositories; and
- higher capital requirements for non-centrally cleared contracts.

1.4.1.1 Europe In **2012** the **European Market Infrastructure Regulation (EMIR)** set the framework for regulating OTC derivatives in Europe. In line with the G20 commitments, EMIR required European financial institutions and their counterparties to measure, report, monitor and clear obligations, and to manage operational risk and counterparty credit risk on OTC derivative transactions not cleared through a clearing house.

1.4.1.2 United Kingdom In **2013** the UK passed the **Financial Services (Banking Reform) Act**, requiring major universal banks to separate and **ring-fence** retail and commercial banking from investment banking.

1.4.1.3 United States The **Dodd-Frank Wall Street Reform and Consumer Protection Act** was introduced in **2010**. It enhanced the role of the **Commodity Futures Trading Commission (CFTC)** to cover regulation of swap trades.

In **2013** the Federal Reserve Board announced that banking entities covered by **Section 619** of Dodd-Frank must fully conform their activities and investments to the statute and regulations by **21 July 2015**. Section 619 — the **"Volcker Rule"** — prohibits insured depository institutions and their affiliates from engaging in short-term proprietary trading of certain securities, derivatives and other financial instruments for the firm's

own account, subject to exemptions including market making and risk-mitigating hedging. It also imposes limits on banking entities' investments in **hedge funds and private equity**.

1.4.1.4 Singapore Singapore aligned with the G20 recommendations through increased standardisation of derivative contracts, increased central clearing, more reporting to trade repositories, and higher capital requirements for non-centrally cleared contracts reflecting their greater counterparty risk.

In adopting a clearing regime, **MAS** considers three factors:

- **a.** The level of systemic risk posed by the product.
- **b.** The depth and liquidity of the market for the product.
- **c.** The availability of OTC CCPs regulated by MAS as **approved clearing houses (ACHs)** or **recognised clearing houses (RCHs)**.

From **1 October 2018**, MAS required all **SGD and USD fixed- and floating-rate swaps** to be cleared through MAS-regulated CCPs, to make derivatives trading safer and reduce counterparty risk. These are the more widely traded interest-rate derivatives in Singapore.

MAS began with the banks holding the largest share of OTC derivatives activity, to address the largest counterparty credit risks first. Banks whose **gross notional outstanding OTC derivatives exceed S**$20*billion* ∗ ∗*mustclearthroughMAS* − *regulatedCCPs*; *thosebanksaccountedfor* ∗ ∗90**20 billion** gross notional outstanding derivatives contracts booked in Singapore for each of the last **four calendar quarters**.

As in the US, EU and Japan, MAS **exempted intra-group transactions**: they do not transfer risk in or out of a corporate group. MAS also exempted **public bodies** — all central banks and governments, plus international multilateral organisations such as the **Bank for International Settlements, the International Monetary Fund and the World Bank** — because their policy functions may require flexibility that clearing obligations would constrain.

Clearing of interest-rate swaps matters because IRS constitute **more than 90% of interest-rate derivative contracts booked in Singapore** by gross notional outstanding, so extending clearing to them significantly reduces systemic risk in the Singapore financial system.

Clearing obligations are more enforceable where both counterparties book and clear in Singapore. They are harder to enforce for cross-border transactions where the foreign regulator does not require counterparties to seek central clearing; the interaction of jurisdictional rules remains an ongoing challenge. The **US and Japan** have commenced mandatory clearing; the **EU** has put a regime in place that has yet to commence.

For transactions traded in Singapore but booked elsewhere — foreign subsidiaries or local branches of foreign banks — MAS maintains oversight through **consolidated supervision** of local banking groups, engages banks regularly on their booking policies and risk controls as part of banking supervision, and assesses risk from **trade data reported to licensed trade repositories**, which includes OTC contracts booked or traded in Singapore.

With the global step-up in regulatory requirements and cost, banks became noticeably more selective about derivatives business, systematically scaling down investment banking activities including proprietary trading, capital-intensive activities and higher-risk exotic and credit derivatives **from 2008 to 2016**. In **2019** OTC derivatives trading recovered to a degree, growing more rapidly than exchange-traded derivatives.

1.4.2 Markets in Financial Instruments Directive II

MiFID II is a set of foundational regulations and directives governing capital markets in the **European Union**. EU-domiciled banks and their branches are subject to it, and so are banks that deal with EU investors, individual or corporate. Singapore hosts many branch operations of EU financial institutions that must meet MiFID II guidelines.

Singapore brokerage firms and local banks are not bound by the EU directive, but they are likely to be called on to support the compliance processes of their European counterparties, which must report a wide range of transactional data. As good corporate citizens in a global financial centre, Singapore institutions and exchanges can be expected to assist EU banks in meeting those requirements.

Most of the review burden falls on senior managers and compliance teams. Much of the additional workflow and documentation should be automated in systems, which requires both operations and sales staff to ensure verification procedures and data are conducted and recorded in line with MiFID II. An investment professional

should know their organisation's obligations, or its decisions on alignment, and their own role under the new rules.

Role or activity
You execute a trade for a local client involving the purchase of EU securities (e.g. BMW, LVMH) from ar
You have EU clients — for example, a local fund manager working for a private bank with EU-based ultra-high-net-wo
You work for an institution that is a market maker for several EU-listed securities and have a trading des
You distribute EU funds in Singapore to APAC clients.
You work for a subsidiary of an EU institution conducting both underwriting and secondary placement.

A Singapore or APAC distributor must not assume that a foreign regime displaces local obligations: product governance, KYC, suitability, disclosure, conflicts and Singapore licensing all continue to apply.

1.5 Structured Products

Structured products combine one or more derivatives with an underlying investment. The underlying can be a single security, a group or basket of securities, or an index. They are popularly sold as **structured deposits, structured notes or structured funds**, offering investors the chance to optimise yield or participate in movement of the underlying.

Structured products are often OTC and are designed to create greater yield for securities the client already holds, or potential participation in the movement of an underlying security, index or basket. Different combinations and outcomes are designed to attract investors, each with its accompanying risks.

In exchange for returns better than the market, investors bear the risk of having to **convert to, or own, securities** if the underlying reaches — or as long as the price does not reach — a particular trigger level. The structure often requires the investor to have **sold put options** to the counterparty, which can result in accumulating more of the underlying as it falls. Investors must be comfortable bearing those inherent risks. Structures can also involve **leverage**, which magnifies both returns and losses.

1.6 The Singapore Derivatives Market

Singapore is a major hub for derivatives trading in Asia. **SGX-DT**, the derivatives arm of SGX, is a Pan-Asian derivatives exchange and the world's most liquid offshore market for the benchmark equity indices of **China, India, Japan and Taiwan**. It is the leading commodities exchange for contracts on **iron ore and rubber**, and provides **Eurodollar, Euroyen, Japanese government bond and stock index** futures trading for Asian investors. It also offers interest-rate futures for trading and hedging — including hedging loans and deposits and interest-rate swaps denominated in **SGD, USD and JPY** — plus intra-commodity, inter-commodity and inter-market trading and arbitrage services.

SGX exchange-traded derivatives volume and revenue, 2015–2019, show contracts rising from 161.2 million to 240.3 million with revenue between SGD 295.7 million and SGD 459.7 million; volume rebounded in the last two years of that period after falling in 2017.

Singapore is the leading FX hub for Asia and the **third largest in the world after New York and London**, benefiting from the synergy of a large FX derivatives business. It lags the major New York and London exchanges in other asset classes. The commencement of two further derivatives exchanges — **ICE Futures Singapore** and **Asia Pacific Exchange (APEX)** — added breadth to the Singapore market.

The **SFEMC** October 2020 semi-annual survey reported average daily turnover in OTC foreign exchange derivatives of **US$91 billion** in FX swaps and options, a decrease on the previous period that coincided with the global downturn caused by the COVID-19 pandemic. Use current statistics for present-day decisions.

1.7 Summary

- A derivative is a financial instrument whose value is linked to the value or volatility of an underlying reference or asset. Derivatives trade on centralised regulated exchanges or bilaterally in the less regulated OTC market.
- Derivatives and structured products let investors hedge, take a directional view without owning the asset, augment profits, change the payoff profile, profit from a fall, secure a right without obligation, arbitrage, change the nature of an asset or liability, and create synthetic positions.

- Futures, forwards, swaps, options and warrants are the basic or "plain vanilla" derivatives. Exotic derivatives combine types or add features. The building blocks are FX, money market interest-rate instruments, fixed income, equities, commodities, and mixes or indices of these.
- Participants use derivatives to hedge and manage risk on existing positions, to speculate, or to capitalise on perceived arbitrage between assets and markets.

Exam cues

- **Five building blocks:** FX, money market, fixed income, equities, commodities. Indices, baskets and funds are extensions, not a sixth class.
- **Numbers:** money market instruments are **>95%** of global exchange-traded futures and options and **>3/4** of OTC trades 2015–2019; maturities **1 day to 1 year**, typically **<90 days**. FX derivatives are the **second largest** category. OTC traded **>5×** exchange-traded value in 2018.
- **Regulatory dates:** EMIR **2012**; UK ring-fencing Act **2013**; Dodd-Frank **2010**, Volcker conformance **21 July 2015**; MAS mandatory clearing **1 October 2018** for SGD and USD fixed/floating IRS, threshold **S$20bn** gross notional over the last **four** quarters, covering banks holding **90%** of OTC contracts; IRS is **>90%** of SGD-booked interest-rate derivatives.
- **VIX:** the CBOE VIX measures perceived S&P 500 volatility from option prices. On **5 Feb 2018** the VIX more than doubled in a day; **SVXY** and **XIV** fell **>80%**, and XIV hit a termination event. ProShares later halved SVXY/UVXY leverage.
- **MiFID II** reaches Singapore desks: T+1 reporting, **15-minute** post-trade transparency, retail/professional/elective-professional classification, target market, inducement disclosure. A foreign regime does not replace Singapore obligations.
- **Trap:** ETFs and ETNs are **not** derivatives, even where they use futures and options internally. And shorting a put is **not** a hedge against downside risk.

Chapter 2 — Futures, Forwards & Swaps

Futures and forwards are the oldest derivatives, and swaps are built from repeated forwards. This chapter covers their mechanics, pricing, settlement and the four swap families the examination tests.

Learning Objectives

The candidate should be able to:

- Discuss the origins and evolution of futures and futures exchanges.
- Explain the economic importance of futures markets.
- Know the difference between futures and other assets and financial instruments.
- Describe the general features of futures contracts.
- Explain the differences between futures and forward contracts.
- Explain specific items within the contract specification of a futures contract.
- Explain and carry out basic calculations on notional value and margin requirements.
- Describe the different types of financial and commodity futures.
- Explain the role of futures exchanges and how they operate.
- Describe the settlement process of futures exchanges, including SGX-DT.
- Show how leverage can impact the performance returns of futures.
- Explain futures pricing models and how arbitrage opportunities arise.
- Explain and apply the concept of basis in futures contracts.
- Explain how to use interest rate swaps, FX swaps, currency swaps, equity swaps and credit default swaps.

2.1 Introduction — Futures and Forwards

Futures are standardised contracts for the purchase and sale of financial instruments or physical commodities on regulated futures exchanges, with the price determined for future delivery. Originally these were non-homogeneous contracts made directly between buyers and sellers — **forward contracts**. As particular specifications of quality, size and delivery time became widely used, the more popular terms standardised. With the desire for a credit-worthy market-maker or counterparty, margin requirements were set in and the futures exchange was born.

A futures exchange is a marketplace for standardised financial and commodity derivatives where buying and selling is contracted for future delivery.

Exchange	Position
CME Group	The world's largest and most diverse derivatives exchange: **3 billion contracts**
ICE (Intercontinental Exchange)	Second largest, built through mergers and acquisitions among exchanges and clea
ICE Futures Singapore (IFSG)	Created from ICE's acquisition of the **Singapore Mercantile Exchange (SMX**
APEX	Asia Pacific Exchange Pte Ltd, a Chinese-investor-backed exchange, granted MAS
Eurex	The biggest futures exchange in Europe, owned by Deutsche Börse.
Euronext	The second largest in Europe, offering access to Belgium, France, the Netherlands
B3 S.A.	Formerly BM&F Bovespa, of São Paulo — the largest futures market serving Sou
SGX	Operates Asia's largest derivatives market in futures, swaps and options, benefitin
HKEX	In addition to its equity and futures exchange, Hong Kong Exchanges and Clearin
Chinese exchanges	The Shanghai Futures Exchange, Dalian Commodities Exchange and Zhengzhou
Others	Japan Exchange Group, Korea Exchange, National Stock Exchange of India, Mos

Given the interconnectivity of today's markets, investors reach futures through multiple platforms and service providers — banks, broking firms and internet-based brokers.

2.1.1 Futures versus Forwards

Forward contracts preceded futures. The differences decide which instrument fits a given exposure.

Futures
Standardised as to quality, quantity, delivery time and place.
The exchange is counterparty, so there is **no counterparty risk between the actual buyer and seller** — it is realise
Marked to market at the end of each trading day; participants must cover margin shortfalls under exchange rules.
Usually offset (closed out) on the exchange before maturity; live contracts settle on the settlement price and procedures a
Globally, most trading now occurs on electronic platforms linked to the exchanges, with pit trading phased out.

Forwards remain more useful where a player needs a customised solution — a position or hedge in an exotic currency, or a delivery period not readily available on futures. Those needs are met through a customised contract with a financial institution or another commercial counterparty.

Understanding both matters because the directional and leverage effects of futures let an investor express the degree of bullishness or bearishness they hold.

2.1.2 Economic Importance and Impact of the Futures Market

The futures market is highly active and central to the global financial marketplace. It is an important source of market information and often indicates market sentiment.

2.1.2.1 Price Discovery The futures market reflects the current price of a contract for a product to be delivered in the future, providing a platform for ongoing **price discovery** — buyers and sellers contracting and influencing prices now, for future delivery, given available information and predictions. How participants absorb and react to information constantly changes prices. That information includes future economic performance, potential political developments, and forecasts of environmental and climatic change that may or may not come about.

2.1.2.2 Risk Management The futures market is where investors manage risk. Because market prices are transparent and readily available, investors know how much they need to buy or sell. For consumers and buyers of commodities, hedging lowers the eventual cost to consumers by reducing the risk that manufacturers raise prices for key supplies because of spot-market volatility. Business managers and financial investors use suitable futures as part of planning to reach their objectives.

2.1.2.3 Augmenting Volatility Participants use futures to leverage long or short positions. In major financial distress in equity, bond or commodity markets, the leveraging impact of futures and options can significantly augment volatility. The downward pricing pressure from shorting futures usually adds selling pressure on the spot market and underlying assets; hedge funds and traders exploit such situations, and the value of underlying assets can fall faster than normal, destroying market value over a very short time.

Computer-driven trading and electronic platforms have quickened the pace of wealth destruction during shocks and extended its ripple into other asset classes and the economy, so governments have had to intervene in both futures and spot markets in extreme situations to dampen that ripple.

2.2 General Features of Futures Contracts

- **a. Standardised.** Futures are standardised contracts for the purchase and sale of financial instruments or physical commodities on a regulated futures exchange, for future delivery on the delivery date.
- **b. Price driven by the underlying.** Price movement is directly influenced by the underlying asset's price and value. The futures price is set by current demand and supply, which are in turn shaped by composite information and market expectations about the future.
- **c. Finite life.** A contract ends on the expiration date, so market direction and time horizon are central considerations.
- **d. Notional value.** Futures carry a notional value depending on contract size or multiple and the market price at any time. At expiration the parties must fulfil their obligations — one to deliver, the other to receive. Unlike out-of-the-money options or warrants, which go to zero at expiry, a futures contract **retains a notional value** at expiration.
- **e. Lower counterparty risk.** The counterparty is the futures exchange, backed by margin requirements.
- **f. Leverage.** The investor posts only a fraction of contract value as initial margin to gain exposure to the whole position, augmenting both gains and losses.
- **g. Daily mark-to-market.** The value is marked at the end of each trading session and any margin shortfall must be covered.
- **h. Close-out usual.** In most executions buyers and sellers do not deliver or receive the underlying, but square off with the exchange to realise gain or loss.
- **i. Contract-specific terms.** Contract size and units depend on the type of futures; they appear in the contract specification or term sheet. Traders combine futures characteristics with the underlying and other derivatives to craft strategies with different payoff and risk profiles.

2.2.1 Calculating Notional Value for Different Types of Futures Contracts

Notional value depends on the futures price and the multiplier for that contract:

$$\text{Notional value} = \text{futures price} \times \text{multiplier (contract size)}$$

The multiplier differs by contract, and by exchange for the same underlying index.

Contract	Setting	Calc
E-mini S&P 500, Jun 2020	Index at 2,358.50 on 31 March 2020; bid/ask 2,361.00 / 2,361.25; multiplier 50	Purc
E-mini Nasdaq 100, Jun 2020	Contract done at 5,429.25 on 4 April; multiple 20	$20 \times$
SGX FTSE China A50, Sep 2020	Trading at 10,300 on 4 April; multiple 1	1×1
EUR/USD futures, Jun 2020	Last traded at 1.07045; contract unit EUR 125,000	EUR

2.2.2 Leverage

- **a. Initial margin.** The small upfront payment that initiates a position; the investor does not pay for the whole contract at trade initiation.
- **b. Maintenance margin.** The minimum the customer must keep on deposit at all times. While net equity is above it, no request for additional funds arises even after an adverse move.
- **c. Additional margin.** Each account is credited or debited at the end of each trading day and checked against the appropriate margin for all open positions. If the balance falls below maintenance level, additional margin is required and the account must be restored to the **initial** margin level immediately or within a stipulated time.

In Singapore, **SGX** publishes maintenance and initial margin for every futures contract on a daily margin schedule. Margins vary when market conditions change significantly — for example, in extreme volatility.

How margins are used: SGX Nikkei 225 The contract is worth **JPY 500 × the Nikkei level**. At 22,500 the contract value is:

$$500 \times 22{,}500 = \text{JPY } 11{,}250{,}000$$

At an initial margin of **6.5%**:

$$0.065 \times 11{,}250{,}000 = \text{JPY } 731{,}250$$

If the index rallies 7% to 24,075, the contract is worth JPY 12,037,500. The profit is:

$$500 \times (24{,}075 - 22{,}500) = \text{JPY } 787{,}500$$

That is more than **100% of the initial margin** — the leverage effect, and one reason futures appeal.

If the index falls 160 points to 22,340, the contract is worth JPY 11,170,000 and the loss is:

$$500 \times (22{,}340 - 22{,}500) = -\text{JPY } 80{,}000$$

The exchange compares contract value to the account daily and adds profits or subtracts losses. The balance must stay above the maintenance margin — **5%**, or:

$$0.05 \times 11{,}250{,}000 = \text{JPY } 562{,}500$$

Here the balance becomes JPY 651,250, still above the JPY 562,500 maintenance level.

2.2.3 Contract Specifications and Term Sheets

A contract specification sets the basic terms and conditions of the investment. It should state:

- description and standard size or denomination of the underlying;
- contract size and value;
- tick size;
- trading hours;
- price limits;
- last trading day;
- settlement basis;
- final settlement price;
- mark-to-market procedures;
- margin call procedures.

Appendix C lists the main SGX-DT contract specifications.

Term	Meaning
Contract standardisation	Each contract has a standardised size fixed in currency and contract size relative to the un
Contract value	Contract size × current price. At an index level of 1,850, one E-mini contract is USD 50 ×
Tick size	The smallest amount the contract price can move; it varies by contract. An E-mini tick is
Price limits	The maximum the price may move in one day from the previous settlement. Set by the ex
Mark-to-market	Contracts are marked daily as a safety measure. At the end of each day the exchange sets
Margin call	Accounts are credited/debited on the day's profit/loss and checked against open positions.

The SiMSCI futures contract

Term	Specification
Contract size	SGD 200 × Singapore MSCI (SiMSCI) futures price
Contract months	2 nearest serial months and 4 quarterly contract months
Trading hours	8.30 am – 5.15 pm (T session); 6.15 pm – 10.55 pm (T+1 session)
Minimum price fluctuation	0.1 index point, worth **SGD 20**
Daily price limits	On a **15%** move either way from the previous settlement, trading at or within ±15% is allow
Last trading day	The **second-last business day** of the contract month. A business day is one on which the
Settlement basis	Cash settlement
Final settlement price	The SiMSCI value computed on the **Special Quotation** methodology applied to each comp

2.3 Types of Futures

Growth in futures began with farmers, miners, manufacturers and businesses needing to manage risk and uncertainty: a futures contract fixes a price for a period, which makes planning easier. The categories are:

- **a. Commodity futures** — agriculture, energy, metals.
- **b. Financial futures** — equity, interest rates, foreign exchange.
- **c. Others** — innovation has created futures in new areas such as weather.

2.3.1 Interest Rates Futures

Banks lending or borrowing over a period are exposed to rate movements and can partially or fully hedge through interest-rate futures or options on them. More broadly, these contracts let investors and traders take positions on how short-term or long-term rates may move across the yield curve, given fiscal and monetary policy and their view of economic statistics. Interest-rate futures are available on most exchanges, especially in the US, UK and Singapore.

2.3.1.1 Short-Term Interest Rates Futures The most popular short-term contracts are **US Treasury bill futures, Eurodollar futures and Euroyen futures**. The underlying is typically:

- **a.** A time deposit in a **Eurocurrency** — a currency lent or borrowed outside its country of origin — where the rate offered for a specified period to a first-class bank determines the futures price for the same period; or
- **b.** A government treasury bill, where the rate at which an investor is willing to invest for the period forms the basis of the futures price.

$$P = 100 \times (1 - R)$$

where

$$R$$

is the annualised interest rate for the period. At

$$R = 3.50\%$$

:

$$P = 100 \times (1 - 0.035) = 96.50$$

2.3.1.2 Long-Term Interest Rates Futures Used to manage rate exposure or to take a market view. A borrower with outstanding debt may protect against rising rates; a fixed-income manager may sell interest-rate futures to protect a bond portfolio. If rates rise, futures prices fall — rates and bond prices move in opposite directions — and the profit on closing the futures offsets the portfolio decline.

$$\left(1 + \frac{d_1 r_1}{360}\right)\left(1 + \frac{d_2 r_2}{360}\right) = \left(1 + \frac{DR}{360}\right)$$

Rearranged to solve for the second-period rate:

$$r_2 = \frac{360}{d_2}\left(\frac{360 + DR}{360 + d_1 r_1} - 1\right)$$

where

$$d_1$$

is days in the first period,

$$d_2$$

days in the second,

$$D$$

days for the entire period,

$$r_1$$

the first-period rate,

$$r_2$$

the second-period rate and

$$R$$

the rate for the whole period.

Worked example. Spot date to 15 September is 45 days; to 15 December is 135 days. The 45-day rate is 1.25% and the 135-day rate is 1.75%. The implied 90-day forward rate is:

$$\left(1 + \frac{45 \times 0.0125}{360}\right)\left(1 + \frac{90 \times r_2}{360}\right) = \left(1 + \frac{135 \times 0.0175}{360}\right) = 1.006563$$

Solving gives

$$r_2 = 2.00\%$$

.

2.3.1.3 US Treasury Futures US Treasuries are among the most liquid financial products in the world, and Treasury futures and options support hedging, income enhancement, duration adjustment, rate speculation and spread trades. CME contracts mirror the **2, 5, 10 and 30-year** tenures. On SGX, **mini-Japanese Government Bond futures** are the popular long-bond contract.

Each US Treasury futures contract has a face value at maturity of **USD 100,000**, except the **2-year and 3-year** contracts at **USD 200,000**. Prices are quoted in points per **USD 2,000** for 2- and 3-year contracts and per **USD 1,000** for all others.

Profit example. A fund manager expects Treasury yields to rise. He sells 10 contracts of Jun 2017 5-year Treasury Note futures at 117.245 and buys them back at 116.000:

$$10 \times (117.245 - 116.000) \times 1{,}000 = \text{USD } 12{,}450 \text{ profit}$$

The 5-year tick is **¼ of 1/32** of a point, worth:

$$0.25 \times 0.03125 \times 0.01 \times 100{,}000 = \text{USD } 7.8125$$

so the gain is

$$(117.245 - 116.000) \times 1{,}000/7.8125 = 159.36$$

ticks, and

$$10 \times 159.36 \times 7.8125 = \text{USD } 12{,}450$$

.

Loss example. On Thursday 7 April 2017 a surprise US missile strike on Syrian air bases pushed yields down and bond prices up; the 10-year yield fell to 2.31%, the lowest since the March rate hike when it was 2.6%. A trader reads the June 2017 10-year T-Note futures as having risen only modestly, from 125 8/32 to 125 12/32, and buys 10 contracts at 125 12/32 expecting more buyers. Prices reverse and the contract ends the day at 124 8/32:

$$10 \times (124\,\tfrac{8}{32} - 125\,\tfrac{12}{32}) \times 1{,}000 = -\text{USD } 11{,}250 \text{ loss}$$

The 10-year tick is **½ of 1/32**, worth USD 15.625, so the loss is **−72 ticks**, and

$$10 \times -72 \times 15.625 = -\text{USD } 11{,}250$$

.

2.3.2 Currency Futures

Currency futures are contracts to buy or sell a specified amount of one currency against another at a set price and future date. Financial institutions, investment managers, corporations and private investors use them to manage currency risk or to profit from rate movements. They were first introduced at the **Chicago Mercantile Exchange in 1972**, after the fixed exchange-rate system and gold standard were discarded.

Currency futures volume tracks the underlying spot market. The BIS Triennial Survey reported global FX turnover of **USD 5.1 trillion per day** in 2016, **5.5% below** USD 5.4 trillion in 2013 (the source text prints "5.1 billon"; read trillion). Part of the decline reflected lower growth in global trade and capital flows, still below pre-crisis levels.

More significant was a structural shift towards using FX for hedging rather than risk-taking, a consequence of reduced proprietary trading and tighter risk controls. Banks consolidated and cut costs over the decade. The number of dealer banks willing to warehouse risk declined while **non-bank market-makers** gained a stronger foothold as liquidity providers, sometimes bypassing traditional bank intermediaries. Despite the overall decline, trading in most FX derivatives — particularly FX swaps — kept growing, and emerging-market currencies, especially the **renminbi**, gained share. Singapore kept the top spot for FX trading, which supports FX-derivative business here.

2.3.2.1 Types of Foreign Exchange Futures Contracts Offered Futures activity mirrors demand in the underlying currency market, so major pairs dominate. On the **CME**, currency futures are grouped into majors against the USD, emerging-market currencies against the USD, cross-rates not involving the USD, and **E-Micro** contracts trading at **1/10th** the size of the regular contract. On **Euronext**, trading concentrates on the more active majors — **EUR/USD, GBP/USD and GBP/EUR**.

On **SGX-DT**, Asian currencies trade against the USD, among themselves, and against the non-deliverable forward for the **renminbi (CNY), Indonesian rupiah (IDR), Taiwan dollar (TWD), Indian rupee (INR), Korean won (KRW), Malaysian ringgit (MYR) and Philippine peso (PHP)**. SGX remains one of few venues offering hedging, trading and investment exposure in these currencies, and Singapore's FX derivatives market benefits from demand for emerging Asian currencies as China, Indonesia and India expand.

2.3.2.2 Interest Rate Parity Theory The theory holds that the forward (futures) premium or discount between two currencies should equal the difference in domestic interest rates for securities of the same maturity, ignoring small transaction costs. Where the relationship is violated, arbitrageurs can earn risk-free profit and their actions force futures and spot rates back into line.

$$F = S \times \frac{1 + R_c\left(\frac{n}{360}\right)}{1 + R_b\left(\frac{n}{360}\right)}$$

where

$$F$$

is the forward (futures) rate,

$$S$$

the spot rate,

$$R_b$$

the annualised base-currency rate,

$$R_c$$

the annualised counter-currency rate and

$$n$$

the number of days.

Worked example. Base currency GBP, counter currency USD. Spot GBP/USD = 1.2422;

$$R_b$$

= 1-year GBP LIBOR = 0.70444%;

$$R_c$$

= 1-year USD LIBOR = 1.79900%.

$$F = 1.2422 \times \frac{1 + 0.01799\left(\frac{360}{360}\right)}{1 + 0.00704\left(\frac{360}{360}\right)} = 1.2557$$

So GBP 1 = USD 1.2557, a forward price of GBP/USD 1.2557.

2.3.3 Equity Index Futures

Equity index futures replicate the future performance of an underlying equity market index, letting investors capitalise on broad market movements without changing their dollar investment in stocks. An index is a representative basket or portfolio of stocks or commodities; an equity index measures the performance of a selected basket representing the whole market. Indices may be **price-weighted** (DJIA) or **market-capitalisation weighted** (MSCI).

Buying the index raises or lowers exposure to specific markets or sectors and is a cost-effective way to gain equity exposure. Most exchanges list futures on their own indices, while some position themselves regionally or globally. SGX has consolidated its position as a regional platform for futures, options and structured warrants on Asian markets, and offers index futures relevant to managers benchmarked to indices such as MSCI, so they can rebalance rapidly, hedge positions and exploit country-specific opportunities. The most successful launch has been the **SGX FTSE China A50 Futures**, which continues to trade heavily.

SGX equity index futures include **MSCI Singapore, Straits Times Index, FTSE China A50, FTSE China H50, FTSE Malaysia, FTSE Philippines, FTSE Taiwan, FTSE Thailand, Nikkei 225, USD Nikkei 225, Mini Nikkei 225 and Nifty 50**.

2.3.3.1 Differences between Equity Index Futures and Stocks

Feature	Equity index futures	Stocks
What is owned	A cash-settled contract on the index	Legal ownership of shares
Settlement	Daily mark-to-market; cash settlement at expiry	T+2 settlement; no daily margin
Leverage	Margin only	Full consideration, or margin financing
Short selling	Direct, without borrowing stock or an uptick rule	Requires stock borrow, with recall and cost risk
Income	No dividend entitlement	Dividends and voting rights

2.3.3.2 Pricing Equity Index Futures

$$F = S\left[1 + r\left(\frac{t}{360}\right)\right] - D$$

where

$$S$$

is the spot index,

$$r$$

the short-term rate,

$$t$$

days to expiry and

$$D$$

dividends paid before expiry.

Index futures do not trade precisely at fair value all the time; the difference between futures and spot is the **basis**. Fair value is normally expected to be positive — futures above spot — because under normal conditions financing costs, reflected in rates such as LIBOR or SIBOR, exceed dividend yields. Loose post-2008 monetary policy has at times reversed this.

If futures rally much above fair value, an arbitrageur buys the equity portfolio and sells index futures, trading a basket or subset of constituents or using electronic systems to trade all or nearly all of them. Buying stocks and selling futures bids stocks up or pushes futures down until equilibrium returns.

2.3.4 Single Stock Futures

A **Single Stock Futures (SSF)** contract tracks the price of the underlying stock. On SGX the buyer agrees to pay a specified price at a predetermined date and the seller is obligated to deliver the value of the stock at that date; SGX SSFs are **cash settled**, with no physical delivery. Portfolio managers use them to protect against adverse moves and to diversify risk using margin financing. SSFs carry **no shareholder privileges**.

Advantages:

- **a.** Leveraged, based on several thousand shares, so the leverage effect amplifies gains and losses.
- **b.** Lower transaction cost than buying the stock, because the outlay is only the margin.
- **c.** For the same reason, offshore investors bear less currency exposure — they invest only the margin, not the full underlying value.
- **d.** Ease of short selling lets investors participate in and profit from market downturns.
- **e.** Once an exchange admits a contract it must ensure liquidity, transparency, fairness and timeliness in market-making with reasonable bid-offer spreads, and investors must be able to close out, typically electronically. The exchange guarantees clearing-house support from designated market-makers.

The main disadvantage is that SSFs can invite short selling of the stock, reducing underlying prices and causing downward spirals in uncertainty and shocks. The market's ability to absorb increased downward volatility is an important consideration before a new contract is introduced.

2.3.5 Commodity Futures

Physical-market storage, transport, quality and delivery-location differences create basis and roll risk. Commodity contracts are grouped as agriculture, energy and metals.

2.3.5.1 Gold and Other Precious Metal Futures In potential military conflicts involving major powers there is a tendency to **flight to safety**. In early April 2017, tension escalated between the US and Syria–Russia over banned chemical weapons and the US missile response, while US–North Korean nuclear-testing tension also rose. Gold futures moved up **4% to USD 1,290.70/ounce** just before Easter; when tension eased the price retraced.

Calculating notional profit for gold futures On 10 April 2017, with US military tension involving Syria, North Korea and Russia, June gold gained 0.39% on escalating concern. At the start of European trading, Jun 2017 gold futures on ICE traded at **USD 1,259.10/Troy ounce**. A trader expecting escalation bought **100 contracts**, each **100 Troy ounces**. Gold rose to USD 1,288.50/ounce before Good Friday, 13 April.

$$100 \times (1{,}288.50 - 1{,}259.10) \times 100 = \text{USD } 294{,}000$$

Conversely, when interest rates rise, gold and other precious metals come under pressure because they yield no interest or dividend; when perceived risk is low, investors prefer risky assets and safe havens fall. Gold is also increasingly intertwined with **gold ETFs**: according to the World Gold Council the largest growth in gold demand in **2016 (+660%)** came from ETFs.

Beyond financial markets, gold has jewellery demand: **China and India** are the largest sources, and more than half of physical gold trading is estimated to come from Asia. Interruption of precious-metal imports creates underlying downward pressure.

Gold futures on the **CME and ICE** are quoted in **100 troy ounces**. In Singapore the **SGX Gold contract is dormant** and not traded.

2.3.5.2 Base Metal and Ferrous Metal Futures Demand and prices depend on economic growth, with underlying demand from major traders, miners and manufacturers wanting physical delivery. Base and ferrous metals rose significantly during China's infrastructure expansion, especially **2002–2012**; China's slowdown **2012–2016** amid overcapacity produced a sharp fall. Expected demand from proposed US infrastructure fiscal spending in **2017–2018** lifted prices again.

Steel pricing is not uniform and US–China trade disputes and anti-dumping allegations disrupt price discovery. The US announced **steel and aluminium tariffs** to protect local manufacturers, which widens the difference between US and non-US exchange prices. Steel comes in different types and grades, and SGX steel quotes from China differ from CME quotes featuring US steel.

Iron ore and steel are key SGX contracts. The **London Metal Exchange** is the largest global metal exchange and remains a main venue for iron ore and steel futures, while Chinese exchanges have grown strongly. Singapore is positioned as a key **clearing centre for iron ore** traded elsewhere, especially China.

SGX iron ore and steel contracts:

- **a.** SGX TSI Iron Ore CFR China (62% FE Fines) Index
- **b.** SGX HRC CFR ASEAN Steel
- **c.** SGX Platts Iron Ore CFR China (Lump Premium) Index
- **d.** SGX MB Iron Ore CFR China (58% FE Fines) Index
- **e.** SGX TSI Iron Ore CFR China (58% FE Fines) Index

2.3.5.3 Crude Oil Futures Crude oil futures remain a focal point for investors because of dramatic price volatility: a high of **USD 155/barrel** just before the Global Financial Crisis, a plunge to **USD 48** in 2009, a rebound above **USD 100** for most of 2010–2013, then new lows of **USD 29.60** in 2015 amid structural change and competition from fracking. Futures may have helped producers lock in higher prices on some occasions, but extended declines offer producers little protection. An **OPEC** agreement to limit supply eventually held crude in a **USD 55–65** range in the later part of 2018, ahead of the Saudi Aramco listing. Crude remains one of the most actively traded instruments.

2.3.5.4 Energy — Natural Gas, Other Refined Products and Coal Futures The volatility of energy futures is relatively well correlated. Like crude, other traditional energy products — **LNG, jet fuel, gas oil, kerosene, coal** — have been volatile over the decade. For downstream refined products there may be a **lag effect** before the full impact of price changes works through inventory. Coal, though environmentally unattractive, remains a viable substitute in many developing economies.

Singapore is a major petrochemical refining base, and SGX lists futures on diverse petrochemical and refined products from regional players, plus **Indonesian coal** futures.

2.3.5.5 Energy — Electricity Futures Energy futures have been available in the US for many years. Multiple production sources (LNG, coal, hydroelectricity) across regions, many users (towns, cities, industries), and the need to optimise non-peak hours against fluctuating demand drove the development of electricity futures.

On **SGX** there are monthly and quarterly electricity futures, still relatively new and serving specific needs of Singapore industrial users. The local market is thermal, with **more than 90% dependent on natural gas**. Generating companies offer to sell into the spot market **every half-hour**, matching prevailing demand, but prices fluctuate during tight supply from plant outages or network constraints. Consumers buying directly from the spot market are exposed to those fluctuations. Electricity futures offset expensive electricity for consumers or secure higher proceeds against cheaper off-peak electricity.

2.3.5.6 Agriculture Futures A wide range of agriculture contracts — **grain, soft commodities, livestock, dairy and forest products** — has existed for decades, driven mainly by the supply and business needs of farmers, producers, wholesale consumers, investors and speculators.

2.4 Contract Settlement and Clearing Process and Orders on the SGX

2.4.1 Modes of Settlement

A buyer or seller can settle a futures position in three ways.

2.4.1.1 Delivery or Hold to Expiry Every contract has an expiry date. On that date the buyer accepts delivery of the underlying and the seller liquidates by delivering at the agreed price. Commodity contracts state physical delivery to an approved warehouse; currencies and some financial instruments such as the S&P 500 call for **cash settlement**. Every contract specifies the last trading day before expiry.

Investors must know the expiry dates, because as expiry approaches **liquidity decreases** as traders roll to the next contract month.

2.4.1.2 Offset Position The simplest and most common route for financial futures. Buying two S&P 500 contracts is offset by selling two; selling two is offset by buying two. To limit overnight risk, many traders exit all positions and are **flat** at the end of each trading day.

2.4.1.3 Roll Position Rather than closing out, the investor closes the expiring contract and simultaneously opens a position in a later month, maintaining exposure and deferring settlement.

2.4.2 Minimum Price Movement for Contracts on SGX-DT — Understanding Tick Sizes

The tick size is three things at once: the minimum movement in the quoted price, the minimum increment between bid and offer, and a dollar amount per contract. Convert between them using the contract multiplier.

2.4.3 Listing of Derivatives Contracts on SGX-DT

SGX-DT lists five categories:

Category	Contracts
FX	USD/SGD, AUD/USD
Equity indices	SiMSCI, Nikkei 225 and the wider Asian roster
Interest rates	Eurodollar, Euroyen LIBOR
Commodities	PLATTS Singapore Fuel Oil 380cst, SICOM TSR20 Rubber
Dividend indices	Nikkei Stock Average Dividend Point Index

2.4.4 Derivatives Clearing Process on SGX-DT

SGX-DC clears:

- **a.** SGX-DT products;
- **b.** OTC commodity trades; and
- **c.** OTC financial derivatives, through trade registration.

Margin cycles run **three times intraday** — late morning, afternoon and post-end-of-day. Settlement covers variation settlement, option premiums, OTC coupons, price alignment interest and clearing fees. **ICE Clear Singapore** clears IFSG products.

2.4.5 Mutual Offset System

Since **1984**, the CME mutual-offset arrangement lets positions opened on one exchange be closed on the other, facilitating global clearing. It applies to specific SGX products — **NK, NU, FEM and FCH** — and is product-specific, not general.

2.4.6 Potential Role of Singapore as a Global Clearing Centre

SGX-DC is a **CFTC-designated clearing organisation**, has been assessed for EU equivalence, and **ESMA** recognises SGX-DC and CDP. Combined with the CME mutual-offset links, this positions Singapore as a high-quality global clearing centre for futures and derivatives; as technology changes how transactions are performed, the clearing process must continue to adapt.

2.5 Futures Pricing Model

2.5.1 Spot Price

The spot price is the current market price of the underlying asset or commodity.

2.5.2 Cost of Carry Model

The relationship between futures and spot prices is summarised by **cost of carry**. Carry includes financial costs — interest on bonds, interest on margin accounts, interest on money borrowed to buy a security — and economic costs such as the opportunity cost of taking the initial position. The **net** cost of carry is all costs of holding the equivalent cash position less the returns that position would have generated: financing at prevailing rates, and dividends or bonuses from shares, or the average return of an index in the case of index futures. In futures the investor deposits only a fraction of position value as margin.

The model assumes arbitrage between cash and futures eliminates all pricing imperfections. For simplicity it assumes:

- **a.** the contract is held to maturity, so a fair price can be derived;
- **b.** there are no transaction costs or margin requirements;
- **c.** there are no restrictions on short selling; and
- **d.** investors can borrow and lend at the same rate.

Once arbitrage has removed distortions, the fair futures price is spot plus the net cost of carry from today to expiry:

$$F_{0,t} = S_0(1 + C_{0,t})$$

2.5.3 Cash-and-Carry Arbitrage

Position	Cash-and-carry (futures too high)	Reverse cash-and-carry (futures too low)
Debt	Borrow to buy the spot asset	Lend the proceeds of a short sale
Physical	Buy and hold the spot asset	Short or borrow the spot asset and deliver
Futures	Sell futures	Buy futures

Forward case. Gold spot is USD 1,275/oz, carry is 2.25% for the year, and the futures price is USD 1,305.

$$F_{0,t} = 1{,}275 \times (1 + 0.0225) = 1{,}303.69$$

Futures at USD 1,305 exceed fair value of USD 1,303.69, so the arbitrage is to sell futures and buy spot. Borrow to buy 100 oz:

$$1 \times 1{,}275 \times 100 = \text{USD } 127{,}500$$

Sell one 100 oz contract at USD 1,305 for USD 130,500. Repay principal plus interest:

$$127{,}500 + (0.0225 \times 127{,}500) = \text{USD } 130{,}368.75$$

Profit:

$$130{,}500.00 - 130{,}368.75 = \text{USD } 131.25$$

— a negligible **0.1%** opportunity. To improve it, the futures price must be higher, the spot lower, and the borrowing cost materially lower.

Reverse case. Spot gold is USD 1,285.

$$F_{0,t} = 1{,}285 \times (1 + 0.0225) = 1{,}313.91$$

Futures at USD 1,305 are below fair value of USD 1,313.91, so buy futures and short spot. Borrow 100 oz of gold at **1.00%** for a year and sell it at spot for:

$$100 \times 1{,}285 = \text{USD } 128{,}500$$

Lend the proceeds at **2.25%**, earning USD 2,891.25, so after a year:

$$128{,}500 + 2{,}891.25 = \text{USD } 131{,}391.25$$

Buy 100 oz one-year futures at USD 1,305 for USD 130,500 to cover the borrowed gold. Surplus:

$$131{,}391.25 - 130{,}500.00 = \text{USD } 891.25$$

Cost of borrowing the gold:

$$128{,}500 \times 0.01 = \text{USD } 1{,}285.00$$

Overall:

$$891.25 - 1{,}285 = -\text{USD } 393.75 \quad (-\mathbf{0.306}\%)$$

So there was **no arbitrage**. To create one, the gold borrowing cost must be much lower or free while the lending rate rises, or the futures price must fall or spot rise.

Arbitrage rarely exists or lasts more than milliseconds: computer-based trading has pushed efficiency past human levels and any differential disappears.

2.5.4 Expectancy Model

Futures prices are expected future spot prices. If more traders expect the futures price to rise than to fall, the current futures price is bid up. What matters is not the cash-futures relationship but the relationship between the **expected spot price at expiry** and the futures price. Both the cost-of-carry and expectations models explain part of futures price movement.

2.6 Basis in Futures Contracts

The difference between futures and spot is the **basis**.

When markets are in **contango** — typically stable or bullish conditions — the basis is usually positive, with futures above spot because of carrying costs such as financing and storage. In **backwardation** — often bearish or stressed conditions — the basis can be negative, with futures below spot.

Because a contract settles at the cash price on expiry, **futures and spot converge as expiry approaches**.

> **Convention warning.** The official guide states on page 46 that contango gives a *positive* basis with futures above spot, and then on page 47 defines and uses basis as *spot minus futures*, which makes contango's basis negative. The two are the same economic relationship described with opposite signs. This guide uses **basis = spot − futures** throughout, so contango produces a negative basis. State your convention in any answer.

2.6.1 Factors Affecting Basis

- **a.** Cost or return of carry.
- **b.** Time to maturity.
- **c.** Yield-curve changes.
- **d.** Relative liquidity of cash and futures markets.
- **e.** Market rates versus administered rates.
- **f.** Expectations of market participants.
- **g.** Coupon or income differences between the hedged asset and the deliverable.

2.6.2 Basis, Contango and Backwardation

Basis risk persists for three reasons: the cash and futures markets are not perfectly correlated, the exposure being hedged may not match the deliverable, and the hedge may be closed before expiry, so convergence is incomplete. A short hedger's effective price is the futures price plus the ending basis:

$$V_t = S_t + (F - F_t) = F + (S_t - F_t)$$

Roll yield arises when a position is rolled from an expiring contract to a later one; under contango the roll costs money, under backwardation it earns.

2.7 Swaps

Swaps are customised OTC agreements to exchange cash flows over several periods on a notional amount. Singapore's FX market is heavily swap-based: forex forwards and swaps were **70–75% of FX turnover in 2016–17**, and currency interest-rate swaps **60–66% of FX derivatives**.

2.7.1 Interest Rate Swaps

One party pays fixed and receives floating, or the reverse. The notional is normally **not exchanged**; the swap rate or spread and the benchmark reset dates matter.

Plain-vanilla example: USD 100 million, paying 5% fixed against six-month LIBOR. With LIBOR at 4.2% in September 2014, the fixed payer's net receipt is:

$$0.5 \times (0.042 - 0.05) \times 100\text{m} = -\text{USD } 0.4\text{m}$$

so Bank B receives USD 2.1m on the fixed leg and pays USD 2.5m floating. With LIBOR at 4.8% in March 2015 the floating leg is USD 2.4m and the net position reverses.

$$\text{fixed payment} = \text{notional} \times \text{swap rate} \times \frac{\text{days}}{360}$$

The **swap spread** is the swap rate less the yield on the government bond of matching maturity. The **LIBOR swap curve** plots swap rates by tenor. A party with an existing swap can enter an **offsetting swap** to reduce or cancel the exposure.

2.7.2 FX Swaps

An FX swap exchanges currencies on a near leg and reverses the exchange on a far leg, managing short-term currency funding. The near-leg rate is the spot rate and the far leg is set by the swap points for the value dates. Example: a firm needing EUR 5 million for six months sells USD for EUR spot and agrees to reverse the exchange in six months at the forward rate, locking the funding cost.

2.7.3 Currency Swaps

A currency swap exchanges **principal and interest** in two currencies. A Singapore bank needing USD swaps SGD principal with a US bank needing SGD, each paying the other's interest, transforming both the currency and the interest exposure for the life of the deal.

2.7.4 Commodity Swaps

A fixed price is exchanged for a floating commodity price, letting a producer or user lock or transform price exposure without repeatedly rolling futures. SGX lists standardised commodity swaps.

2.7.5 Equity Swaps

One party receives the **total return** of an equity or index and pays a fixed or floating rate, creating synthetic equity exposure without ownership.

Four-period example on a notional of USD 200 million against the S&P 500, where the manager pays the index total return plus 2% and receives 4% fixed:

Period	Index level	Index return	Swap cash flow (USD m)	Portfolio value (USD m)	Reported return
31 Jul 2017	2,470.20	—	—	400.00	—
Period 1	2,569.00	+4%	−2.00	414.00	+3.50%
Period 2	2,363.48	−8%	+10.00	390.88	−5.58%
Period 3	2,623.47	+11%	−9.00	424.88	+8.70%
Period 4	2,754.64	+5%	−3.00	443.12	+4.28%

The manager's realised return each period is the portfolio return plus the swap cash flow, which is why a falling index produces a positive swap flow.

2.7.6 Credit Default Swaps

A protection buyer pays a spread, usually in basis points quarterly, to a protection seller. On a defined **credit event** the seller pays under cash or physical settlement. Typical notionals are **USD 10–20 million**.

Premium arithmetic: at **160bp** on **USD 20 million**, quarterly:

$$160\text{bp} = 1.60\% \;\Rightarrow\; 20\text{m} \times 1.60\% = \text{USD } 320{,}000 \text{ p.a.}$$

so each quarter is one quarter of the annual amount, **USD 80,000** — equivalently 40bp of notional per quarter.

2.7.6.1 CDS Spreads The CDS "spread" is not the same as a benchmark spread; it is the premium for credit protection on the reference entity. Spreads widen with credit deterioration — the PIIGS (Portugal, Ireland, Italy, Greece, Spain) episode is the standard example.

2.7.6.2 Credit Event The ISDA-defined events are bankruptcy, failure to pay, restructuring, obligation acceleration, obligation default, repudiation/moratorium and (in some definitions) government intervention. Debt restructuring is the most negotiated of these.

2.7.6.3 Cash Settlement The protection seller pays the difference between par and the market price of the deliverable obligation after the event.

2.7.6.4 Physical Settlement The protection seller pays par and receives the deliverable obligation, so the buyer delivers the defaulted bond and receives face value.

Naked CDS creates large synthetic credit exposure. The AIG episode during the Global Financial Crisis showed how leverage, opacity and counterparty concentration amplify credit risk: AIG had sold protection far in excess of its capital and required a public rescue.

2.8 Summary

Futures are standardised, margined, daily-settled and cleared; forwards and swaps are usually customised OTC. Pricing is driven by carry and parity, but basis, liquidity, margin and credit constraints matter. Swaps exchange exposures rather than principal ownership and are powerful but documentation- and counterparty-dependent.

Exam cues

- **Notional** = price × multiplier. **Margin** = initial margin % × notional. Learn the Nikkei example: 22,500 × JPY 500 = JPY 11,250,000; 6.5% = JPY 731,250; a 7% index rise is a 107% return on margin.
- **Short-rate quote**

  $$P = 100(1 - R)$$

  , so rates up → price down. Long-rate pricing chains the two periods to the whole period and solves for

  $$r_2$$

  (the 45/135/90-day example gives **2.00%**).

- **US Treasury:** face **USD 100,000** except 2- and 3-year at **USD 200,000**; 5-year tick **USD 7.8125** (¼ of 1/32), 10-year tick **USD 15.625** (½ of 1/32).
- **Interest-rate parity:**
$$F = S \times (1 + R_c n/360)/(1 + R_b n/360)$$
; the GBP/USD example gives **1.2557**.
- **Arbitrage:** cash-and-carry when futures is **too high**, reverse when **too low**. The reverse gold case loses **USD 393.75 (−0.306%)** once gold borrowing at 1% is counted.
- **Basis = spot − futures.** Contango puts futures above spot, so under this definition contango gives a **negative** basis. Some texts define basis the other way round; state your convention before using it. Convergence forces basis to zero at expiry.
- **CDS:** 160bp on USD 20m is USD 80,000 a quarter, USD 320,000 a year.
- **Trap:** a futures contract retains notional value at expiry; an out-of-the-money option or warrant goes to zero. And the notional is never exchanged in an interest-rate swap.

Chapter 3 — Futures Strategies

Who trades futures, how orders behave, which spread isolates which relationship, and how to size a hedge. This is the most calculation-heavy chapter after Options.

Learning Objectives

The candidate should be able to:

- Understand and explain the participants and their roles in the futures market.
- Explain the various types of futures orders, futures trades and trading strategies.
- Describe and explain the different types of spread trades.
- Explain how to identify and manage financial risks using futures.
- Describe and demonstrate how to develop and manage hedging programmes for various cash positions.
- Describe and demonstrate how to develop and manage hedging for short-term and long-term interest rate risks.
- Explain how to manage a hedge programme for equity risk exposures.
- Describe and demonstrate arbitrage strategies using short-term and long-term interest rate futures.

3.1 Market Participants

The main participants are:

- **a.** Hedgers
- **b.** Speculators
- **c.** Market makers
- **d.** Proprietary trading firms
- **e.** Arbitrageurs
- **f.** Portfolio managers
- **g.** Hedge funds
- **h.** Central banks, sovereign funds and regulators

As volatility rises, participants with different motivations turn to derivatives to hedge or to maximise return. The same firm can act as speculator, hedger and arbitrageur at once, and knowing which group is driving the market helps decide whether to follow or oppose the direction. Growth in hedge funds and alternative assets has driven innovative derivative strategies, and growth in wealth management has pushed institutions to offer structured products built from futures and other derivatives.

3.1.1 Hedgers

Hedgers hold a position in the underlying asset or commodity and use futures to reduce or limit the risk of an adverse price change. Producers such as farmers sell futures on the crop they grow to hedge a price fall, which makes long-term planning easier. Industrial users buy futures to fix input costs — a food processor fixing corn or wheat, an airline hedging fuel, a jeweller hedging gold and silver — so they can plan on a fixed cost for core supplies and stabilise the price passed to customers.

3.1.2 Speculators (Individual Traders)

Many speculators are individuals trading their own funds, expressing a view on a market's direction. Electronic trading has levelled the field by improving access to price and trade information; speed and ease of execution, with modern risk management, give individuals access to markets and strategies once reserved for institutions.

3.1.3 Market Makers

Market makers are trading firms contractually obliged to provide liquidity, continually quoting both bids and offers, usually in exchange for reduced trading fees. Electronic market makers now supply much of the liquidity that lets large transactions occur without moving price substantially. They profit from capturing the **spread** — the small difference between bid and offer across many transactions — or by trading related futures markets where they see price or profit opportunities.

3.1.4 Proprietary Trading Firms

Prop shops profit directly from their traders' activity. They supply education and capital so traders can execute many trades a day, gaining more leverage than they would on their own account and access to research and strategies developed by larger institutions.

Some large banks retain proprietary units. The most famous loss is **Barings in 1995**, where a single proprietary trader's excessive risk-taking on **Nikkei 225 futures** accumulated losses **twice the bank's capital**. Poor controls and supervision led several banks to close proprietary units, but losses from distressed debt, CDOs, CDS and **Delta One** instruments continued to be reported well after the crisis.

> **Delta One.** Desks provide OTC derivative structures pegged one-to-one against an underlying, usually an index. The bank trades the underlying and related derivatives — sometimes swaps with other banks — seeking arbitrage profit or profit from holdings moving against the direction of the structure sold. Trades are typically very large because margins are extremely fine, and because most are OTC they are hard to monitor or regulate.

Financial institutions also hedge their sale of derivatives to institutional and private clients through futures and options, and ETF and ETN issuers manage daily offsetting futures and options positions against their long and short baskets through the proprietary desk.

3.1.5 Arbitrageurs

Arbitrageurs seek riskless profit from disequilibrium between cash and futures markets. They take no directional bet, instead exploiting price differences between related markets or markets with a fixed relationship — the gap between implied and market value. They rely on computer systems and models, because windows close very fast: computers spot the opportunity and arbitrage it away. Arbitrageurs improve liquidity and keep the pricing relationship between futures and the underlying consistent, and they usually work for market makers and proprietary firms.

3.1.6 Portfolio Managers

A portfolio or investment manager invests or hedges the assets of a mutual fund, ETF or closed-end fund, implementing the strategy and managing day-to-day trading. Futures are often used to raise or lower overall market exposure **without disrupting a portfolio balance that took significant effort to build**.

3.1.7 Hedge Funds

The deep, liquid futures market lets hedge funds execute large transactions and adjust exposure without changing security holdings. A hedge fund is a managed portfolio using advanced strategies to maximise return, absolutely or against a benchmark. The name is historical: early funds shorted the market to offset bear-market risk. On occasion hedge funds have exploited market weakness by shorting futures collectively, adding sharp downward pressure to both futures and spot.

3.1.8 Sovereign Funds, Central Banks and Regulators

Collective selling by speculators and hedge funds has sometimes forced governments to intervene in spot and futures markets to counterbalance speculative forces and stabilise the market. In severe cases, tightening liquidity, capital flows and trading activity has been used to break the momentum in both markets.

During the **1997 Asian Currency Crisis**, several Asian currencies and equity markets were driven down under persistent heavy selling, but the speculation was broken when some governments intervened in spot and futures markets and introduced **capital and currency exchange controls**, locking speculators into short currency positions they could not unwind while incurring interest cost.

Day to day, many governments manage their currency within a trading band and participate in spot and futures markets to influence its direction.

3.2 Types of Orders on the SGX

Most order types are common across futures and stock exchanges.

Order type	Behaviour
Limit order	Carries a specified price and quantity — the price the trader will pay, or wants
Market-to-limit (MTL) order	Has a validity time and can be entered as Fill-and-Kill (FAK), Fill-or-Kill (FO
Market order	Entered while orders are continuously matched, it matches at the best possible
Stop order (limit, MTL or market)	Hidden until its trigger is met, then converted to a limit, MTL or market order
Market-if-touched (MIT) order	An order to buy or sell a contract **below** or **above** the market, held until the t
Session state order (SSO)	Triggers entry of an order into the book only when the market transitions into

3.2.1 Other Types of Orders — Packs and Bundles

Packs and bundles were introduced by **CME Group** for Eurodollar contracts; NYSE Euronext LIFFE offers them for STIR Euribor and Sterling futures. They buy or sell a series of futures representing a segment along the yield curve, creating or liquidating positions along the curve and gaining exposure to longer-term rates **without legging risk** — the risk of failing to complete one leg. The advantage is execution at a single price: no multiple orders per contract, lower overall cost, and no unfilled legs.

- A **pack** buys a predefined number of contracts in **four consecutive delivery months**. Eurodollar packs are the simultaneous purchase of an equally weighted, consecutive series of four Eurodollar futures, executed in a single transaction so partial fills cannot occur.
- A **bundle** buys a predefined number of contracts in **each consecutive quarterly month for two or more years**, across the standard **2, 3 or 5-year** periods. There are **no 1-year bundles** because those are effectively packs. The first contract is typically the first quarterly contract in the strip, but a bundle can start at any quarterly contract.

Packs and bundles are quoted on the **average net change** in value of all constituent Eurodollar contracts since the prior settlement, in increments of **¼ of 1 basis point (0.01%)**.

3.3 Types of Trades and Strategies

- **a. Outright trades.** Buy or sell a single futures contract, wagering the underlying rises or falls. Familiar and easy to understand.
- **b. Hedging.** Sell a futures contract to offset a cash position. A holder of a large US equity portfolio who does not want to sell for tax reasons but fears a decline can sell S&P 500 index futures.
- **c. Basis trades.** An arbitrage strategy taking opposing long and short positions in two securities to profit from convergence. A trader could be long 10-year US Treasury **bond futures** and short physical 10-year Treasury **bonds**, expecting undervalued futures to appreciate relative to the overpriced bond. Making the trade worthwhile requires substantial **leverage** to enlarge the positions.
- **d. Spread trades.** See 3.4.

3.4 Spread Trades

A spread normally needs less margin than two unrelated outright positions but is not risk-free: correlation can break, liquidity can vanish, and one leg may fill without the other. Types:

- **Intra-commodity / intra-delivery** — same commodity and market, different delivery months.
- **Inter-commodity** — related commodities, e.g. crude against refined products.
- **Intra-market** — related contracts on the same exchange.
- **Inter-market** — similar contracts on different exchanges.
- **Calendar / horizontal** — same underlying, different months.

For a calendar spread, long the near month and short the deferred is generally a bullish view on the near-versus-deferred relationship; short near and long deferred is bearish. Profit depends on the **change in the spread**, not simply on whether the underlying rises.

3.4.1 Calendar Spread

With the S&P E-mini index at 2,344.00 on 19 April 2020, June at 2,335 and December at 2,330, a long June / short December spread requires initial margin of **USD 70,050 + USD 69,900 = USD 139,950** — margins are not netted across the legs.

Scenario	Index move	June	December	Spread P&L
1	+4%; June +6% to 2,475; Dec +5% to 2,447	Gain	Loss	**+USD 70,000 −**
2	+2.5%; June +3% to 2,405; Dec +3.5% to 2,412	Gain	Loss	**+USD 35,000 −**
3	−4% with a circuit breaker; June −7.5% to 2,160; Dec −9% to 2,120	Loss	Gain	**−USD 87,500 +**

Read the results carefully: scenario 3 makes money on a **falling** market because the deferred leg falls further, so the spread widens. The spread, not the direction, is what is being traded.

A currency example: E-micro EUR/USD, 12,500 EUR per contract, tick 0.0001 USD. Selling the near leg at 1.1810 and buying the far at 1.1866, then closing at 1.1750 and 1.1828, gives **+USD 7,500 and −USD 4,750, net +USD 2,750**.

3.4.2 Butterfly Spread

Four legs across three maturities with weights **+1, −2, +1**. It targets curvature or relative richness and is more market-neutral than an outright.

3.4.2.1 Buying a Butterfly **Gold butterfly.** On 6 January 2020 the April, June and August contracts stood at 1,176.3, 1,179.0 and 1,181.7; by 21 April they were 1,287.4, 1,289.1 and 1,292.4. The two spreads widened to **+1.00 and +0.60**, a net **+1.60**. With 100 oz per contract and 100 contracts:

$$100 \times 1.60 \times 100 = \text{USD } 16{,}000$$

The same +1.60 recurred on 24 April after the French election.

3.4.2.2 Selling a Butterfly **Eurodollar sell-butterfly** on 8 February 2020: short 1 Jun-20 at 98.8000, closing at 98.7350; long 2 Dec-20 at 98.5325, closing at 98.5550; short 1 Jun-21 at 98.3250, closing at 98.3800. The results are **+0.0875, −0.0325** on two contracts, and **−0.0550**, giving a net **+0.0550 = 5.5 ticks**. One basis point is one tick worth **USD 25**:

$$5.5 \times 25 = \text{USD } 137.50$$

3.4.2.3 The Double Butterfly A butterfly is bought and another sold in staggered fashion, with the second and third legs of the first butterfly being the first and second legs of the second.

Stage	Near	Mid	Mid	Distant
Buy first butterfly	Buy 1	Sell 2	—	Buy 1
Sell second butterfly, matched to the two mid-term sells	—	—	Buy 2	Sell 1
Overall structure	**Buy 1**	**Sell 3**	**Buy 3**	**Sell 1**

3.4.3 Condor Spread

A condor also combines a bear and a bull spread, but the four expiry dates are **all different**, with equally distributed delivery months and a ratio of **+1 : −1 : −1 : +1**. Buying a condor means buying the nearest month, selling the next, selling the next, and buying the farthest. It typically has a wider but flatter profit zone than a butterfly.

A Eurodollar condor on 8 February 2017: May 98.8450, June 98.8000, September 98.6675, Mar-21 98.4350. By 20 April they stood at 98.7875, 98.7300, 98.6250 and 98.4650. The near-to-mid move is **+1.25 ticks**, the mid-to-distant **+7.25 ticks**, net **+8.5 ticks**:

$$8.5 \times 25 = \text{USD } 212.50$$

The tick value is derived from

$$0.01 \times 0.01 \times 1{,}000{,}000 \times \frac{90}{360} = \text{USD } 25$$

3.4.4 Other Spread Trades

3.4.4.1 TED Spread The **TED spread** compares Treasury bill and Eurodollar (LIBOR) pricing. It is a **credit and liquidity** indicator, not a pure interest-rate measure: as the spread widens, default risk is perceived to rise and investors prefer safer assets; as it narrows, perceived default risk falls. Three-month T-bill futures were delisted from the CME, so the spread is now measured between the T-bill rate and 3-month USD LIBOR.

3.4.4.2 Inter-Product or Inter-Commodity Spread Takes advantage of the value differential between two or more related commodities or financial products influenced by similar factors. A trader may look at the spread between two financial products — US T-bills and Eurodollar — as rates move, or between two related commodities such as **gasoline and heating oil** as crude moves. Rather than taking exposure to one product, the trader exploits the greater sensitivity and reaction of one relative to the other.

3.5 Hedging and Portfolio Management

Forwards and futures were created to hedge the future delivery price of a commodity. The same principle hedges or partially hedges a fall in portfolio value.

3.5.1 Identifying and Measuring Risk

Hedging reduces risk: an adverse move is contained or profit protected, but a favourable move has capped profit. Hedging does not eliminate price risk — it converts price risk into **basis risk** between the underlying and the hedging instrument, confining the final price to a determinable range.

In portfolio management, hedging is used to:

- **a.** smooth cash flows;
- **b.** simplify financial planning;
- **c.** reduce working capital requirements;
- **d.** allow more efficient product pricing; and
- **e.** manage inventory more efficiently.

Before hedging, evaluate whether the likelihood and impact justify the cost and whether the hedge is dependable. Some subjectivity remains. Six questions structure the decision:

Factor	Question
1. Probability of negative change	What is the probability of the price or rate change? What is the likelihood o
2. Probable size of negative change	What is the likely size of the change? What was the past record?
3. Overall impact of negative change	What is the risk value of not hedging — small or large?
4. Transaction costs	What are the brokerage, financing, initial and variation margin costs? How d
5. Percentage of value to be hedged	What percentage of principal does the transaction cost represent?
6. Assumed basis risk	What is the basis risk for the hedge?

The overall assessment is subjective and not fully quantifiable, but traders estimate a **risk value** and a **hedging cost**: if risk value exceeds hedging cost, hedge.

Worked risk-value decision A treasury exposure has a **45%** probability of an adverse change of **10–11.5%**, giving a risk value of **5.18%** of principal (0.45 × 11.5%). The hedge costs, on a principal of **USD 1 million**:

Cost	Calculation	Amount
Brokerage	Given	USD 15.00
Initial margin	1,500 × 12% × 3/12	USD 45.00
Additional margin	45 ticks	USD 33.75
Total hedging cost		**USD 93.75**

As a percentage of principal:

$$\frac{93.75}{1{,}000{,}000} = 0.009375\%$$

Adding an assumed basis risk of **3.0%**, total cost is **3.009375%**, which is **below** the **5.18%** risk value, so the decision is to **hedge**.

3.5.2 Developing an Effective Hedge Program

- **a. Risk exposure.** Evaluate exposure in assets and liabilities by group, maturity or profit centre.
- **b. Objective.** Set the hedge objective as an acceptable risk level or tolerance, measured by beta, volatility, value at risk or another risk unit.
- **c. Hedgeability.** Determine it by price correlation analysis, using **cross hedges** where necessary — hedging a security with a positively correlated instrument with similar price movement.
- **d. Hedge vehicle.** Choose an instrument with a high correlation to the target security.
- **e. Target.** Determine the rate or price to lock in.

3.5.3 Determining the Hedge Instrument

Find an instrument highly correlated with the target. Where the target differs from the futures contract, evaluate the yield relationship and correlation between the two, measured by regression. Examples:

- **a.** A long-term government bond is better hedged with **30-year T-bond futures** than 3-month Eurodollar futures, because long-bond rates correlate more highly with 30-year T-bond rates.
- **b.** T-bills are better hedged with **matching T-bill futures** than Eurodollar futures: in distress, T-bill futures prices rise faster than Eurodollar futures and the Eurodollar hedge becomes ineffective.
- **c.** US T-bond futures hedge the **interest-rate** risk of US corporate bond portfolios; **credit quality and sector risk are not hedged**.
- **d.** Economic factors hit sectors differently. In distress, funds switch from corporate to Treasury bonds, widening credit spreads and pushing corporate prices down relative to governments. Basis risk on a Treasury-futures hedge of corporate bonds is therefore greater, and the hedge less effective.

Futures and exchange-traded options carry four limitations as hedges:

- **i.** round numbers of contracts;
- **ii.** margin maintenance;
- **iii.** time horizon and expiry dates; and
- **iv.** no exact hedge instruments.

3.5.4 Determining the Target Rate

An investor may want to lock in a currency rate, interest rate or asset price. A **minimum variance hedge** held to the futures delivery date locks in the futures rate. If the hedge is lifted **before** delivery, the hedger cannot be assured of a spot or futures rate, because the basis changes.

$$\text{Target rate for the hedge} = \text{futures rate} + \text{target-rate basis}$$

For a long cash exposure hedged with a short futures position:

$$V_t = S_t + (F - F_t) = F + (S_t - F_t) = \text{initial futures price} + \text{ending basis}$$

3.5.5 Current and Anticipated Positions

3.5.5.1 For a current cash position **a. Weak form cash hedge.** The portfolio's time horizon is not known, so the goal is to minimise variance in the rate of return for a specified holding period.

b. Strong form cash hedge. The horizon is finite or known; the goal is to protect the portfolio by minimising variance in expected total return for a given investment period. This is **immunisation** — matching asset and liability durations. A cash and futures portfolio is created and maintained with the same interest-rate sensitivity as a **zero-coupon bond whose initial maturity equals the investment period**. The zero-coupon bond's value and rate sensitivity are tracked continuously: whenever the cash portfolio's sensitivity falls below the zero-coupon bond's, futures are **bought** to raise it; whenever it exceeds the zero-coupon bond's, futures are **sold** to reduce it.

3.5.5.2 For an anticipated cash position **a. Weak form anticipated hedge.** The goal is to minimise variance in the rate of return on an asset for a specified holding period, based on cash flows to be received at an **unknown** date. Futures are bought with the same interest-rate sensitivity as the asset to be acquired. Uncertain timing reduces effectiveness, though futures can still narrow the range of outcomes.

b. Strong form anticipated hedge. A known amount of cash will be received on a **certain** date; the goal is to minimise variance of the acquisition price. The hedge requires acquiring future delivery of the anticipated dollars of bonds, and because margin calls on the long futures position must be financed, the appropriate hedge ratio is **not an exact dollar match**.

3.5.6 Hedge Ratio

The hedge ratio is the expected movement in the value of the cash instrument given a movement in the futures contract; formally, the ratio of futures to spot that minimises or eliminates risk. Equal dollar exposure does not create an optimum or **delta-neutral** hedge.

$$h = \frac{\Delta S}{\Delta F}$$

where, for the hedged position,

$$V = S + h \times F$$

and

$$V$$

is the value of the hedged position,

$$S$$

the security price and

$$F$$

the futures price. For a complete or delta-neutral hedge,

$$\Delta V = 0$$

:

$$h = -\frac{\Delta S}{\Delta F}$$

Number of contracts:

$$\text{contracts} = \frac{-h \times \text{exposure value}}{\text{contract size}}$$

Worked example. A **USD 100 million, 180-day** loan. A 1% rate move on a 180-day basis changes value by:

$$0.01 \times \frac{180}{360} \times 1{,}000{,}000 = \text{USD } 5{,}000$$

per million, and a 90-day Eurodollar contract moves by 100bp × USD 25 = **USD 2,500**. So:

$$h = \frac{5{,}000}{2{,}500} = 2$$

and the hedge needs:

$$2 \times \frac{100\text{m}}{1\text{m}} = \textbf{200} \text{ contracts}$$

Sell **200** contracts for the 180-day exposure. A 90-day exposure would need 100. Two further points: holding a stack of 90-day contracts against a 180-day exposure creates **basis risk at the roll**, and margin calls arise before the hedge pays off. The ratio must be recalculated when the stack is rolled.

Strips and Stacks A **strip** uses sequential futures delivery months matched to the exposure periods — three-month, six-month, nine-month — so each resets against its own contract. A **stack** concentrates the whole exposure in one or deferred contracts and rolls them; it is cheaper and more liquid but carries roll and basis risk. Choose between them by **time horizon**: an interpolative hedge uses a contract expiring beyond the exposure and interpolates; an extrapolative hedge uses a contract expiring before it.

3.5.7 Strips and Stacks

See the strip and stack definitions above: a strip matches sequential delivery months to the exposure periods, a stack concentrates the exposure in one contract or a deferred month and rolls it. Strips avoid roll risk but carry more contracts and lower liquidity per month; stacks are cheaper and more liquid but expose the hedger to basis at each roll.

3.5.8 Hedging Long-Term Interest Rate Risk

Compare **dollar interest-rate sensitivity**, not face value:

$$\text{hedge ratio} = \frac{\text{PVBP of the hedged security}}{\text{PVBP of the cheapest-to-deliver bond}} \times \text{conversion factor of the CTD}$$

where PVBP is the change in price for a one basis point change in yield.

Worked example. A fixed-income manager, Alice Tan, hedges a **USD 20 million** bond position with Treasury futures. The CTD conversion factor is **0.91**, the CTD's PVBP at the settlement date is **0.6895**, and the PVBP of the bond to be hedged is **0.5954**. Assuming a fixed yield spread between the two bonds:

$$\text{hedge ratio} = \frac{0.5954}{0.6895} \times 0.91 = 0.7858071 \approx 0.79$$

$$\text{contracts} = -0.7858071 \times \frac{20{,}000{,}000}{100{,}000} = -157.16$$

so **sell 157 contracts**.

3.5.9 Hedging Equity Risks

An equity portfolio carries two risks: **systemic or market risk** and **specific or non-market risk**. Equity index futures hedge only market risk. A manager offsets market exposure by selling futures when long in the market — a **short hedge** — or buying futures when short — a **long hedge**.

Two factors decide the number of contracts:

- **i. Beta.** Measures portfolio risk or volatility relative to the index. The manager must decide whether the portfolio should be more or less volatile than the index, usually through regression on the historical betas of the constituent stocks.
- **ii. Modified Portfolio Value (MPV).** The value of the actual portfolio multiplied by the **weighted beta** — the sum of each stock's beta multiplied by its percentage share of the portfolio. This forms the base hedge.

$$N = \frac{V_P}{F \times T} \times \beta$$

where

$$N$$

is the number of contracts,

$$V_P$$

the current portfolio value,

$$F$$

the current futures quote,

$$T$$

the value per tick and

$$\beta$$

the portfolio beta. The formula assumes the futures maturity is close to the hedge maturity.

Worked example. John Lim hedges an equity portfolio over three months with 3-month futures. The S&P index is **1,000** in September, December S&P 500 futures are **1,010**, the portfolio is worth **USD 5,050,000**, its beta is **1.50**, and each contract is worth **USD 250**:

$$N = \frac{5{,}050{,}000}{250 \times 1{,}010} \times 1.50 = 20.0 \times 1.50 = 30.0$$

so **sell 30 December contracts**.

December leg. The index falls to **900**, futures to **902**, and the portfolio falls 15% to **USD 4,292,500**. The futures gain:

$$(1{,}010 - 902) \times 30 \times 250 = \text{USD } 810{,}000$$

Total fund value:

$$4{,}292{,}500 + 810{,}000 = \text{USD } 5{,}102{,}500$$

3.5.10 Structuring the Hedge

Contract month, timing and maturity sensitivity must be chosen against the exposure before the hedge is executed. Match the futures maturity as closely as the exposure allows, keep margin headroom for variation calls, and decide the lift date in advance.

3.5.11 Managing the Hedge

Timing, contract month and maturity sensitivity must be adjusted as the hedge progresses. A hedge with **90% effectiveness leaves 10% residual standard deviation**: with a 10bp standard deviation on the basis, the hedged rate lands within ±10bp about **67%** of the time. Lift-date basis and cross-hedge parameter error are the main sources of that residual. Adjust the contract month as the exposure rolls, re-run the ratio when basis moves, and decide in advance at what level the hedge is lifted.

3.5.12 Further Examples of Hedging

Case	Setup	R
Yankee Trading Co.	CAD/USD spot 0.9125, April futures 0.9235; on 1 April spot 0.9485	C
Fixed deposit hedge	5 January, USD 25m; Eurodollar March 97.75 (2.25%); 5 March 98.16 (1.84%)	D
LMN Corp.	27 July, Eurodollar September 98.62 (+1.38%); 10 September 99.40 (−0.60%)	D
Floating-rate note	USD 10m, 3-month LIBOR + 1%; LIBOR 1.85% rising to 2.25%; Eurodollar 98.15 to 97.75	E
Nikkei hedge	JPY 530m, beta 1, June-18 futures 21,455, contract JPY 500	**4**
130/30	Long 130%, short 30%	N

3.6 Portfolio Management Techniques

Futures implement asset allocation, tactical rebalancing, cash equitisation, duration management, synthetic equity and bond exposure, market overlays and loss control. They are cheaper, more liquid and more margin-efficient than trading every cash security, but add basis, roll, leverage, margin and tracking risk. Separate market risk from security-specific risk: an index hedge will not perfectly protect a concentrated portfolio.

3.6.1 Examples of Portfolio Management Applications

a. Asset allocation switch. A USD 50 million portfolio moving from 50/50 to 40/60 equity/bond is implemented with bond futures, avoiding the cost and disruption of selling 50% of the equity book.

b. Synthetic money market. A corporate bond portfolio is converted to synthetic short-term exposure by selling T-bond futures, without liquidating the credit positions.

c. Jessie Goh. A USD 50 million portfolio with 10% into a short-term instrument at an 8% coupon. She buys July at 101 with 2.00 accrued, so the price is 103.00. In September the futures are **108 13/32 = 108.406**, conversion factor 1.0, accrued 3.33, giving proceeds of **111.736** and a gain of **8.736**, or **8.48%**. The hedge ratio:

$$N = \frac{5{,}000{,}000}{108{,}406.25} = 46.12 \rightarrow \text{sell } 46 \text{ contracts}$$

Loss control also uses futures to defer loss realisation across reporting periods.

3.7 Arbitrage

Arbitrage strategies include cash-and-carry, reverse cash-and-carry, bond stripping and reconstitution, futures-versus-IRS, FRA-versus-futures and option parity.

3.7.1 Arbitrage Strategies

3.7.1.1 Cash versus Futures Fifty days at 1.00/1.0625, 91 days, and 141 days at 1.25/1.3125 give

$$r_2 = 1.351\%$$

. The theoretical futures price is **98.65** against a market **98.72**:

$$(9{,}872 - 9{,}865) \times 25 = \text{USD } 175$$

3.7.1.2 FRA versus Futures The interbank FRA trades at a 1–3bp spread against a 1bp futures spread, and **value-date mismatch** leaves residual basis and fixing risk. Avoid calendar events and data releases falling between the two dates.

3.7.1.3 Futures versus Interest Rate Swaps IMM-dated, quarterly-fixing swaps align with futures dates. A speculative variant takes only the partial leg to express a curve view.

3.7.2 Arbitrage using Options

Put-call parity violations are arbitraged with conversion and reversal strategies, which Chapter 4 covers.

Arbitrage is risk-free only in the idealised model. Value dates, fixing dates, margin, liquidity, bid-ask spread, funding, delivery options, basis and counterparty constraints turn it into **relative-value trading**.

3.8 Summary

Know why each participant trades — hedge, speculate, make markets or arbitrage — what each spread isolates, how to compute the hedge ratio, and how margin and mark-to-market change cash needs. A hedge reduces a targeted risk; it does not remove all risk.

Exam cues

- **Hedge ratio for a rate exposure** is the ratio of dollar sensitivities, then scaled by exposure over contract size. The **180-day** USD 100m case needs **200** contracts; a 90-day exposure needs 100.
- **Bond hedge:** PVBP ratio × conversion factor. **0.5954 / 0.6895 × 0.91 = 0.79**, and 0.79 × 20m/100,000 = **157 contracts to sell**.
- **Equity hedge:**

$$N = V_P/(F \times T) \times \beta$$

. The John Lim case is 5,050,000/(250 × 1,010) × 1.5 = **sell 30**; with the index at 900 and futures at 902 the hedge recovers USD 810,000.
- **Butterfly weights +1, −2, +1 through three maturities; condor +1, −1, −1, +1 through four.** A double butterfly nets **buy 1 / sell 3 / buy 3 / sell 1**.
- **Eurodollar tick = USD 25**; one basis point is one tick. The sell-butterfly earned 5.5 ticks = **USD 137.50**; the condor 8.5 ticks = **USD 212.50**.
- **Order types:** MTL leaves the unfilled balance at the last execution price; a market order in a non-continuous state takes the **Equilibrium Price**; MIT sells are placed **above** the market while stop sells are **below**; SSOs **cannot be GTC** and vanish at end of day.
- **Packs are four consecutive months; bundles are 2, 3 or 5 years** — there is no 1-year bundle, because that is a pack. Quoted in **¼ basis point** increments.
- **TED spread** is a credit and liquidity indicator, not a pure rate spread. It narrowed in measurement after 3-month T-bill futures were delisted.
- **Trap:** equal dollar exposure does not give a delta-neutral hedge, and an equity index hedge removes **market** risk only — specific risk and credit spread remain.

Chapter 4 — Options

Options separate rights from obligations, which is why they can be priced, hedged and combined far more finely than futures. This chapter builds from payoff arithmetic through the Greeks to the strategy matrix the examination tests.

Learning Objectives

The candidate should be able to:

- Understand the basic concepts of options and how they are traded.
- Explain the features of call and put options.
- Describe the major factors affecting the price sensitivity of options.
- Explain the rights, obligations, benefits and risks of using options.
- Calculate breakeven, profit and loss from an option contract.
- Explain how traders use options strategies.
- Explain neutral, bullish and bearish strategies based on the investor's outlook.
- Explain how to use options to hedge an underlying position.
- Describe options on equities, bonds, interest rates, currencies, futures and swaps.

4.1 What is an Option?

An option is a contract between the **option writer** (seller/issuer) and the **option holder** (buyer) giving the holder the right to buy from, or sell to, the writer at a contracted strike price within a specific period. There are two types: **calls** and **puts**.

The exchange-listed market dates from **1973**, when the Chicago Board Options Exchange listed calls on **16 stocks**; before that, options were OTC only. Singapore's options market remains largely OTC, which makes counterparty scrutiny important.

Call option:

- **a. Buyer.** Has the right but not the obligation to buy the underlying at the strike within the period. On exercise the holder pays the strike and takes possession.
- **b. Writer.** Receives the premium and must deliver the underlying if assigned.

Put option:

- **a. Buyer.** Has the right but not the obligation to sell the underlying at the strike.

- **b. Writer.** Receives the premium and must take delivery at the strike if assigned.

Exotic features include compound options, and underlyings extend to real estate, weather, volatility and ETFs.

4.2 Style of Option

4.2.1 Style of Option

European options are exercisable only at expiry; **American** options at any time up to expiry. OTC contracts may use **Bermudan** or event-dependent exercise. US listed equity options conventionally expire on the **third Friday** of the month, moving to the Thursday if that Friday is a holiday.

4.2.2 Strike Price

The contractual price at which the underlying is bought or sold on exercise.

4.2.3 Expiration Date

The last date on which the option may be exercised. Listed series follow monthly cycles coded **JAJO** (January, April, July, October), **FMAN** (February, May, August, November) and **MJSD** (March, June, September, December).

4.2.4 Class and Series

A **class** is all options of the same type on the same underlying; a **series** is all options of the same class with the same strike and expiry.

4.3 Option Price

$$\text{option price (premium)} = \text{intrinsic value} + \text{time value}$$

4.3.1 Intrinsic Value

For a call, exercise has value when the market price exceeds the strike; for a put, when the strike exceeds the market price.

$$IV_{\text{call}} = \max(0,\ S - X) \qquad IV_{\text{put}} = \max(0,\ X - S)$$

Moneyness	Call	Put
In-the-money (ITM)	$S > X$, positive intrinsic value	$X > S$, positive intrinsic value
At-the-money (ATM)	$S = X$, zero intrinsic value	$X = S$, zero intrinsic value
Out-of-the-money (OTM)	$S < X$, zero intrinsic value	$X < S$, zero intrinsic value

Call example. Strike

$$X = 50$$

, premium

$$P = 6$$

, underlying

$$S = 54$$

. Intrinsic value is **4** and time value **2**. If the underlying is 48 at expiry, intrinsic value is 0.

Put example. Strike

$$X = 50$$

, premium

$$P = 9$$

, underlying

$$S = 44$$

. Intrinsic value is **6** and time value **3**. If the underlying is 53 at expiry, intrinsic value is 0.

4.3.2 Time Value of Options

Time value is generally positive and falls to zero at expiry, so at expiration the option is worth exactly its intrinsic value:

$$\text{option price} = \text{intrinsic value as time value} \to 0$$

The longer the time to expiry, the greater the time value. An option that looks worthless today — ATM or OTM, zero intrinsic value — may be in the money by expiry.

4.3.3 Comparison of Call versus Put

Criterion	Call buyer	Call seller	Put buy
Obligation	Pays the premium	Delivers the underlying if assigned	Pays the
Right	May buy at the strike	—	May sell
Maximum gain	$U_{Mkt} - U_{Strike} -$ premium, theoretically unlimited	The premium	U_{Strike} -
Maximum loss	The premium	Theoretically unlimited	The pre
Counterparty risk	Bears the writer's	Bears the buyer's	Bears th
Opportunity cost	—	Forgone upside on a covered call	—

4.4 Factors Influencing Premiums

Change, all else constant	Call value	Put value
Underlying price rises	Rises	Falls
Strike rises	Falls	Rises
Time to expiry rises	Generally rises	Generally rises
Volatility rises	Rises	Rises
Risk-free rate rises	Rises	Falls
Dividends rise	Falls	Rises

4.4.1 Underlying Price

A call gains as the underlying rises; a put loses.

4.4.2 Strike Price

A higher strike makes a call cheaper and a put dearer.

4.4.3 Time to Expiration

More time means more chance of finishing in the money, so both calls and puts carry more time value.

4.4.4 Volatility of the Underlying

Higher volatility widens the distribution of outcomes and raises the value of both calls and puts, because the holder's loss is floored at the premium while the upside is open.

4.4.5 Interest Rates

The interest forgone on the cash needed to buy the underlying — the opportunity cost, measured against a risk-free alternative such as Treasury bills — is embedded in the option. The higher the rate, the greater that opportunity cost, so there is a **positive relationship between interest rates and call value**; puts move inversely.

4.4.6 Expected Dividends

For equity options, a dividend reduces the share price when it goes ex-dividend, which lowers call value and raises put value.

4.5 Payoff Profiles

A payoff profile shows maximum gain, maximum loss and breakeven as a function of the underlying at expiry.

4.5.1 Buying a Call

An investor bought a call a month ago with premium **USD 4**, strike **USD 100**, and the share is now **USD 110**.

Payoff to the call buyer:

$$C_T = S_T - X \text{ if } S_T > X \text{ (ITM)}; \quad C_T = 0 \text{ if } S_T \leq X$$

Item	Result
i. Payoff and profit	Payoff is **USD 10** $(110 - 100)$; profit after premium is **USD 6**
ii. Breakeven	The share must rise above the strike by the premium: **USD 104**
iii. Maximum gain	Theoretically unlimited
iv. Maximum loss	The premium paid, **USD 4**; if $S_T < X$ exercise does not occur and the option expires worthless
v. Obligations	Apart from paying the premium, none unless the buyer exercises, and then to pay the strike

4.5.2 Selling a Call

An investor sells a call with premium **USD 4**, strike **USD 100**, share at **USD 90**.

Payoff to the writer:

$$C_T = -(S_T - X) \text{ if } S_T > X; \quad C_T = 0 \text{ if } S_T \leq X$$

Item	Result
i. Payoff and profit	Below USD 100 there is no reason to exercise, so payoff is zero and profit is the **USD 4 premiu**
ii. Breakeven	**USD 104** — identical to the buyer's
iii. Maximum gain	The **USD 4** premium, when the share is at or below the strike
iv. Maximum loss	Theoretically unlimited
v. Obligations	Must deliver if assigned

4.5.3 Buying a Put

An investor buys a put with premium **USD 4**, strike **USD 130**, share at **USD 120**.

Item	Result
i. Payoff and profit	Payoff **USD 10** $(130 - 120)$; profit **USD 6**
ii. Breakeven	**USD 126** $(130 - 4)$
iii. Maximum gain	**USD 126** at an underlying of zero, being the strike less the premium
iv. Maximum loss	The premium, **USD 4**
v. Obligation	None except on exercise

4.5.4 Selling a Put

An investor sells a put with premium **USD 4**, strike **USD 130**, underlying at **USD 140** initially and **USD 120** at expiry.

Payoff to the writer is the mirror image of the holder:

$$P_T = -(X - S_T) \text{ if } S_T < X; \quad P_T = 0 \text{ if } S_T \geq X$$

Item	Result
i. Payoff and profit	Zero above USD 130, with the **USD 4** premium as maximum profit; below the strike the profit e
ii. Breakeven	**USD 126**
iii. Maximum gain	The **USD 4** premium
iv. Maximum loss	**USD 126**, when the underlying falls to zero
v. Obligation	Must take delivery at the strike if assigned

4.6 Put-Call Parity

4.6.1 Definition

For European options on the same underlying, strike and expiry, with consistent units and dividends correctly treated:

$$C + \mathrm{PV}(X) = P + S$$

The relationship assumes no arbitrage, a frictionless market, known dividends, the same exercise style and no early exercise.

4.6.2 Arbitrage

A violation is traded as a **conversion** or **reversal**. If

$$C + \mathrm{PV}(X) > P + S$$

, sell the call, buy the put and buy the stock. If the left side is cheaper, reverse the trade. Funding, dividends, exercise style, transaction costs and short-sale constraints determine whether the arbitrage is real.

4.6.3 Synthetic Securities

Synthetic position	Built from
Long stock	Long call + short put at the same strike (plus the parity bond)
Short stock	Long put + short call
Long call	Long underlying + long put
Long put	Short underlying + long call
Short call	Short underlying + short put
Short put	Long underlying + short call

Each synthetic reproduces the cash flows of the position on the left, so a trader can choose whichever route is cheaper after funding and transaction cost.

4.7 Valuation

4.7.1 Black-Scholes

The model takes six inputs — spot, strike, time to expiry, interest rate, dividends and volatility — and returns a theoretical premium. **Binomial trees** handle early exercise and American features. The examination does **not** require model calculations, but does require explanation.

4.7.2 Volatility

Volatility is the annualised standard deviation of daily price changes. **Historical** volatility is measured from past prices; **implied** volatility is backed out of the observed option price and represents the market's forward-looking estimate. The two can diverge.

4.7.3 The Option Greeks

The Greeks measure sensitivity of the premium to each input.

Greek	Measures	Typical sign for a long option
Delta	Change in option price per unit change in underlying	Positive for calls, negative for puts
Gamma	Change in delta per unit change in underlying	Positive
Theta	Time decay	Negative
Vega	Sensitivity to implied volatility	Positive
Rho	Sensitivity to the interest rate	Positive for calls, negative for puts

4.7.4 Delta

$$\delta = \frac{\partial V}{\partial S}$$

A delta of **0.50** means a **USD 2** move in the underlying changes the option price by **USD 1.00**. Delta approaches **1** for deep in-the-money options near expiry and **0** for deep out-of-the-money options.

Delta hedging. A writer short **1,000 calls** with delta **0.642** loses **USD 642** on a one-unit rise; hedging requires buying **642 units** of the underlying.

4.7.5 Gamma

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \delta}{\partial S}$$

Gamma is largest at-the-money and close to expiry, when delta changes fastest. A high-gamma book requires frequent rebalancing.

4.7.6 Theta

Time decay, normally negative for a long option. At-the-money options lose the most dollar value per day: a one-day theta of **−0.0250** is **−2.5 cents a day**.

4.7.7 Vega

Sensitivity to a one-percentage-point change in implied volatility; a vega of **0.090** means the premium changes by 9 cents per volatility point. Vega is larger at-the-money and for longer-dated options. Implied volatility is obtained by inverting the pricing model against the observed price.

4.7.8 Rho

Sensitivity to a one-percentage-point change in the interest rate; a rho of **0.060** means the premium changes by 6 cents per point. Call rho is positive and put rho negative, because higher rates raise the opportunity cost of holding the underlying.

4.7.9 Practical Hedging Problems

Delta hedging is dynamic and imperfect. Rebalancing costs, jumps through levels, liquidity, changing volatility, discrete monitoring and model error all remain. A market maker manages the whole Greek book, not just delta.

4.8 Exchange versus OTC Options

Exchange options	OTC options
Standardised strikes, expiries and sizes	Tailored strike, date, notional, barrier, settlement currency
Clearing house counterparty	Bilateral counterparty and documentation risk
Transparent prices and secondary liquidity	Valuation and liquidity risk; CSA collateral terms matter

4.9 Option Contract Specifications

Read the whole term sheet, not just the ticker. An SGX equity index option specification illustrates what to check:

Term	Specification
Ticker	`CS - Calls`
Contract size	One MSCI Singapore index future
Contract months	2 nearest serial months plus the quarterly cycle (Mar/Jun/Sep/Dec)
Trading hours	T session 8.15–8.30 opening (cancel/match), 8.30–17.15; T+1 18.00–18.15 opening, 18.15–02.00; last tr
Last trading day	Second-last business day of the expiring month
Tick size	0.1 index point, worth **SGD 20**
Strike intervals	**5 index points**
Exercise style	European or American, per the series
Settlement	Cash settled; in-the-money options **auto-exercised**
Price limits	Daily limit of **5%** up or down, with circuit breakers
Position limit	**10,000 contracts** net on the same side, all months combined

4.10 Options on Other Underlyings

4.10.1 Equity Index Options

Index options were listed on the **CBOE in 1983**. They are usually cash settled, and the payoff is multiplied by the contract multiplier.

Contract	Index value	Multiplier	Style
SPX (S&P 500 options)	Full index value	**USD 100**	European
XSP (Mini-SPX)	1/10th of the index	**USD 100**	European

4.10.2 Bond Options

Bond options are mostly OTC, with expiries set well before bond maturity. A bank buying a call at a strike of **USD 102** with the bond at **USD 101** gains protection against a rise in price — that is, a fall in yield.

4.10.3 Interest Rate Options

Caps, floors and collars are the interest-rate equivalents of calls, puts and combinations. CME lists options on US Treasuries and Eurodollar; ICE on Euribor; SGX on the 10-year Mini-JGB and Euroyen (TIBOR).

4.10.3.1 Options on Eurodollar Futures Options on short-rate futures are quoted on **100 − yield**. One tick is:

$$\text{USD } 1{,}000{,}000 \times \frac{1}{10{,}000} \times \frac{90}{360} = \text{USD } 25$$

A quarter tick is **USD 6.25** in the spot month and a half tick **USD 12.50** in other months.

Mid-curve example. A 1-year Eurodollar mid-curve option on the Dec-2019 futures at **96.93**, American style, tick value USD 25, expiry June 2019, is struck on a ladder from 9,637.50 to 9,750.00. Call prices run 51.50, 42.00, 33.25, 26.25, 20.25, 15.50, 12.00, 9.25, 7.25 and 5.75; put prices 4.00, 6.75, 10.50, 15.75, 22.50, 29.75, 38.75, 48.25, 58.75 and 69.75.

Hedge arithmetic. A borrower wanting to cap the rate at 2.75% buys the strike-97.250 option (that is, 100 − 2.750) at a cost of **9.25bp**, so the realised rate is:

$$2.7500\% - 0.0925\% = 2.6575\%$$

A lender buying the put at **48.25bp** achieves:

$$2.7500\% + 0.4825\% = 3.2325\%$$

4.10.3.2 Interest Rate Caps A cap is a series of call-like options on a reference rate, setting a maximum. A **SGD 200 million 3-year cap at 2.00%** on 3-month SIBOR, struck in March 2018 when SIBOR was 1.375%, pays when SIBOR fixes at **2.125%** in June 2019:

$$\frac{90}{360} \times (2.125\% - 2.00\%) \times 200{,}000{,}000 = \text{SGD } 62{,}500$$

4.10.3.3 Interest Rate Floors A floor is a series of put-like options setting a minimum rate, protecting a lender or depositor against falling rates.

4.10.3.4 Interest Rate Collars A collar buys one option and sells another, reducing or eliminating premium at the cost of capping the benefit. A borrower buys a cap and sells a floor; the sold floor funds the bought cap.

4.10.4 Currency Options

A currency option involves **two** currencies, not one price. A USD/JPY call is economically the same exposure as a JPY/USD put. Large notionals and a 24-hour market heighten monitoring and counterparty risk.

4.10.5 Options on Futures

An option on a futures contract confers the right to enter a futures position at the strike. Such options are typically more liquid than options on the cash instrument, but they introduce a **delivery problem** when cross-hedging. All SGX options are futures options, American style, with in-the-money positions auto-exercised. The **SGX Nifty 50** option covers 50 large, liquid NSE stocks representing about 70% of free float, is cash settled on the average weighted prices of the components, and benefits from the CME mutual-offset arrangement.

4.10.6 Swaptions

A **swaption** is an option to enter a swap at a fixed rate on a future date. A **payer** swaption gives the right to pay fixed — it protects against rising rates. A **receiver** swaption gives the right to receive fixed — it protects against falling rates. The terms that matter are premium, expiry, notional, the fixed rate and floating index, payment frequency and day count.

4.11 Basic Option Strategies

4.11.1 Call Buying Strategies

Calls are bought for four reasons: capital investment, taking risk exposure, extracting cash from a position, and hedging.

Hedging a short. An investor is short at **USD 12.00** and buys a call at strike **12.40** for **0.80**. Breakeven is **11.20**; maximum loss **USD 1.20**; maximum gain **USD 11.20** if the share falls to zero.

4.11.2 Call Writing Strategies

Covered call. Buy the stock at

$$S_0 = 12.00$$

, write a call at

$$X = 12.40$$

for

$$C_0 = 0.80$$

.

$$\text{Profit} = S_T - \max(0, S_T - X) - S_0 + C_0$$

Item	Calculation	Result
Breakeven	$S_0 - C_0 = 12.00 - 0.80$	**11.20**
Maximum gain	If $S_T > X$: $X - S_0 + C_0 = 12.40 - 12.00 + 0.80$	**1.20**
Maximum loss	If $S_T = 0$: $0 - 12.00 + 0.80$	**−11.20**

The call will not be exercised once the shares have fallen, so the writer keeps the premium but bears the full fall in the stock.

4.11.3 Put Buying Strategies

Protective put. Stock at **USD 12.00**, put at strike **11.60** for **0.80**. Breakeven **12.80**; maximum loss **USD 1.20**; upside retained without limit above the breakeven.

4.11.4 Put Writing Strategies

A put writer earns premium in exchange for a conditional obligation to buy the underlying. A **bull ELN** works this way: a one-month note issued at **USD 9,900** against USD 10,000 face, a vanilla yield of 5% p.a., ABC spot **USD 10** and a put struck at **USD 9**.

- If
$$S_T \geq 9$$
, the note returns USD 10,000: a gain of 1.01% for the month, or **12.82% p.a.**
- If
$$S_T < 9$$
, the note delivers **1,111 shares**. At **USD 8.80** those are worth USD 9,776.80, a loss of **1.24%**.

The uplift over the vanilla note — **7.82% p.a.** — is exactly the premium on the embedded short put.

4.12 Market Driven Strategies

4.12.1 Neutral Strategies

Used when the investor expects little movement. Premium is collected by selling options, and the risk is a break out of the range.

4.12.2 Bullish Strategies

Long call, bull call spread, bull put spread. The spread versions cap both gain and loss.

4.12.3 Bearish Strategies

Long put, bear put spread, bear call spread.

4.12.4 Creating Spreads

A spread pairs a long and a short option of the same type, differing in strike or expiry. **Debit** spreads pay premium — bull call, bear put. **Credit** spreads receive premium — bull put, bear call.

4.12.5 Straddle

A long straddle buys a call and a put at the **same strike**. With ABC at **USD 20** in September, buying the October 20 call and the October 20 put produces three outcomes at expiry:

Outcome	Result
$S_T = 20$	Both expire worthless; the loss is the whole cash outlay
$S_T > 20$	The call is in the money; profit is $S_T - 20 -$ outlay, potentially unlimited
$S_T < 20$	The put is in the money; profit is $20 - S_T -$ outlay, with a maximum payout of USD 20 at zero

The strategy is neutral on direction and **long volatility**. Maximum loss is the cash outlay, occurring precisely at the strike.

4.12.6 Strangle

A strangle buys a call and a put at **different** out-of-the-money strikes. With ABC at USD 20, buy the October 25 call and the October 15 put. It costs less than a straddle but needs a larger move. The **short** strangle does the reverse: selling the call for USD 2 and the put for USD 1 collects **USD 3**. At an underlying of 14 the put loses USD 1 while the call gains USD 2; at 26 the call loses USD 1.

4.12.7 Bull Call Spread

ABC at **USD 21**. Buy the October 20 call at **1.50** and sell the October 25 call at **0.75**, for a net debit of **0.75**.

Item	Result
Breakeven	**20.75**
Maximum profit	**4.25** (the 5-point strike gap less the 0.75 debit)

4.12.8 Bull Put Spread

ABC at **USD 24**. Buy the October 20 put at **0.75** and sell the October 25 put at **1.25**, for a net credit of **0.50**. Maximum loss is **4.50**; at an underlying of 23 the result is **−1.50**.

4.12.9 Bear Call Spread

ABC at **USD 18**. Buy the October 20 call at **1.50** and sell the October 15 call at **3.50**, for a net credit of **2.00**. Maximum loss is **3.00**.

4.12.10 Bear Put Spread

ABC at **USD 18**. Buy the October 15 put at **1.00** and sell the October 20 put at **3.00**, for a net debit of **2.00**. Maximum gain is **3.00**.

4.12.11 Other Market Driven Strategies

A three-by-three matrix of outlook against expected volatility selects the structure: the direction (bullish, neutral, bearish) fixes which legs to hold, and the volatility view fixes whether to be long or short premium.

View	Structure	Exposure
Strongly bullish	Long call, bull call spread	Rise
Strongly bearish	Long put, bear put spread	Fall
Large move, direction unknown	Long straddle or strangle	Long volatility
Range-bound, low volatility	Short strangle, iron condor	Short volatility, tail risk
Neutral at one strike	Butterfly	Limited central profit
Neutral over a wider zone	Condor	Wider zone, lower maximum

4.13 Hedging

4.13.1 Protect an Equity Portfolio

A protective put floors the downside at the strike after the premium. With ABC at USD 20, selling the USD 30 call for 1.50 is a **covered call**: the premium is earned but the upside above 30 is surrendered — the opportunity cost of the strategy.

4.13.2 Manage Currency Exposure

An importer with USD 2 million payable, USD/JPY at 100, buys a USD call / JPY put struck at 105. The exporter's mirror position is set out below.

Spot at expiry	Unhedged proceeds	Hedged with a 105 put at JPY 1 premium
100	100	**99**

Spot at expiry	Unhedged proceeds	Hedged with a 105 put at JPY 1 premium
110	110	**109**
90	90	**94**

The hedge floors the rate at 94 while retaining the upside above 105, less the JPY 1 premium.

4.13.3 Zero-Cost Options

A **zero-cost collar** buys a put and sells an out-of-the-money call so the premiums roughly offset.

NVDA costless collar. Spot **USD 156.00**. Sell 1,000 January-2020 **USD 160** calls at **USD 15.30**; buy 1,000 January-2020 **USD 155** puts at **USD 15.30**. Net outlay is zero.

- Below 155, say 125: the put pays USD 30 against a stock loss of USD 31 per share, so a residual loss of **USD 1,000** across the position.
- Between 155 and 160: neither option pays.
- Above 160: the stock is called away; upside is capped at the call strike.

The collar is not free of risk — the residual USD 1,000 is the gap between the two strikes — and it assumes no dividends and no interest over the life.

4.13.4 Summary of Options Strategies

The full strategy map crosses the market view — yield, bullish, neutral, protective, bearish — against the structure, and records whether each is a debit or credit, its maximum gain and its maximum loss. Learn the payoff of every single leg before combining them; the combination is only the sum of the legs.

4.14 Summary

Options separate rights from obligations. Price is intrinsic plus time value, and volatility, time, rates and dividends drive the premium. Learn payoff, breakeven and maximum loss for each leg before combining legs. Parity explains synthetic positions and the Greeks explain risk management.

Chapter 4 Appendix — Additional Spreads (non-examinable)

Long Call Butterfly

With strikes at **USD 10, 20 and 30**, buy one in-the-money call, sell two at-the-money calls and buy one out-of-the-money call. Maximum gain is USD 10 less the net outlay; maximum loss is the outlay.

Long Put Butterfly

Strikes at **USD 15, 25 and 35**, the mirror construction. Maximum gain is USD 10 less the outlay.

Vertical, Horizontal and Diagonal Spreads

A **vertical** spread uses the same expiry with different strikes. A **horizontal** or calendar spread uses the same strike with different expiries. A **diagonal** spread differs in both strike and expiry.

Condor Spread

	Long	Short
Calls	Buy the low strike, buy the high strike	Sell the two middle strikes
Puts	Buy the low strike, buy the high strike	Sell the two middle strikes

Ratio Hedge

A 2:1 spread: two options sold against one bought. Upside is limited while downside can become very large.

Seagull Strategy

A three-legged structure with zero net outlay. With USD/JPY at **113**, sell the 109 put, sell the 115 call and buy the 111 put. At a spot of 110 the position can be closed for a profit.

To **re-centre**, buy back the 115 call and 109 put for **−1.50%**, then sell the 112 call and 107 put for **+1.35%**, for a net **−0.15%**, while retaining the in-the-money 111 put.

Protecting or Re-centring a Short Strangle

A three-month roll and adjustment sequence: as the market moves against one leg, buy back the threatened option and sell a further out-of-the-money option to fund it, keeping the position short premium but re-centred on the new range.

Appendix Strategy Summary

The appendix closes with a grid recording, for each structure, the market view, debit or credit, maximum gain and maximum loss.

Exam cues

- **Premium = intrinsic + time value.** Call IV = max(0, S − X); put IV = max(0, X − S). Time value is zero at expiry.
- **Breakeven:** long call **X + premium**; long put **X − premium**. Buyer and writer share the same breakeven.
- **Maximum loss:** long call and long put lose the premium. A naked short call is unlimited; a short put loses down to a zero underlying, so its maximum loss is X − premium.
- **Payoff arithmetic to memorise:** call premium 4, strike 100, share 110 → payoff 10, profit 6, breakeven 104. Put premium 4, strike 130, share 120 → payoff 10, profit 6, breakeven 126, maximum gain 126.
- **Parity**
$$C + \mathrm{PV}(X) = P + S$$
; long stock = long call + short put.
- **Greeks:** delta 0.50 → USD 1 move per USD 2 underlying; short 1,000 calls at delta 0.642 needs 642 units to hedge; gamma peaks ATM near expiry; theta −0.0250 is −2.5 cents a day; vega 0.090 and rho 0.060 are per one-point moves.
- **Eurodollar option tick = USD 25** (1,000,000 × 1/10,000 × 90/360). The 2.75% cap costing 9.25bp realises 2.6575%; the 48.25bp put realises 3.2325%.
- **Strategies:** bull call ABC 21 → long 20 @ 1.50, short 25 @ 0.75, debit 0.75, breakeven 20.75, max profit 4.25. Covered call 12.00/12.40/0.80 → breakeven 11.20, max gain 1.20, max loss −11.20.
- **Trap:** shorting a put is not a hedge — it is a bullish obligation. A collar is not riskless, and a covered call sells away the upside above the strike.

Chapter 5 — Structured Warrants and Daily Leveraged Certificates

Warrants are options issued by a bank and listed on an exchange; DLCs are daily-reset leverage certificates. Both are Specified Investment Products, so the suitability gate comes before the payoff.

Learning Objectives

The candidate should be able to:

- Understand the differences between company-issued warrants and structured warrants.
- Explain the features, benefits and risks of structured warrants.
- Identify and compute the elements in the valuation of structured warrants.
- Understand how adjustments for corporate actions are made.
- Identify exotic warrants — index, currency-translated, basket and yield-enhanced.
- Understand the uses, benefits and risks of Daily Leveraged Certificates, and the representative's responsibilities towards clients trading them.

5.1 Introduction

Besides futures and options, **Specified Investment Products (SIPs)** are products with structures, features and risks that may be more complex, including structured warrants, **daily leveraged certificates (DLCs)** and synthetic exchange-traded funds (ETFs).

Under MAS guidelines enhancing safeguards for retail investors, an investment representative must assess whether a client has the relevant knowledge and experience before the client can invest in a SIP. Investors wishing to trade SIPs complete a **Customer Account Review (CAR)** with their broker or representative. The qualifying criteria are **educational qualifications, work experience and investment experience**; the investor must satisfy **one of the three**.

5.2 Introduction to Structured Warrants

A structured warrant is issued by a third-party financial institution and gives exposure to the performance of an underlying for a fraction of its price. Like an option, it lets the investor profit from a bull market (call warrants) or a bear market (put warrants).

The value of a warrant is influenced by the underlying, which may be:

- **a.** individual stocks;
- **b.** equity indices;
- **c.** investment funds; or
- **d.** commodities.

Structured warrants have a **short finite lifespan, usually under a year and mostly 3–6 months**. A "structured" call warrant holder has the right to buy the underlying at a predetermined price on or before expiry, depending on the exercise style. Some are listed and traded on the SGX-ST platform. Different designs serve investors with different objectives and risk profiles: a **zero-strike warrant** — essentially a synthetic stock — has an exercise price of zero, so it is always in the money, and its cash settlement and warrant price equal the underlying's closing price.

5.2.1 European versus American Style Call and Put Warrants

European-style warrants are exercisable only at expiry; **American-style** warrants throughout their life. **SGX-listed structured warrants are European style**, and for the CM-SIP examination assume European style throughout. Warrants are less popular in the US, where American options dominate.

A **European call warrant** gives the right to purchase the underlying at a fixed strike **at expiry**. If the price at expiry is above the strike, the investor earns the difference. A call buyer holds a **bullish** view. If the price rises towards the strike during the warrant's life the warrant's value rises, but there is **no exercise before expiry**; if the price is volatile and falls back below the strike, the warrant's value falls but no loss is realised.

A **European put warrant** gives the right to sell at a fixed strike at expiry. If the strike exceeds the market price at expiry, the investor profits. A put buyer holds a **negative** view. Settlement at expiry is in **cash**, with typically no delivery of the underlying.

- **Call settlement:** (settlement price of the underlying at expiry − exercise price of the warrant) ÷ conversion ratio.
- **Put settlement:** (exercise price of the warrant − settlement price of the underlying at expiry) ÷ conversion ratio.

5.2.2 Structured Warrants versus Standard Company Warrants

Standard company warrants, often issued as a sweetener for a bond issue, are covered in the CM-EIP examination. Structured warrants differ in five ways:

#	Difference
a.	Structured warrants may be **call or put** warrants; company warrants are **call only**. Structured warrants therefore w
b.	Structured warrants have **shorter durations, usually not more than 6 months**.
c.	Exercise causes **no dilution** of company earnings, because no new shares are issued. Structured warrants are **cash s**
d.	They are issued by **third parties** and provide **no new capital** to the underlying company.
e.	**Designated market makers** act as counterparties, so investors are exposed to **issuer and market-maker counte**

5.2.3 Benefits of Structured Warrants

Warrants are created by financial institutions for profit, but in response to demand for alternative exposure to asset-price movement. The benefits are:

- **a.** Call warrants give **leverage** to the upside at a fraction of the cost of the underlying.
- **b.** Put warrants give **leverage** to the downside, offering portfolio managers some protection against a falling portfolio.
- **c. Losses are limited** to the value of the warrants purchased.
- **d.** They give **access to diverse markets** — equities, baskets and select indices — without a large commitment.
- **e.** Because they are **paid for upfront**, there are **no margin calls**, unlike options and CFDs.

Structured warrants are popular and actively traded in **Singapore, Hong Kong, London and Mumbai**. They are less popular in the US, where American options are preferred.

5.3 Key Terms

5.3.1 Conversion Ratio

The **conversion ratio** is the number of warrants needed to buy or sell one unit of the underlying. If the ratio to buy ABC shares is **5:1**, the holder needs five warrants per share.

5.3.2 Exercise or Strike Price

The conversion may also be expressed as an **exercise price**, in addition to or instead of a ratio: the price at which the holder can buy or sell the underlying.

5.3.3 Expiry Date

The expiration date is the last day on or before which the warrant must be exercised.

5.3.4 Issue Price

The price at which the structured warrant is initially sold or issued by the issuer.

5.3.5 Physical Delivery or Cash Settlement

Warrants may settle by physical delivery or in cash. On exercise, a call holder receives the underlying (or cash) from the issuer; a put holder delivers it. The settlement style is fixed at issue, and **SGX-ST structured warrants are cash settled**.

5.3.6 Gearing

Structured warrants are priced at a fraction of the share price, so the investor can trade more warrants than the underlying for the same outlay. A small gain in the underlying can produce a larger gain in the call warrant, and a fall a larger loss.

Gearing indicates how many more warrants can be bought for a given capital than shares. At **8× gearing**, the investor gets eight times the underlying exposure for the same money.

5.4 Factors Affecting the Price of a Structured Warrant

The warrant price is the speculative value investors will pay. Seven factors drive it.

5.4.1 Life of the Warrant

More time means more chance an out-of-the-money warrant becomes attractive as the underlying moves. As life shortens the speculative value decays. On the day before expiry the warrant is worth only its **intrinsic value**; the day after expiry it is **worthless**.

5.4.2 Performance of the Underlying Asset

Call warrants are worth more as the underlying rises; put warrants as it falls.

5.4.3 Exercise Price Level

A higher exercise price makes a call **less likely** to be exercised, so it is worth less. A higher exercise price makes a put **more likely** to be exercised, so the put is worth more.

5.4.4 Price Volatility of the Underlying Asset

The more volatile the underlying, the more likely the warrant will appreciate within a given period, so investors pay a larger premium. The market prices the warrant on the underlying's volatility; **implied volatility** is that market quotation.

Warrant buyers are **long volatility**; writers are **short volatility**. Holding everything else constant, buying at **7% implied volatility** and selling at **8%** produces a profit, because higher volatility means a higher warrant price. Conversely a seller at 7% gains if volatility falls below 5%.

5.4.5 Dividends

Warrant holders are not entitled to dividends, but a dividend declaration has a **mild effect**: call warrants lose slightly and put warrants gain slightly.

5.4.6 Interest Rate

A higher interest rate raises the hedging cost for call warrants written by the issuer, who must delta hedge by buying the shares, so **call warrant prices rise**. Hedging put warrants becomes cheaper at higher rates, so **put warrant prices fall**.

5.4.7 Leverage

Warrants offering greater leverage through the conversion ratio command higher prices.

Factor	Call warrant	Put warrant
Life of the warrant	Longer → higher	Longer → higher
Exercise price	Higher → lower	Higher → higher
Underlying asset price	Higher → higher	Higher → lower
Volatility	Higher → higher	Higher → higher
Dividend expectations	Mild negative	Mild positive
Leverage (conversion ratio)	Greater → higher	Greater → higher
Interest rate	Higher → higher	Higher → lower

Potential gains from in-the-money warrants An investor may either exercise an in-the-money warrant or sell it in the market before maturity. Realised gains include transaction costs.

Scenario	Call warrants
	Underlying rises above the exercise price
Gain from selling the warrant	Prevailing warrant price **less** sales transaction cost, less the purchase price, less t
Gain from exercising the warrant	(Current share price sold − call exercise price) × conversion ratio, **less** the purch

5.4.8 Naming Convention on the SGX for Structured Warrants

The SGX code identifies the underlying, issuer, type, style and expiry. Two real examples are `DBS MB ECW220627` and `HSI 21000MBEPW220530`. Decode them as follows:

Element	Meaning
Leading text	The **underlying** — `DBS` for DBS shares, `HSI` for the Hang Seng Index
Number after an index name	The **exercise level** of an index warrant, e.g. `21000`
Two-letter code	The **issuer** — `MB` is Macquarie Bank, `VT`/`BK`/`GS` others
`e` prefix	**European** style; no prefix means American
`CW` / `PW` / `DC`	**Call warrant**, **put warrant**, **discount certificate**
Six digits	Expiry as **YYMMDD** — `220627` is 27 June 2022

Element	Meaning

5.5 Trading Structured Warrants

5.5.1 Interpreting the Trading Name

Read the name before the price. The code fixes the underlying, issuer, direction, style and expiry, and the term sheet fixes the ratio and settlement. Inferring any of these from the ticker alone is an error.

5.5.2 Investing in Call and Put Warrants

DBS MB ECW1710020. DBS traded at **SGD 18.16** on 6 December 2016. The warrant had strike **X = SGD 16.50**, warrant price **WP = SGD 0.325**, conversion ratio **n = 8:1**, last trading day 25 September 2017 and expiry 2 October 2017. Initial gearing was:

$$\frac{18.16}{0.325 \times 8} = 6.98\times$$

The subsequent path shows both the leverage and the decay:

Date	DBS price	Move	Warrant price	Move
6 Dec 2016	18.16	—	0.325	—
3 Jan 2017	17.32	−4.63%	0.245	**−24.62%**
18 Jan 2017	18.20	+0.22%	0.305	**−6.15%**
2 Aug 2017	22.19	+22.19%	0.665	**+104.62%**

Holding **8,000 warrants** gave values of **SGD 1,960**, **SGD 2,440** and **SGD 5,320** at those dates; selling out produced a gain of **SGD 2,720**. Holding to maturity and settling cash:

$$(22.19 - 16.50) \times 1{,}000 - 2{,}600 = \text{SGD } 3{,}090 \ (\mathbf{118.8}\%)$$

The breakeven share price for a new buyer at 0.665 is **SGD 21.82**, being the warrant price expressed per share plus the strike:

$$0.665 \times 8 + 16.50 = 21.82$$

Note the second row: the share rose 0.22% and the warrant still fell 6.15%, because time decay outweighed the move. That is the central risk of a short-dated warrant.

5.5.3 Investing in Put Warrants

The Hang Seng Index stood at **27,854.9** on 9 August 2017, up **25.8%** year to date from 22,150.40. A manager wanting downside protection screened SGX and HKSE put warrants. Three SGX candidates moved as follows:

Warrant	Price at screen	Price at review
HSI 26200 MB EPW170928	0.133	0.049
HSI 26388 VT	0.076	0.062
HSI 27000 MB	0.122	0.108

Thirteen HKSE warrants were also screened — from UBS, SG, JP Morgan, Hang Seng Bank and Goldman Sachs. The chosen contract was **28585 GS-HSI @EP1802B**: underlying **27,757.09**, strike **27,200**, warrant price **HKD 0.205**, conversion ratio **8,000**, last trading day 21 February 2018 and expiry 27 February 2018. Gearing was:

$$\frac{27{,}757.09}{0.205 \times 8{,}000} = 16.93\times$$

Scenario	Index	Warrant price	Move	Maturity settlement
1	25,814.09 (−7%)	0.31	**+51.2%** (+HKD 840)	HKD 1,385.91 against HKD 1,640 outlay, so −
2	24,981.38 (−10%)	0.36	**+75.6%** (+HKD 1,240)	+HKD 578.62
3	29,422.51 (+6%)	0.11	**−46.3%** (−HKD 760)	Worthless at expiry

Scenario 1 is the instructive one: the warrant rose 51.2% on the screen, yet holding to maturity still lost money, because near maturity the price decays towards intrinsic value and the buyer's breakeven was **24,720**. Scenario 2's breakeven was **25,560**, a fall of 7.9% from spot.

5.5.4 Delta

Warrant delta measures the change in warrant price per unit change in the underlying, scaled by the ratio:

$$\delta = n \times \frac{\partial WP}{\partial S}$$

Delta is about **0.5** at the money, approaching **1** deep in the money. The DBS call carried a delta above **90% (95%)**; the GS put carried **−45.33%**.

Statistic	DBS call	GS put
Effective gearing	**6.65×**	**7.58×**
Gearing	7×	16.73×
Moneyness	29% in the money	0.871% out of the money

5.5.5 Effective Gearing

Effective gearing combines delta and gearing and approximates short-term percentage sensitivity far better than gearing alone:

$$\text{effective gearing} = \delta \times \text{gearing}$$

For the DBS call,

$$0.95 \times 7 = 6.65\times$$

. For the GS put,

$$-0.4533 \times 16.92 = -7.67\times$$

; the statistics box alongside the same example prints −7.58×, so treat the two as the same magnitude with a rounding or basis difference and quote whichever figure a question supplies.

5.5.6 Implied Volatility

Implied volatility is backed out of the observed warrant price; it is the market's current pricing of the underlying's future volatility, not a forecast, and it moves independently of realised volatility.

5.6 Valuation

5.6.1 Intrinsic Value

$$IV_{\text{call}} = \max\left[0, \frac{S-X}{n}\right] \qquad IV_{\text{put}} = \max\left[0, \frac{X-S}{n}\right]$$

5.6.2 Conversion Price

$$\text{call conversion price} = X + n \times WP \qquad \text{put conversion price} = X - n \times WP$$

5.6.3 Premium

The **warrant premium** is the price above intrinsic value. The dollar premium is:

$$\text{call} = n \times WP + X - S \qquad \text{put} = n \times WP - X + S$$

DBS: intrinsic value **SGD 0.2075**, conversion price **SGD 19.10**, premium **5.18%**. **GS:** with the underlying at 24,981.38, intrinsic value **SGD 0.2773**, conversion price **SGD 25,560**, premium **−2.32%** — a negative premium, because the put was out of the money with little time left.

5.7 Roles of the Holder and Issuer

The **listing document** sets out the terms and the contractual rights and obligations of both parties.

- **Holder's right and obligation.** The right, but not the obligation, to buy or sell a stated number of shares at the exercise price within the period. The holder's only obligation is to pay the exercise price if it chooses to exercise.
- **Issuer's right and obligation.** To deliver the underlying and receive the exercise price on a call exercise, or receive the underlying and pay the exercise price on a put exercise. Most, if not all, SGX-ST structured warrants settle in cash rather than by delivery.

5.8 Role of Designated Market Makers in Structured Warrant Trading

An issuer that **commits to make a market** in the warrants it issues is **exempt from the minimum placement and holder-size requirements** for listed equity securities, and the minimum issue size is **SGD 2 million** (previously SGD 5 million).

The listing document states whether the issuer has made that commitment. A designated market maker (**DMM**) appointed by the issuer must then provide competitive bid and offer prices during trading hours as prescribed by SGX, creating adequate liquidity. The **maximum spread** between the DMM's bid and offer, and the **minimum lot size**, are set out by the issuer in the listing document. DMMs may be large international banks and brokers able to react to changing conditions and generate demand.

The spread and quote size are part of the investor's liquidity risk: a DMM that is not obliged to quote, or is quoting wide, may not offer an exit at a fair price.

5.9 Settlement of Warrants

Structured warrants trade and settle like ordinary shares and company warrants listed on SGX-ST.

5.9.1 Physical Settlement

To exercise a call warrant the holder submits an **exercise notice with cash payment** of the exercise price to the **warrant agent**. The agent informs **CDP** and the issuer; CDP credits the holder's account with the underlying securities and debits the issuer's.

For a put warrant, CDP debits the holder's account for the securities and credits the issuer's, and the warrant agent makes a cash payment on the issuer's behalf to the holder. Whichever the type, **exercise expenses are payable by the holder** to the warrant agent.

5.9.2 Cash Settlement

Settlement is the in-the-money amount per warrant multiplied by the number of warrants, paid in cash rather than by delivery. This is the normal route for SGX-ST structured warrants.

5.9.3 Settlement Procedures

SGX uses **Asian-style expiry**. In the "Ready" and "Unit-Share" markets the last trading day is **at least three business days before expiry**, and the settlement price is the **arithmetic average of official closing prices over the five market days before expiry**. For the DBS warrant above, the last trading day was **25 September 2017** against an expiry of **2 October 2017**.

The averaging makes the final payoff **path dependent** and prevents a single closing print from setting the settlement.

5.10 Corporate Actions

Dividends, special dividends, buybacks and cancellations, rights and bonus issues, and splits can all adjust the strike and conversion ratio so the product stays economically comparable. The adjustment factors are:

Event	Factor
Dividend / special dividend	$\frac{P - SD - ND}{P - ND}$
Buyback or cancellation	$\frac{A}{B}$ and $\frac{P - CD}{P}$
Split	$\frac{A}{B}$

The **new strike = old strike × factors**, and the **new conversion ratio = old conversion ratio × factors**. Read the issuer's adjustment announcement rather than assuming the terms.

5.11 Exotic Warrants

5.11.1 Index Warrants

Settled on exercise by cash. The cash received is the difference between the exercise level and the current or settlement index level, multiplied by a multiplier converting index points into cash.

For call warrants on MSCI Singapore Free Index futures:

$$\text{Cash settlement} = \max\left\{0,\ \text{SGD } 1.00 \times \left[\frac{\text{Settlement Reference Level} - \text{Exercise Level}}{n}\right]\right\}$$

The settlement reference level is based on the underlying index futures contract's final settlement price as determined by the derivatives exchange. Because the index has already been adjusted for corporate actions, **index warrants normally need no adjustment**.

5.11.2 Currency Translated Warrants

The settlement and trading currency — the investor's home currency — differs from the underlying asset's currency. For a structured warrant on the HSI listed on SGX-ST, the underlying and reference currency is **HKD** while settlement is in **SGD**.

$$\text{Settlement amount} = \max\left\{0,\ \text{HKD } 1 \times \left[\frac{\text{Reference Level} - \text{Exercise Level}}{n}\right] \times \text{Exchange Rate (SGD/HKD)}\right\}$$

5.11.3 Warrants with Yield Enhancement

Designed to give an attractive yield based on the current share or spot price. For marketing, **capped warrants** are often grouped with other yield-enhancement securities as **discount certificates**.

In a capped warrant the **cap strike** is the maximum price achievable. Upside exposure is sacrificed in return for buying the underlying at a discount to market. At maturity, if the closing price is **at or above** the exercise price the holder receives cash equal to the exercise price — the cap; if **below**, the holder receives the value of the underlying.

Worked example. Issue price of the warrant **SGD 5.25**, which is **89.0%** of the reference spot; reference spot **SGD 5.90**; underlying ABC Ltd; conversion ratio **n = 1**. The cap is **SGD 5.45**, being **92.4%** of spot. At or above 5.45 the holder receives 5.45; below it the holder receives the underlying value, so a fall below 5.25 produces a loss despite the discount.

5.11.4 Basket Warrants

The warrant references a basket of underlyings. Correlation, the weights and the composition of the basket matter, and a worst-of basket pays on the weakest constituent.

5.11.5 Commodity Warrants

Commodity warrants typically carry **longer tenors** than commodity futures or options. The terms must verify the **benchmark** used for settlement and the delivery-location, storage and transport arrangements, because a commodity's price is location- and grade-specific.

5.11.6 FX Warrants

FX warrants give directional exposure to a currency pair. A USD/JPY call and a JPY/USD put express the same view; the quotation convention must be checked. Commodity currencies such as **AUD and CAD** carry additional sensitivity to their export commodities.

5.12 Daily Leveraged Certificates

5.12.1 DLC Features

DLCs are exchange-traded, short-term, high-risk products delivering a **daily** multiple of an underlying's movement. Long and short certificates are commonly issued at **3× or 5×**, with terms permitting leverage up to **7×**; the current term sheet controls the multiple.

Feature	Specification
Board lot	**100**
Currency	SGD
Leverage	commonly 3× or 5×, terms permitting up to **7×**
Maximum tenure	**3 years**
Trading and settlement	SGX/CDP, **T+2**
Fees	brokerage around **0.001%** (0.1bp), clearing around **0.004%** (0.4bp), plus a management fee cov
Airbag threshold	**20%** for a 3× product, **10%** for a 5× product
Origin	European constant-leverage products and factor certificates; introduced on **SGX in 2017**

The naming convention identifies the product, leverage and underlying, for example `DLC SG 5x Long HSI 200714` with code **CIUW**.

5.12.1.1 Daily Reset and Compounding Daily reset means a DLC is **not** a simple multiple of the underlying's total return. In a volatile or mean-reverting market, repeated daily losses erode capital even if the index ends where it started.

The HSI in August 2017 shows the effect:

Date	HSI	Index move	DLC price	DLC move
4 Aug	27,576.59	—	3.11	—
7 Aug	27,699.35	+0.45%	3.11	+1.63%
8 Aug	27,854.91	+0.56%	3.22	+3.54%
9 Aug	holiday	—	—	—
10 Aug	27,462.73	−1.06%	3.00	**−6.83%**
11 Aug	26,890.96	−2.08%	2.70	**−10.00%**

Over the period the index fell **2.49%** while the DLC fell **11.66%**. As a rule of thumb, an 8–10% index move can produce a 47–61% move in a 5× DLC.

The five-day compounding arithmetic is stark. At **+2% a day** for five days the index gains **10.41%**, while a 3× long gains **33.82%** and a 3× short loses **33.82%**; a 5× long gains **61.05%** and a 5× short loses **61.05%**.

5.12.1.2 Designated Market Maker Obligations A DLC DMM must quote within a maximum bid-offer spread of **10 ticks or SGD 0.20, whichever is greater**, and for a minimum quantity of **10,000 DLCs**. Quote obligations may **stand down** in six circumstances: in the **five minutes before the pre-open**; when the **underlying is suspended or subject to price limits**; when the **DLC is suspended or limited**; in **exceptional volatility**; on **technical failure**; or where the DMM is **unable to hedge or unwind**.

5.12.1.3 The Airbag The airbag suspends the product when the underlying moves adversely by the stated threshold, then restarts it from an observed reset level after a **30-minute suspension**.

Worked example. Prior close **SGD 2.00**, index 1,000. At 11:00 the index is **900**, a **10%** fall, and the 5× DLC is **SGD 1.00**, a 50% fall. The airbag triggers a suspension from **11:00 to 11:30**. A further 5% fall in the index is 25% of SGD 1.00, so the DLC reopens at **SGD 0.75**. Without the airbag it would have been **SGD 0.50**. A further 10% fall takes it to zero. The post-trigger observation window is **15 minutes**.

5.12.2 DLC Benefits

Eight benefits define the product: exchange trading and transparency; defined leverage; long and short access from one instrument; no need for a margin account; capital limited to the purchase price; **no implied volatility effect**, because the payoff is delta-one rather than optional; **no time decay** in the option sense; and a single, simple underlying.

5.12.3 DLCs at Expiry

Unlike leveraged ETFs, DLCs are not reverse-split to keep the price in a range, so **they can go to zero and stay there**. At 2% a day against a 5× product, the price path is roughly 2.50 → **0.01** over 50 days. The Dow Jones fell around **50%** from October 2007 to February 2009; a leveraged product held through such a path loses a multiple of that.

5.12.4 Comparisons

Product	Typical maximum leverage
CFD	up to **20×**
DLC	**3–7×**
Warrant	**5–50×**
Leveraged ETF	up to **3×**

Compared with CFDs, DLCs are exchange traded and have no OTC financing or margin account in the same form. Compared with warrants, they target a daily multiple rather than an option payoff. Compared with leveraged ETFs, their issuer, reset mechanics and the absence of reverse splits differ.

5.13 Summary

Warrants embed options issued by a bank; the price equals intrinsic value plus time value and is shaped by the conversion ratio, gearing, volatility, dividends, rates, time, liquidity and issuer credit. DLCs amplify daily returns and reset, making path dependency and compounding the central risks. SIP/CAR suitability and disclosure are part of the product analysis, not an administrative add-on.

Exam cues

- **Suitability first:** warrants and DLCs are SIPs, so a **CAR** must be completed and the investor must meet one of the three criteria — education, work experience or investment experience.
- **Cash settlement formulas:** call = (S − X)/n; put = (X − S)/n. Gearing = share price ÷ (warrant price × n). Effective gearing = delta × gearing.
- **Conversion price:** call = X + nWP; put = X − nWP. The DBS example: 0.665 × 8 + 16.50 = **SGD 21.82** breakeven.
- **Seven price factors:** life, underlying price, exercise price, volatility, dividends (mild), interest rate and leverage. Calls rise with the underlying, volatility, rates and gearing and fall with dividends; puts invert the spot, dividend and rate signs.
- **Naming:** underlying, then index exercise level, then two-letter issuer, `e` for European, `CW`/`PW`/`DC`, then YYMMDD.
- **DMM relief:** a market-making issuer is exempt from placement and holder-size rules, with a **SGD 2 million** minimum issue size (was SGD 5 million).
- **Asian expiry:** last trading day at least **three business days** before expiry; settlement is the average of the **five** closing prices before expiry.
- **DLC numbers:** board lot **100**; up to **7×**; maximum tenure **3 years**; airbag **20%** at 3× and **10%** at 5×; a **30-minute** suspension; DMM spread **10 ticks or SGD 0.20**, minimum **10,000** certificates.

- **Compounding:** five days at +2% gives +10.41% on the index but +33.82% on a 3× long. A 2.49% index fall produced an 11.66% DLC fall.
- **Trap:** a warrant can rise 51.2% on the screen and still lose money when held to maturity, because of decay. And a leveraged product is never a buy-and-hold instrument — the daily reset, not the total return, is what is delivered.

Chapter 6 — Barrier Options, Binary Options and Callable Contracts

Exotic features are sold as cheaper or higher-yielding alternatives to vanilla exposure. They are not cheaper — the cost is displaced into a condition, a path dependency or a termination right. This chapter is about locating that displacement.

Learning Objectives

The candidate should be able to:

- Explain how an exotic option differs from a vanilla option.
- Distinguish knock-out from knock-in barriers and single from double barriers.
- Explain the effect of a barrier on the premium and on the path dependence of the payoff.
- Explain binary and basket payoffs.
- Explain the mandatory call event and the Category N / Category R distinction for CBBCs.
- Compute a CBBC's theoretical price, gearing and residual value.

6.1 Introduction

Exotic derivatives extend a vanilla option by changing one or more of its features:

- **a. Exercise events** — conditions under which the option lives or dies.
- **b. Path dependency** — the payoff depends on the route the price took, not just the level at expiry.
- **c. Payoff shape** — digital, range, or otherwise discontinuous rather than linear.
- **d. Conditional events or trigger points** — if the price touches or crosses a trigger, validity is affected and the final payoff changes.
- **e. Underlying assets** — extending beyond traditional classes and indices to alternative investments such as real estate, and to market indicators such as volatility.
- **f. Currency features** — producing a payoff in a different currency, as in a Quanto structure.

Before recommending one, explain how the payoff is computed, what observation triggers action, whether the option can terminate early, whether the premium can be lost, and who bears counterparty risk.

6.2 Barrier Options

Barrier options are among the most popular exotic structures banks sell. Clients find them easy to understand because the concept is "game-like", following an **if-then-or-else** pattern.

The payoff depends not only on whether and by how much the option is in the money at exercise, but also on whether the underlying has:

- **a.** touched the barrier at any time during the option's life;
- **b.** been above or below the barrier at expiry, or throughout the life; or
- **c.** stayed within two barriers, or outside two barriers, at expiry or throughout.

In a **knock-out** (or kick-out) option, touching the barrier **deactivates** the features and rights. In a **knock-in** (or kick-in) option, touching the barrier **activates** them.

Investment professionals must review the terms to establish exactly how the payoff is calculated, what is payable once the barrier event occurs, and what happens if it does not — because the payoffs differ depending on the price path. Features and payoffs of the options inside structured products are frequently **negotiable and customisable**.

A **barrier** is a price trigger level which, when reached, produces an event: a payment of interest, a higher interest rate, a currency conversion, a sale or purchase of stock. Depending on the underlying, the trigger

may reference a stock price, an index level, an FX rate, a money-market or bond interest rate, or a weighted average of a basket.

Barrier options are **path dependent**, because they are activated or cancelled when a set trigger is reached. Intuitively they are **cheaper than their vanilla counterparts**, because the investor risks being knocked out, or never knocked in.

6.3 Knock-Out Barrier Options

A knock-out option ceases to exist when the barrier is touched. Two barrier configurations matter.

6.3.1 Single Barrier Knock-Out

Investor	Position	Consequence
A	Call with strike $8.00 * *, knock - out * *$**9.00**, spot **$7.00**	Maximum payoff $8.99 - 8.00 -$ price, breakeven
B	Put with strike $13.50 * *, knock - out * *$**12.00**, spot **$15.00**	Maximum payoff $13.50 - 12.01 -$ price, breakeve

The economic decomposition is a **visible vanilla call plus an embedded sold call at the barrier**. That is why the option is cheaper: the investor has sold the issuer the payoff beyond the barrier.

Contrast this with a bull or bear **spread**. In a spread the distant leg is a separately priced short option. In a knock-out, breaching the barrier **extinguishes the whole option**, so the investor loses the **full intrinsic value** accumulated to that point, not merely the capped excess.

6.3.2 Double Barrier Option

A **double knock-out call** carries two barriers. With strike $8.00 * *, spot * *$**7.80**, lower knock-out $7.00 * *and upperknock - out * *$**9.00**, there are three outcomes:

- the underlying stays inside 7.00–**9.00** and the call pays at expiry, with a maximum of

$$8.99 - 8.00 = \$0.99$$

  ;
- the underlying touches **$7.00** and the option is extinguished;
- the underlying touches **$9.00** and the option is extinguished.

Because it can be knocked out at **both** ends, a double barrier option is **cheaper than a straight call and put combined** with the same specifications. Note that launching the option in the money — spot at or above the strike — is the more popular configuration.

6.4 Knock-In Barrier Options

For knock-ins the reverse applies: the option becomes **activated** if the price touches the knock-in barrier. The terms become active from the moment of the touch **through to expiry, even if the price falls back**. A **rebate** is usually paid to the buyer at expiry if the knock-in barrier is **never** reached — compensation for the option never coming to life.

6.5 Summary and Application of Barrier Options

Unlike regular European calls and puts, barrier options are **path dependent**.

- **Knock-in.** The option only comes into existence when the condition is met, typically when the barrier is touched at any time. Once breached, the underlying call or put becomes live and then behaves as an ordinary European option.
- **Knock-out.** The underlying European option is **extinguished** when the barrier is touched. Once breached, the entire structure expires worthless.

Type of barrier option	When the barrier is exceeded
Knock-out	Deactivates the option; the underlying option is extinguished
Knock-in	Activates the option; the underlying option becomes live

6.5.1 Advantages of Barrier Options

- **a. Cheaper premium** than standard options, because the knock-out may terminate early, or the knock-in may never activate. A **double** knock-in has a **higher** knock-in probability than a single knock-in.
- **b.** Leverage on capital for a given view.
- **c.** A **wider selection** of option products for diverse market views.
- **d.** Knock-in barriers suit **large** expected moves; knock-out barriers suit **small sideways** moves.

6.5.2 Disadvantages of Barrier Options

- **a. High risk of losing the premium**, either because a knock-out occurred or because no knock-in ever occurred.
- **b.** They are **OTC** products. The investor pays upfront and does not trade on an exchange, so bears **issuer counterparty risk**.

Barrier features are used heavily by banks in FX structures. A typical structured product embeds a knock-out option:

Term	Value
Current spot	**112.00**
Strike price	**112.00**
Knock-out barrier	**119.00**
Minimum yield	**0%**
Knock-out rebate	**2%**
Money-market rate	**1.5%**
Maturity	**3 months**
Participation rate	**100%**
Minimum size	**USD 500,000**

The client has money-market deposits, and the interest is invested in a knock-out barrier option with a **2% rebate** to make the downside more attractive. The option is active at inception. As long as spot trades **below 119**, the client participates in the movement from 112 to 118.99; at or above 119 the structure knocks out and the rebate applies.

6.6 Barrier Features in Structured Products

6.6.1 Barrier Capital Preservation Certificates

Often called a **shark's fin**, the structure combines a zero-coupon principal component with an **up-and-out call**. It suits an investor expecting a rise that will not breach the barrier.

Six characteristics define it:

- **a.** Capital preservation is defined as a percentage of **nominal**, for example 100%.
- **b.** Capital preservation refers to the nominal **only at maturity**, and **not to the purchase price**.
- **c.** The product's value may fall **below** its capital-preservation level during its life.
- **d.** Participation in positive underlying performance continues **until knock-out**.
- **e.** A **rebate** may be paid following knock-out.
- **f.** Upside is **capped**.

6.6.2 Barrier Reverse Convertibles

The barrier reverse convertible enhances yield when the underlying is expected to be flat or to move slightly. It offers **conditional** capital preservation through a **kick-in barrier** set a specific percentage below the share price at issue. It is commonly called a reverse convertible note and comprises a note, money-market instrument or bond, plus a **put option sold by the investor to the issuer**.

Investment view:

- **a.** neutral on the underlying's direction;
- **b.** expects flat to very slightly moving prices with **falling volatility**;

- **c.** expects the underlying **not** to breach the barrier during the life; and
- **d.** is looking for income or enhanced yield.

Characteristics. By selling the knock-in put the investor takes **downside risk**: if the underlying finishes **below the strike at expiry and the kick-in level was breached**, the investor **loses capital**. Principal is at risk once the barrier is touched.

The payout, with the strike and finish expressed as levels, is:

$$\text{Payout} = \min\left\{100\%,\ 100\% - \frac{1}{\text{Strike}} \times \max\left[0,\ \text{Strike} - \frac{S_T}{S_0}\right]\right\}$$

The **coupon is paid regardless**; what changes is whether principal is returned in full or converted into a devalued underlying.

6.6.3 Bonus Certificates

A bonus certificate sets a barrier below spot and a bonus level above it. If the barrier is not breached, the investor receives at least the bonus; if it is, protection and bonus disappear.

Investment view:

- **a.** neutral or generally bullish that the underlying will not trade below the barrier.

Characteristics:

- **i.** Exposure similar to direct investment, with a **minimum return (the bonus)** if performance is flat or slightly negative — available **so long as the underlying stays above the barrier**.
- **ii.** If the underlying is **below the barrier**, the investor has **1:1 downside exposure**.
- **iii.** The barrier is usually observed **continuously** throughout the term.
- **iv.** Investors **forgo dividends**, which partly fund the issuer's purchase of the put.
- **v.** If caps are introduced to limit upside, the certificate prices cheaper or the bonus can be increased.

Risks: a loss equal to the negative performance of the underlying; surrendered dividends; and issuer credit risk.

The issuer has four levers: the level of participation and any cap; the strike or bonus level; the knock-out barrier level; and maturity. The trade-offs are real. Lowering participation and capping upside allows a **higher bonus** or a **further** barrier. Lowering the bonus towards the initial price may allow a **closer** barrier or **shorter** maturity. Lengthening maturity raises investor risk, so a **higher bonus** may be required to sell it.

6.6.4 Barrier Capital Preservation Certificates (Straddle)

This product contains **two knock-out options** — a barrier call and a barrier put. The investor receives a **capped** return at maturity if **no knock-out event occurs**. There are two possible knock-out events, one for each option; if either occurs, the investor receives the agreed capital amount instead.

6.6.5 Positioning of the Barriers

Barrier placement sets the economics. A **closer** barrier is **cheaper** but **easier to trigger**; a **wider** barrier costs more but preserves more vanilla-like participation. The investor must be told which side of spot the barrier sits on, whether it is monitored continuously or on observation dates, and whether a rebate applies.

6.7 Binary Options

A **binary** or digital option pays a fixed amount if a condition is satisfied and zero, or another fixed amount, otherwise. **One-touch** pays if the barrier is touched; **no-touch** pays if it is never touched; **double-no-touch** or range structures pay if neither boundary is breached.

Binary payoffs are **discontinuous** and highly sensitive to barrier proximity, volatility, time and the **monitoring convention**.

A **one-touch** example: EUR/USD spot **1.08**, strike **1.04**, 3 months, maximum **10%**, minimum **1%**, money-market rate **2.00%**, minimum **USD 100,000**.

A **range currency deposit**: range **1.06–1.10**, 3 months, maximum **8.25%**, minimum **2.00%**, money-market rate **1.50%**, minimum **USD 100,000**.

A **range accrual** paying on observations inside the range:

$$\frac{60}{90} \times 8.25\% + \frac{30}{90} \times 2.00\% = 6.17\%$$

The trap is that **88 of 90 days** inside the range with **one** day outside does not deliver close to 8.25% if the terms count day by day — read whether the condition is observed **at expiry or throughout**.

6.8 Basket Options

A basket option references several assets. **Correlation** drives the premium: imperfect correlation can make a basket cheaper than buying each single-asset option, but it creates correlation, composition, rebalancing, dividend and concentration risk. A **worst-of** basket pays on the weakest constituent and is especially correlation-sensitive.

6.9 Knock-Out Warrants

Knock-out warrants combine exchange-traded warrant leverage with a barrier that can terminate the product before its stated expiry. A breach may leave the warrant worth zero or a residual amount under the terms.

Six characteristics matter: exchange trading; a barrier that terminates early; delta close to one before termination; a **flattening skew** in pricing; issuer and DMM dependence; and settlement on the terms rather than on the expiry date. Compare the trigger, settlement, issuer, DMM, ratio and monitoring rules against a standard warrant.

6.10 Callable Bull/Bear Contracts

CBBCs are exchange-traded leveraged long or short products, usually linked to an equity or index. A **bull** profits from a rise; a **bear** from a fall. Delta is often close to 1 before the call event.

6.10.1 Price Movement

A bull CBBC moves with the underlying, up or down, until the call event. Because delta is near 1 and the conversion ratio is fixed, the percentage move in the CBBC is a multiple of the percentage move in the underlying.

6.10.2 Mandatory Call Event

The **mandatory call event (MCE)** is triggered when the underlying reaches the **call price**:

- for a **bear**, spot rises to or above the call price;
- for a **bull**, spot falls to or below the call price.

On an MCE the product terminates immediately. It is not a stop-loss the investor controls.

6.10.3 Gearing

$$\text{gearing} = \frac{\text{underlying price}}{\text{CBBC price} \times \text{conversion ratio}}$$

A conversion ratio of **10:1** means **10 CBBCs per share**.

6.10.4 Hedging

The issuer hedges the CBBC by trading the underlying. Because delta is close to 1, the hedge ratio is close to the conversion ratio, and the issuer's hedge must be unwound on an MCE.

6.10.5 Category N and Category R CBBCs

Category	Call price relative to strike	Residual value after MCE
Category N	Call price **equals** the strike	**No residual value** — the product is terminate
Category R	Call price is **above** the strike for a bull (in the money)	Residual value **may** be paid, but **may be zero**

Category R is not a guarantee of a positive residual. The call price, strike and conversion ratio are all specified in the term sheet.

6.10.6 Theoretical and Market Pricing

$$\text{bull CBBC} = \frac{(\text{underlying} - \text{strike}) + \text{financial cost}}{\text{conversion ratio}}$$

$$\text{bear CBBC} = \frac{(\text{strike} - \text{underlying}) + \text{financial cost}}{\text{conversion ratio}}$$

The **financial cost** comprises funding cost, the dividend adjustment and the issuer's margin.

6.10.7 Residual Value on MCE

For a Category R CBBC the residual value is direction-specific:

$$\text{bull residual} = \frac{\text{settlement price} - \text{strike}}{\text{conversion ratio}} \qquad \text{bear residual} = \frac{\text{strike} - \text{settlement price}}{\text{conversion ratio}}$$

The **settlement price** follows the MCE observation rules: a **bull** uses the **minimum** and a **bear** the **maximum** traded price between the MCE and the next session, subject to the product terms.

6.10.8 Valuation at Maturity

At maturity the same direction-specific formula applies, using the **closing** settlement price rather than the post-MCE observation.

6.10.9 Capital Adjustments

Corporate actions adjust the strike, call price and conversion ratio to keep the product economically comparable. **Regular dividends do not** require adjustment, because they are already reflected in the financial cost.

6.10.10 Other SGX Issues

The issuer publishes a **listing document** stating the terms, including the call price and the observation method. Read it rather than inferring from the ticker.

6.10.11 CBBC Examples

Category R bull. Strike $10.00 * *, callprice * *$**11.00**, conversion ratio **10:1**, financial cost **10% p.a.**, 12 months, spot **$12.00**.

$$\text{price} = \frac{(12.00 - 10.00) + \text{financial cost}}{10} = \$0.30$$

Scenario	Outcome	CBBC price	Return
1 — after 6 months the underlying is **$15.00**	Residual rises with the underlying	**$0.55**	The underlying gaine
2 — MCE at **$10.50**	Residual $(10.50 - 10.00)/10 = \$0.05$	**$0.05**	A loss of **$0.25**, or -
3 — at maturity the underlying is **$15.00**	$(15.00 - 10.00)/10$	**$0.50**	**+66.67%**

The size of scenario 2's loss is the point. The underlying fell only 12.5% from 12.00 to 10.50, but the CBBC lost 83% of its value — that is the leverage, and the MCE removed any chance of recovery. Recompute the percentage against the **purchase price of $0.30** and state your base.

Category R bear. Strike $11.00 * *, callprice * *$**10.00**, conversion ratio **1:1**, spot **$9.00**.

$$\text{price} = \frac{(11.00 - 9.00) + \text{financial cost}}{1} = \$3.10$$

Scenario	Outcome	CBBC price	Return
1 — after 6 months the underlying is **$7.50**	Residual falls as the underlying falls	**$4.05**	The underlying fell 16
2 — MCE at **$10.50**	Residual $(11.00 - 10.50)/1 = \$0.50$	**$0.50**	A loss of **$2.60**, or **−**
3 — at maturity the underlying is **$6.50**	$(11.00 - 6.50)/1$	**$4.50**	**+45.16%**

6.11 Risks of CBBCs

- **MCE and early termination** — the product can end before the investor's view plays out.
- **Residual-value uncertainty** — a Category R residual may be zero.
- **Issuer credit risk.**
- **Liquidity and DMM spread** — quote obligations can stand down.
- **Gap risk** — the underlying can jump through the call price.
- **Leverage** — magnifies both directions.
- **Financing cost** — accrues against the position.
- **Underlying suspension** — trading halts suspend the product.
- **No margin misconception** — there is no margin account, but the capital at risk is the whole purchase price.

6.12 Similarities and Differences

Feature	Barrier option	Binary	Warrant / knock-out warrant	CBBC
Typical venue	OTC	OTC or exchange	Exchange	Exchang
Payoff	Conditional vanilla or rebate	Fixed, discontinuous	Leveraged option	Leverag
Main trigger	Barrier touch or cross	Event, range or touch	Product barrier, if any	Mandat
Path dependent	**Yes**	**Yes**	Only with a barrier	**Yes**
Key extra risk	Path and counterparty	Discontinuity and monitoring	Issuer and time value	Early te

6.13 Summary

Exotics exchange a lower premium or a higher coupon for **conditionality**. The barrier level, its direction, the monitoring convention, any rebate, knock-in or knock-out status and the settlement basis determine the payoff. CBBCs add a mandatory call event and the Category N / Category R distinction; an investor can lose most or all of the capital quickly.

Chapter 6 Appendix — Average Price (Rate) Options (non-examinable)

An **average price** or **Asian** option settles against the arithmetic mean of the underlying over a stated sampling period rather than a single point:

$$\text{Payoff}_{\text{call}} = \max\left\{S_{\text{avg}} - X,\ 0\right\} \qquad \text{Payoff}_{\text{put}} = \max\left\{X - S_{\text{avg}},\ 0\right\}$$

Averaging **reduces** the option's sensitivity to a spike near expiry, so an average price option is **cheaper than the equivalent vanilla**. The longer the averaging window, the larger the discount. Sampling may be daily, monthly or quarterly.

Structured products use them to neutralise **tail-end spikes**: an underlying that gaps on the observation date cannot move the settlement as much when that date is one of many in the average. The trade-off is that the investor also gives up the benefit of a favourable spike.

Exam cues

- **Path dependence is the defining feature.** A barrier does not change the final level; it changes whether the option is alive at that level. Two assets finishing at the same price can produce different outcomes.
- **Knock-out kills, knock-in births.** Knock-out extinguishes the option and can lose the **full intrinsic value** already accumulated — a sharper loss than a spread's capped excess. Knock-in activates the terms from the touch **through expiry even if price falls back**, and pays a **rebate** if never activated.
- **Double barriers are cheaper** than a straight call plus put, because they can be knocked out at both ends. A **double knock-in** has a **higher** probability of activation than a single.
- **Barrier capital preservation is nominal at maturity only** — not the purchase price — and the product's market value may fall below that level during its life.
- **BRC payout:** min(100%, 100% − (1/Strike) × max[0, Strike − S_T/S_0]). The coupon is paid regardless; only principal is at risk once the kick-in is breached.
- **Bonus certificate:** bonus applies only while above the barrier; below it, **1:1 downside**; dividends are forgone.
- **CBBC:** call price is where the MCE fires — bull falls to it, bear rises to it. **Category N** has **no** residual; **Category R** may have residual but may be zero.
- **CBBC arithmetic:** bull price = ((underlying − strike) + financial cost)/ratio. R-bull at 12.00 strike 10.00 ratio 10 = $0.30 * *; afteranMCEat10.50theresidualis * *$**0.05**.
- **Binary trap:** a range accrual observes **days**, so 88 of 90 days inside does not pay the maximum.
- **Trap:** an investor can lose 83% on a CBBC when the underlying moves only 12.5%, and there is no margin call to warn them — the whole purchase price is at risk from the outset.

Chapter 7 — Structured Deposits and Other Structured Products

A structured deposit is a bank deposit whose return is produced by an embedded option. The principal may be preserved, but the yield is not, and the conversion obligation is real. This chapter teaches the decomposition and the suitability test.

Learning Objectives

The candidate should be able to:

- Explain why structured products are treated as complex investments.
- Apply the seven suitability criteria to a structured product.
- Distinguish capital-preserving from yield-enhancing deposits.
- Decompose a structured deposit into deposit plus option legs.
- Compute a participation rate, a range-accrual payout and an accumulator's obligation.
- Identify the worst case, not only the headline yield.

7.1 What are Structured Products?

Structured products combine a deposit, note, fund, warrant, ILP, ETF or ETN with one or more derivatives. They can customise a market view, enhance yield, preserve capital at a stated maturity, provide participation, a discount or leverage, hedge an exposure, change currency, or rebalance without directly buying the underlying.

These are **complex investments**. Their risks and payoff structures are less easily understood and they are usually offered to more experienced investors. Losses from OTC derivatives and structured products have historically been the subject of litigation by retail and high-net-worth individuals against banks, with clients claiming they did not understand, were not informed, or were misinformed about the risks. Verification of a client's risk profile and investment understanding, supported by **documentation and acknowledgement**, has therefore become standard practice.

From the institution's side, the growth of derivatives over 30 years helped offset narrowing trading commissions on traditional asset classes. OTC structured products raised transaction volume and generated higher returns than trading traditional investments would have.

7.1.1 Client Suitability for Structured Products

Seven criteria must be considered:

Criterion	What it requires
a. Risk appetite	The investor's **ability and willingness** to take risk — both, not either.
b. Investment experience and knowledge	Confirm the investor can understand the features and risks. The investo
c. Liquidity needs	Review liquidity needs and **highlight the consequences of early red**
d. Investment horizon	Value may only be fully realised at maturity. Some products have a hori
e. Product risk	Features can dramatically augment the payoff. Some products offer capi
f. Restrictions	Structured products are complex because they embed derivatives. Only
g. Product disclosure	Full disclosure of potential risks. Investors must also understand how re

7.1.2 Risks in Structured Products

Risk	Description
Market risk	Adverse movement in the underlying, its rate, its spread or its volatility
Credit and counterparty risk	Issuer, SPV, swap counterparty or reference entity failure
Liquidity risk	An illiquid bespoke structure, or a lock-up, forcing a discounted exit
Early termination risk	Unwinding before maturity incurs break costs; principal preservation may apply only a
Structure risk	The combination itself can produce losses the investor did not anticipate
Currency risk	The underlying and the settlement currency may differ

7.1.3 Trade-off between Features

Features are traded against one another. Raising one benefit is paid for elsewhere.

Feature change	Effect on headline return or yield	Why
Sell a call	Return rises	The investor has sold upside participation
Sell a put	Return rises	The investor has taken a conditional obligation to buy a falling a
Buy a call	Return falls	The investor is paying for upside participation
Buy a put	Return falls	The investor is paying for downside protection
Longer maturity	Return rises	More time means more optionality sold, and more issuer and rat
Higher leverage	Return rises, and so does loss	The exposure is multiplied in both directions

The strike's **proximity to spot** is the second lever. A strike close to spot makes the sold option more likely to finish in the money, so it is worth more premium — and more likely to cost the investor.

7.1.4 Participation Rate

A participation rate states how much of the underlying's upside the investor receives, once the cost of the principal component and the option has been accounted for.

$$\text{participation rate} = \frac{\text{discount sum}}{\text{call premium}}$$

Worked example. An S&P 500 call costs $24.00 * *. Twozero - couponbondsof face * *$**100.00**, both with **5 years** to maturity:

Parameter	Bond A	Bond B
Discount rate	5.00%	7.00%
Present value	**$78.35**	**$71.30**
Discount sum (face − PV)	**$21.65**	**$28.70**
Call premium	$24.00	$24.00
Participation rate	$21.65/24 = \mathbf{90.21}\%$	$28.70/24 = \mathbf{119.58}\%$

The higher-yielding Bond B frees more cash to buy the call, so it buys more participation — but the investor has taken a weaker credit to get it.

7.1.5 Purpose and Uses of OTC Structured Products

For investors, OTC structured products offer:

- **a. Customisation.** They meet needs standard investments cannot, and clients can negotiate criteria and trade-offs with the counterparty bank. They can also improve portfolio efficiency where a wide variety of instruments already exists.
- **b. Combining outcomes that cannot be bought separately.** An investor wanting upside participation with some principal preservation cannot achieve it by buying securities directly, because the principal is exposed. A bond plus a call option on the securities preserves capital while retaining the upside — which is exactly the structured product.
- **c. Optimising yield and return.** In low-rate periods investors seek higher yield. Nothing is free: with the promise of higher return come risks and trade-offs.
- **d. Access** to markets, underlyings or structures that are otherwise unavailable or uneconomic.
- **e. A hedge** with a payoff profile tailored to a specific exposure.
- **f. Currency transformation**, producing a return in a different currency.
- **g. A discount entry** into an underlying the investor is willing to own.
- **h. Leverage** where the investor accepts amplified loss.

7.1.6 Types of Structured Products

Capital preservation
FX and precious metals — currency deposit with call or put (± knock-out, no-touch/lock-out, call or put spread), kno
Equity
Fixed income
Funds, ETFs and others

7.2 Structured Deposits with Capital Preservation

A structured deposit is a bank deposit whose principal is repaid in full at maturity, with or without interest. It is governed by the Banking Act and, after the Deposit Insurance Act 2005 was replaced, by the **Deposit Insurance and Policy Owners' Protection Schemes Act 2011**. MAS has issued the **Guidelines on Structured Deposits (FAA-G09)**. Accredited and institutional investors may be exempt from parts of the regime.

Structured deposits are **not necessarily deposit-insured**, and early withdrawal may cost principal or return. The bank's credit standing matters.

From a risk perspective, structured deposits are generally **less risky than direct investment in the underlying**, because the bank must repay the full principal at maturity, or on early redemption where that feature exists. They are **riskier than ordinary deposits**, because the return depends on another instrument's performance; in some scenarios the investor gets back only the principal with no return.

7.2.1 Structured Deposits

Feature	Detail
Type of client	Risk-averse, low risk tolerance
Purpose	Seeking higher yield than a deposit with some capital preservation
Common underlyings	Foreign-currency investment such as USD/JPY or EUR/USD
Benefits	Slightly higher yield
Risks and losses	No capital loss if held to maturity; **loss of interest** (zero or lower rate is possible); counterpar
Obligations	To provide the specified amount of deposit for the structure
Structure types	Deposit with standard call or put; deposit with standard option and knock-out; structured dep

7.2.2 Deposit with Standard Call and Put

A deposit plus a sold option. USD/JPY spot **112.43**, 3 months, strike **113.50**, minimum yield **0.00%**, maximum **unlimited**, money-market rate **1.75%**, minimum **USD 250,000**.

If spot finishes at **115.00**, the payoff on the sold call, in JPY terms, is:

$$(115.00 - 113.50) \times 250{,}000 = \text{JPY } 375{,}000$$

At the prevailing rate that is about **USD 3,260.87**, or **1.3% a quarter — 5.2% annualised**. The maximum is described as unlimited because the sold call has no cap; the deposit's principal is preserved but the currency gain is not capped.

7.2.3 Standard Option with Knock-Out

Adding a knock-out at **109.50** with a minimum yield of **0.50%** reduces the cost of the option, so the headline yield can rise. The trade-off is that a touch of 109.50 extinguishes the enhancement and leaves the investor with the minimum.

A **step-down** variant shifts the whole payoff down by a stated amount, **2.00%** in the source example, once the barrier is touched. The lock-out or no-touch component is priced off the distance to the strike and the volatility.

7.2.4 Bull Call (or Put) Spread

USD/JPY spot **112.00**, 3 months, spread **114.00–118.00**, minimum **1.00%**, maximum **14.50%**, money-market rate **1.75%**, 100% participation. The implied option cost is about **0.75% p.a.**

The spread's theoretical maximum would be 15%; the product offers 14.5%, and the 0.5% difference is the spread capture retained by the bank.

7.2.5 One-Touch Lock-in

EUR/USD spot **1.18**, strike **1.12** (about 5% below), 3 months, maximum **12.00%**, minimum **0.75%**, money-market rate **1.75%**, minimum **$200,000**.

As long as **1.12 is reached at any time** during the three months, the 12% return is locked in. The client is **sacrificing 1%** against the money-market rate of 1.75% to buy the one-touch option. The notional capital is guaranteed, and the bank can provide the 12% because it holds or hedges the position as counterparty.

7.2.6 Range Currency Deposit (Double Barrier Binary Option)

A range currency deposit suits an investor who believes the pair will stay within a range. Inside the range the investor earns a better yield than the money market; a breach at either end lowers the yield.

Example. EUR/USD spot **1.09**, upper barrier **1.10**, lower barrier **1.06**, 3 months, maximum **7.25%**. The client earns 7.25% annualised as long as the pair stays within **1.06–1.10**.

The four levers that raise the yield are: **narrow** the range; **skew** it so spot sits nearer one barrier; **extend** maturity; or rely on **higher volatility**. Each makes the range more likely to be breached.

7.2.7 Range Accrual Currency Deposits

Rather than all-or-nothing, a range accrual pays the high rate on the **observations** inside the range and the low rate outside:

$$\text{payout} = P_1 \frac{n}{N} + P_2 \frac{N - n}{N}$$

Example. 88 of 90 days inside, with a 7.25% inside rate and a 2.00% outside rate:

$$\frac{88}{90} \times 7.25\% + \frac{2}{90} \times 2.00\% = 7.13\%$$

Accrual may be **weekly** or **monthly** rather than daily, which changes the payoff materially.

7.3 Yield-Enhancement Deposits

A yield-enhancement deposit gives up some or all capital protection to buy a higher coupon. The investor is selling an option, and the coupon is the premium.

Feature	Detail
Type of client	Higher risk tolerance, seeking yield
Purpose	Maximise yield, accepting conversion or principal risk
Common underlyings	A currency pair, with the alternate currency as the conversion asset
Benefits	A materially higher coupon than a deposit
Risks	Conversion into a weaker currency; loss of principal in the alternate currency; counterparty and
Obligations	To accept delivery of the alternate currency at the strike if the condition is met
Structure types	Standard DCI, knock-out DCI, knock-in DCI, accumulators, decumulators

7.3.1 Standard Dual Currency Investments

A DCI has existed in the local market for **20–25 years**. It consists of:

- **a.** a **short** position in a call on the reference currency, or a short position in a put on another currency; and
- **b.** a **long** position in a money-market deposit or bond denominated in the reference currency.

The option premium received lets the DCI offer a higher rate than the money market.

- **a.** If the exchange rate **strengthens or stays flat** and does not cross the strike at expiry, principal plus interest is repaid in the **reference currency**.
- **b.** If the exchange rate **weakens through the strike**, principal plus interest is repaid in the **alternate currency** — a conversion into a weaker currency. The risk is being forced to convert and then watching that currency fall significantly.

Worked example. The client holds USD as reference currency and believes GBP/USD will be flat or the USD will strengthen. Invests in a standard GBP/USD DCI: spot **1.37**, strike **1.34**, return **10.00% p.a.** against a money-market rate of **1.75%**.

- If GBP/USD $>$ **1.3400**, the client receives the initial investment plus 10.00% interest **in USD**.
- If GBP/USD $<$ **1.3400**, the client receives the initial investment plus interest **converted at 1.3400 into GBP**, and may incur an FX loss.

The headline 10% is roughly 8.25 percentage points above the money market. That gap is the option premium — the price of the conversion obligation.

7.3.2 Knock-Out DCI (Disappearing)

Strike **1.1000**, knock-out **1.1500**, spot **1.1200**, 3 months. The knock-out DCI pays **8.50%** against **9.00%** for the standard version, money-market rate **1.75%**.

The knock-out **disappears** the conversion feature if the barrier is touched, which protects the investor — so the yield is **lower** than the standard DCI. The further the barrier from spot, the less likely the protection is triggered, and the closer the yield moves to the standard version.

7.3.3 Knock-In DCI (Appearing)

Spot **1.1150**, EUR put / USD call strike **1.1000**, knock-in **1.0900**, 3 months, maximum **9.00%**, money-market rate **1.50%**.

The conversion feature only **appears** once 1.0900 is touched. Until then the investor has an ordinary deposit. Because the conversion risk is conditional, the structure can pay more than a standard DCI with the same strike — the investor is being paid for accepting a risk that may not materialise.

7.4 Structured Products

Every structure should be decomposed into a deposit or bond plus long or short option legs. Step-up structures alter the coupon over time, often with a callable feature. Bull and bear spreads cap the maximum payoff. A one-touch may lock in a rate or convert after a touch.

7.4.1 Accumulators

An accumulator buys the underlying on a regular schedule at a **discounted** price, and is typically **long in-the-money calls plus short out-of-the-money puts**. Below the strike the investor must keep buying as the price falls; above the strike but below the knock-out they buy at the discount; at or above the knock-out the note terminates.

Because the ITM calls cost more than the OTM puts, an **up-and-out knock-out barrier** is placed to reduce the cost of the calls — which caps the gain. For a **leveraged** accumulator more than one OTM put is sold, which lets the knock-out barrier be pushed further from the strike or the strike reduced further. Those barriers lower the cost but can work against the client.

The risks are high because the investor is **obliged to accumulate at prices above the prevailing market** as it falls. Step-up features multiply the quantity — commonly **2× to 5×** — once the price falls below a level, so the client accumulates more of a falling asset. Some banks offer **margin financing**, which magnifies gains and losses and makes the product riskier for both client and bank. Unwinding or offsetting is costly, and when credit concerns arise a bank may **lower its allowable lending value**, forcing a client to cover or sell at a substantial loss.

There is **no capital guarantee**. Gains are capped by the knock-out while losses continue below the strike.

7.4.2 Decumulators

A decumulator is the **reverse** of an accumulator. The investor is a **writer of call options** to the bank, taking on the obligation to **sell** a certain amount of the underlying on a regular basis at the contracted price or exchange rate.

Investors buy decumulators expecting a slightly **bearish** market. There is usually a knock-out level that terminates the obligation to sell once the price reaches it. For currencies, a decumulator is almost the mirror image of an accumulator, since every transaction involves a currency pair, and so decumulators are **less popular** than accumulators.

7.4.3 Currency Notes and Baskets

Currency notes and basket structures are covered with the other structured notes in Chapter 8.

7.5 Summary

Structured deposits may improve yield or customise currency and market exposure, but they can return less than a plain deposit, convert currency at an unfavourable level, be hard to exit, and expose the investor to bank credit risk. The client must understand the **worst case**, not only the headline yield.

Chapter 7 Appendix — Step-Down and Step-Up Digital Options (non-examinable)

Deposits with a Standard Option and a Step-Down Option

Strike **114.50**, knock-out or step-down level **109.00**, minimum **2.00%**. Once the barrier is touched, the whole payoff **shifts down by 2.00%** — so the investor keeps a positive return rather than falling to zero. The step-down is a partial protection: the investor retains something, but the headline is no longer available.

Step-Up One-Touch Digital Option

Rather than a single all-or-nothing payout, the payout steps with the level reached:

Level		Payout
EUR/USD	**1.04**	**14.00%**
1.04–1.06		**7.50%**
1.06–1.08		**3.50%**
$>$ 1.08		**1.00%**

With spot at **1.08** and a money-market rate of **1.75%**, the structure pays more the further the pair moves in the investor's favour, and never pays nothing. The stepped payoff is a ladder of digital options, each struck at a different level.

Exam cues

- **A structured deposit is a bank deposit**, but it is **not necessarily deposit-insured**, and early withdrawal may cost principal or return. Return depends on another instrument; the worst case may be principal back with **zero interest**.
- **Suitability has seven criteria:** risk appetite (ability **and** willingness), experience and knowledge, liquidity needs, horizon, product risk, restrictions (RNF-notified representatives), and disclosure.
- **Feature trade-offs:** selling a call or put raises yield; buying one lowers it; longer maturity raises yield; higher leverage raises both return and loss; a strike closer to spot makes the sold option dearer and more likely to cost the investor.
- **Participation rate = discount sum / call premium.** Bond A: 21.65/24 = **90.21%**; Bond B: 28.70/24 = **119.58%**.
- **DCI is a short call on the reference currency plus a long money-market deposit.** The premium is the yield uplift above the money-market rate — the 10.00% against 1.75% example.
- **Knock-out DCI pays less than the standard** (8.50% vs 9.00%) because the knock-out protects the investor; **knock-in DCI** can pay more because the conversion risk is conditional.
- **Range accrual:** payout = inside rate × days inside/total + outside rate × days outside/total. 88 of 90 days at 7.25%/2.00% gives **7.13%**.
- **Accumulator:** 1× above the strike to the knock-out, and multiplied — commonly **2× to 5×** — below the strike. Gains are capped by the knock-out while losses continue. A decumulator is the mirror, sold as a call writer.
- **Trap:** the discounted strike is compensation for an obligation, not a bargain. Below the strike the investor must keep buying a falling asset, and with step-up gearing must buy **more** of it.

Chapter 8 — Structured Notes

A structured note is debt plus a derivative. The investor's claim is on the issuer or SPV, not on the underlying. This chapter covers the note families the examination tests — range accruals, inverse floaters, ELNs and CLNs — and how to find the real exposure inside each.

Learning Objectives

The candidate should be able to:

- Define a structured note and identify the issuer or SPV claim.
- Explain the capital-preservation spectrum across note types.
- Compute a range accrual payout and an inverse floater coupon.
- Explain why an ELN is economically a short put, and compute the breakeven.
- Explain worst-of structures and correlation risk.
- Explain credit-linked notes, first-to-default baskets and bond-linked notes.
- Distinguish ETNs, index-linked notes, participatory notes and access notes.

8.1 What is a Structured Note?

A structured note is a **debt or debenture obligation** whose return is linked to an underlying — equity, index, currency, commodity, rate, credit reference or basket. The investor owns a claim on the **issuer** or an **SPV**, not the underlying asset.

Under Singapore's offering rules a prospectus may be required unless an exemption applies — for example an offer to accredited or institutional investors, or a minimum investment of **S$200,000**. The accredited-investor thresholds are:

Investor	Threshold
Individual — net personal assets	more than **S$2 million** (primary residence net of mortgage, up to a cap)
Individual — financial assets	more than **S$1 million**
Individual — income	at least **S$300,000** in the last 12 months

Investor	Threshold
Corporation	net assets more than **S$10 million**

Read the prospectus, the **Product Highlights Sheet** and the term sheet. The note may be distributed directly by a bank or issued through an SPV, which may be off balance sheet and may give only the legal and asset claim described in its documents.

8.2 Issuing Structured Notes

The principal component can be a zero-coupon bond, a corporate or AAA bond, or a deposit-like instrument. The return component can be a long or short option, a digital, a range, a swap, a CDS or another derivative.

Issuance, hedging, funding, collateral, valuation, fees, tax, maturity mismatch and issuer or SPV recourse together determine the investor's actual risk. A note of five years' maturity funded by a shorter-dated hedge is exposed to roll and funding risk even if the reference asset behaves as expected.

8.3 Types of Structured Notes

Capital preservation	Yield without capital preservation	Capital gain / participation
Equity — capital-protected notes	Bull and bear ELNs, worst-of ELNs, accumulators	Participation notes, twin-wi
Bonds and credit	CLNs, BLNs, first-to-default baskets	—
Interest rate	Range accrual notes, inverse floaters, step-up callables	—
FX	Dual currency notes, range currency notes	—

8.4 Yield Enhancement and Some Capital Preservation

Feature	Detail
Type of client	Moderate risk tolerance, seeking yield with nominal protection
Purpose	Higher yield than a deposit, with nominal capital preserved
Common underlyings	An interest-rate benchmark or an index
Benefits	Enhanced coupon
Risks	The coupon can be zero; the note's value can fall with rates and volatility during its life; **no n**
Obligations	To provide the specified amount
Structure types	Various range accrual notes

8.4.1 Range Accrual Notes

A **RAN** pays a target return if a reference index stays within an agreed range, and less or nothing if it does not. The **principal is not affected**.

The reference can be a stock index such as the DJIA, or an interest-rate benchmark such as **LIBOR, SIBOR or the Singapore Swap Offer Rate (SOR)**. Where the reference is 3-month LIBOR, interest accrues at the higher rate only on days the rate trades inside the range; outside it, no interest accrues.

Interest is accrued and calculated **daily** for days inside the range. The payout therefore depends on observed movements of the index.

$$\text{Payout}_{\text{inside}} = P_1 \times \frac{d}{N} \qquad \text{Payout}_{\text{outside}} = P_2 \times \frac{N-d}{N} \ \text{(or zero)}$$

where

$$N$$

is the total number of observations in the period,

$$d$$

the number of observations inside the range,

$$P_1$$

the payout inside and

$$P_2$$

the payout outside.

The observation period may be daily, weekly or monthly, though **daily** is usual, and it is more common for the payout outside the range to be **zero**.

SOR-linked worked example. SGD 100,000 invested for one year at **5% p.a. inside a 1%–1.5% SOR range and 0% outside**, semi-annual, ACT/360, fixing on 22 June and 22 December 2013.

- First half: **110 of 182** days inside →

$$5\% \times \frac{110}{182} = 3.02\%$$

- Second half: **160 of 183** days inside →

$$5\% \times \frac{160}{183} = 4.37\%$$

The annualised return over the year is about **3.68%**, not 5%. The gap is the whole point of the structure: the headline rate is conditional on the path.

8.4.2 Callable Daily RAN (VMRAN)

A callable daily range accrual note lets the issuer **call** the note after a **lock-out period**, on designated coupon dates. Calling creates **reinvestment risk**: the investor loses the attractive coupon and may only be able to reinvest at lower rates.

8.4.3 Inverse Floaters

The coupon moves **inversely** to a reference rate, so the note performs as rates fall:

$$\text{coupon} = \max\left[X\% - \text{leverage} \times \text{floating-rate index},\ \text{minimum coupon}\right]$$

A floored version stops the coupon falling below a stated minimum, at the cost of a lower maximum elsewhere. Leverage makes the rate sensitivity nonlinear: at leverage 2, a 1% rise in the index cuts the coupon by 2%.

8.4.4 Callable Step-Up Notes

Coupons step up on a schedule, but the issuer may **call**. A typical issue might be an **American bank, BBB+, USD 20 million**, traded **30 May 2018**, minimum **USD 25,000** against a **USD 1,000** denomination, maturing **23 May 2024** over six years, with semi-annual coupons stepping **3.00%, 4.00%, 5.00% and 6.00%**.

The investor bears **issuer call and reinvestment risk**, rates risk, and the risk that the step-up is never received because the note is called first.

8.5 Yield Enhancement without Capital Preservation

8.5.1 Equity Linked Notes

An **ELN** is generally a funded bond or deposit plus a **short put** (bull ELN) or **short call** (bear ELN), and may embed a strangle or a worst-of structure. A bull ELN pays a higher yield if the equity stays above the strike; below it, the investor may receive shares or cash at the strike and bear the downside, with principal falling to zero if the underlying collapses.

Six key terms: underlying and initial reference spot; strike and conversion ratio; **knock-in level** where applicable; **cash or physical settlement**; observation dates; and the **fixing date, normally two business days from maturity**. The examination focuses on bull ELNs.

8.5.2 Bull ELNs

Apple worked example. Spot **USD 188.59** on 11 May 2018; strike **USD 174.52**, which is **92.54%** of spot; six-month put premium **USD 6**; minimum **USD 1 million**; money-market rate **2.51688%**; six months to **16 November 2018**; valuation 14 November.

Conversion shares:

$$\frac{1{,}000{,}000}{174.52} = 5{,}730 \text{ shares}$$

Premium received is **USD 34,380**, so the deposit of USD 1,034,380 grows to **USD 1,047,309.75**. The six-month return is **4.730%**, or **9.68% p.a.** Issue price is **95.49%**, that is **USD 954,900**.

Scenario	Spot at valuation	Outcome	Return
1	**+12%** to USD 211.22	Above the strike, so cash settlement	**9.68% p.a.**
2	**−5%** to USD 179.16	Still above the strike, cash settlement	**9.68% p.a.**
3	**−10%** to USD 169.73	Below the strike, put exercised, shares delivered	A **−2.74%** book loss, or **USD 27,**

Scenario 3 is the essence of the product: the investor was paid 9.68% p.a. to accept the risk of owning Apple 10% lower. The note is economically a **short put**, so the maximum loss is nearly the whole investment if the stock falls to zero.

8.5.3 Worst-of ELNs

A worst-of ELN pays on the **weakest** underlying. Two examples show the mechanics.

Worst-of ELN #1. Three stocks A, B and C with initial spots **13.34, 12.43 and 3.12**; current spots **13.26, 12.21 and 3.09**; strikes **14.00, 13.05 and 3.28**; knock-in levels **11.87, 10.92 and 2.76**. Issue date **15 May 2018**, valuations **15 June, 16 July and 14 August 2018**, investment **SGD 200,000**, coupon **11.35% p.a.**

The critical insight is that the **worst performer need not be the knock-in toucher**. Conversion occurs at the strike, giving:

$$\frac{200{,}000}{13.26} = 15{,}082 \text{ shares}$$

plus a cash residual.

Worst-of ELN #2 (variable cumulative coupon). USD 200,000, two years, the call price is **100% of the trade-date close** and the **conversion price is 90%**. Stock A is **30.00 / 27.00**, Stock B **55.00 / 49.50**. Calls are semi-annual, and the coupon steps **8%, 16%, 24%, 32%** if the laggard is at or above its call price.

Scenario 1 at the first call date: Stock A at **29.50** (−1.67%) is the laggard and below its call price, so the note **pays zero** and continues. Scenario 2 at the second call date: Stock A **32.00** (+6.67%) and Stock B **56.50** (+2.73%); Stock B is the laggard but above its call price, so the note is **called at 100%** and the client receives **USD 232,000** — the **16%** return. Scenario 3 at the final date: Stock A **32.00**, Stock B **65.00**; Stock A is the laggard but above its call price, so principal plus the **cumulative 32%** coupon is paid — **USD 264,000**.

Scenario 4 is the loss case. The note is not called, and the laggard closes **below its call price but above its conversion price**. A variable coupon is calculated from the laggard's spot over its conversion price. Stock A at **29.16** against a conversion price of **27.00**:

$$\left(\frac{29.16}{27.00} - 1\right) \times 100 = 8.00\%$$

so the client receives **USD 216,000**, being principal plus an 8% coupon. Scenario 5 — Stock A at **26.00**, below the conversion price — delivers shares instead:

$$\frac{200{,}000}{27.00} = 7{,}407 \text{ shares}$$

worth USD 192,593 at 26.00, a loss of **USD 7,407** or **−3.70%**, with **no coupons** received.

8.5.4 Risks of ELNs

The investor is short a put on the underlying, so bears the full downside of the weakest reference; receives no dividend; bears issuer and SPV credit risk; and faces liquidity risk because the note is bespoke. Worst-of structures add **correlation risk** — several names deteriorating together — and the coupon forgone while uncalled.

8.6 Credit Linked Notes

A **CLN** investor effectively **sells credit protection** through a CDS on a reference entity or basket, and receives an enhanced coupon in return.

On a defined **credit event** the payout may be reduced, securities delivered or principal lost. The events are:

- bankruptcy;
- failure to pay;
- restructuring;
- obligation acceleration;
- obligation default;
- repudiation or moratorium; and
- where the definitions provide for it, government intervention.

The investor bears **both** the issuer or SPV credit risk **and** the reference-entity credit risk. A typical structure is **five years at 4.00% p.a.** on a reference entity, with the issuing bank rated **single-A**.

The 2014 Credit Derivatives Definitions reformed the event definitions and the settlement mechanics. A CLN can be cash settled — paying par less the recovery value — or physically settled by delivering a qualifying obligation.

8.6.1 First-to-Default CLN

A **first-to-default** basket pays out on the **first** reference entity to default. That amplifies both **correlation** and **concentration** risk: with highly correlated names, one systemic shock can trigger several at once, and the protection the investor has sold is triggered by the weakest link. The coupon is higher than for a single-name CLN precisely because the probability of *some* name defaulting is higher than the probability of any *particular* name defaulting.

8.6.2 Risks of CLNs

Two distinct credit risks: the **issuer or collateral** risk — the note is only as good as the issuer's ability to pay, and any collateral must actually be available and enforceable — and the **reference-entity** risk, which is the credit protection the investor has sold.

8.6.3 Bond-Linked Notes

A **BLN** embeds a **short put on a bond**. The investor receives an enhanced coupon and, if the bond's price falls below the strike, takes delivery of the bond. A BLN differs from a CLN in four ways: the trigger is a **price** fall rather than a **credit event**; the deliverable is the bond itself; there is no ISDA credit-event determination; and the credit exposure is to the bond issuer's market price rather than to a defined default.

8.6.4 Risks of BLNs

Issuer risk plus bond price risk, driven by rates, credit spread and liquidity. A BLN on a long-dated bond carries substantial duration risk — a rise in yields can move the bond price far more than the coupon the investor received.

8.7 Market Access

8.7.1 Exchange-Traded Notes

An **ETN** is **unsecured listed debt** tracking an index. The investor has no ordinary ownership, no guaranteed coupon, and **central issuer credit risk** — the instrument is a promise by the issuer, not a pool of assets. An ETN therefore differs from an ETF in four ways: no segregated assets; no creation and redemption against a portfolio; no custodial protection; and credit exposure to the issuer.

8.7.2 Index-Linked Notes

An index-linked note links participation, a minimum and possibly a cap to an index, normally subject to the issuer's claim.

Worked example. Issuer ABC Structured Products, rating **A**; STI at **3,278** on 18 May 2014; denomination **SGD 1,000**; investment **SGD 1 million**; maturity **15 May 2022**, eight years. Redemption is the **greater** of:

- **100% + 4%**, or
- **85% + average performance**,

where the average is the arithmetic mean of the anniversary closes.

Scenario	Average performance	Redemption
1	**+5%**	85% + 5% = 90%, less than 104%, so the investor receives the **104%** floor — **SGD 1,**
2	**−5%**	85% − 5% = 80%, below the floor, so the investor again receives **104%** — **SGD 1,040**

The structure protects the downside at a cost: eight years of capital for a total 4% return. The investor has sold the issuer a long-dated call and received a floor in exchange.

8.7.3 Participatory Notes

A **participatory note (P-note)** gives exposure to a market that a foreign investor cannot access directly — Indian equities are the classic case — through a note issued by a registered intermediary rather than by acquiring the shares. The investor takes the intermediary's credit risk and has no shareholder rights.

8.7.4 Access Notes

An **access note** provides exposure to a market, asset or strategy through a note rather than direct ownership. The terms and the issuer's recourse control the outcome.

8.8 Investors and Suitability

8.8.1 Suitability for Retail

Assess liquidity and lock-up, risk and return, capital-protection conditions, the underlying scenario, term, fees and charges, currency, issuer or SPV, collateral, secondary market, early termination, tax and concentration. Explain that a **high coupon is compensation for selling a put, a call or credit protection** — not a free enhancement.

8.8.2 Sales-Process Factors

Five factors determine whether the sale is sound: **liquidity** (can the client exit, and at what cost); **risk and return** (including the worst case); **terms** (the payoff formula, the triggers and the settlement); **fees and charges** (including any embedded in the issue price); and **documentation** (prospectus, PHS and term sheet, with the acknowledgement retained).

8.9 Summary

Structured notes are debt plus derivatives. Ask whether principal is actually protected, **when, in which currency and by whom**. Callable, range, inverse-floater, ELN and CLN products shift market, rate, equity or credit risk into the investor's return; issuer or SPV credit and liquidity are as important as the reference asset.

Chapter 8 Appendix — Spread CDRANs, RAN Variations and PHS Content (non-examinable)

Spread CDRANs

A spread range accrual pays on the **differential** between two points on a curve — for example the **30-year against the 10-year**. It pays while the curve is positive and fails on inversion. The investor is expressing a view on curve shape, not on the level of rates.

Variations of RANs

- A **daily range accrual equity-linked note** observes an equity index rather than a rate. It typically carries **unprotected principal**, because the equity index can move far outside a range that a rate index would rarely breach.
- A **basket accrual** requires **all** constituents to be inside their ranges, which materially lowers the probability of accrual and therefore raises the potential coupon.

Documentation

A prospectus and PHS are prepared for each issue, though the requirement is **exempt** for offers to institutional or accredited investors.

PHS Contents

A PHS should be at most a few pages, set in no smaller than **10-point** type, and must cover seven mandatory sections: the **product summary**; **suitability**; **key features**; **key risks** — business, legal, market, credit and liquidity; a **prominent warning** of possible loss of all principal, in bold or italic; **definitions**; and **contact information**. The framework applies to offers of debt securities, hybrid instruments and equity securities.

Exam cues

- **A note is a claim on the issuer or SPV**, not on the underlying. An SPV may be off balance sheet with no parent guarantee. Prospectus exemption threshold: $\mathbf{S}200,000 * *; accreditedindividual : ** > S\mathbf{2m}$ net personal assets, $>\mathbf{S}1m * *financialassets, or * *S\mathbf{300{,}000}$ income; corporation $>\mathbf{S\$10m}$ net assets.
- **RAN:** payout =
$$P_1 d/N + P_2(N-d)/N$$
, usually with
$$P_2 = 0$$
. The SOR example: 110/182 at 5% plus 160/183 at 5% gives about **3.68%**, not 5%. The principal is unaffected; the **coupon** is conditional.
- **Inverse floater:** max[X% − leverage × index, floor]. At leverage 2, a 1% index rise cuts the coupon **2%**.
- **A bull ELN is a short put.** Apple: strike 174.52 (92.54%), 5,730 shares, premium USD 34,380, **9.68% p.a.** if above the strike, **−2.74%** if 10% below.
- **Worst-of:** the **worst performer need not be the knock-in toucher**. Variable coupon = (spot/conversion price − 1). Stock A at 29.16 against 27.00 gives **8.00%**.
- **CLN:** the investor **sells credit protection** and bears issuer risk **and** reference risk. Seven credit events. **First-to-default** pays on the first name, amplifying correlation and concentration.
- **BLN versus CLN:** a **price** trigger and bond delivery, not a credit event and ISDA determination.
- **ETN versus ETF:** the ETN is **unsecured debt** with issuer credit risk and no segregated assets.
- **Index-linked note:** redemption is the greater of 100% + 4% or 85% + average performance — the floor caps the upside at a modest total return.
- **Trap:** a high coupon is never free. It is the premium for a sold put, a sold call or sold credit protection, and the investor's maximum loss on a worst-of or an ELN is close to the whole investment.

Chapter 9 — Structured Funds

A structured fund is a collective investment scheme whose payoff is produced by assets, a formula, or a swap. Unlike a note, it publishes a NAV — and unlike a deposit, its capital is not a bank obligation. This chapter

covers fund mechanics, CPPI, the prospectus, and ETFs.

Learning Objectives

The candidate should be able to:

- Define a structured fund and distinguish it from a traditional fund, a tracker and a note.
- Explain total return swap and zero-plus structures.
- Compute the CPPI cushion, multiplier and risky allocation, and explain cash-lock.
- Explain the roles of manager, trustee and custodian, and their reporting clocks.
- Identify the contents of a fund prospectus and PHS.
- Distinguish physical from synthetic ETF replication and their risks.
- Explain tracking error, NAV premium and discount, and TER.

9.1 Structured Funds

A structured fund combines securities and/or derivatives to produce a specified risk-return profile, capital-preservation objective, participation formula, yield or cost structure. It is a **collective investment scheme**, governed by the applicable Code on Collective Investment Schemes, the offering documents, and licensing and custody requirements.

Capital preservation normally applies **only at maturity** and is subject to the fund, the issuer and the strategy performing. It is **not** deposit insurance and not a guaranteed NAV. A structured fund must **publish its NAV**, which distinguishes it from a note or a deposit.

Key governance points are a **licensed manager**, a **fiduciary trustee or custodian**, NAV calculation and reporting, valuation and pricing policies, disclosure of fees and risks, audit, and investor dealing terms. Conflicts arise where the manager, issuer, swap counterparty, custodian or distributor are affiliated.

9.1.1 Common Terminology

Term	Meaning
Asset allocation	The division of a portfolio among asset classes
AUM	Assets under management
Closed-end fund	A fund with a fixed number of units, not continuously issuing or redeeming
Collective investment scheme (CIS)	A pooled arrangement in which investors contribute to a common portfolio
CPPI	Constant proportion portfolio insurance — a formulaic floor mechanism
Cushion	Portfolio value less the floor value. With SGD 100 of assets and an SGD 80 fl
Custodian	Holds the fund's assets, segregated from the manager's
Drawdown	The fall from a previous peak
Excess return	Return above the benchmark
Index fund	A passively managed fund tracking a benchmark index
ISIN	The international securities identification number
Management fee	The annual fee for managing the fund
Minimum subscription amount	The smallest initial investment accepted
Multiplier	In CPPI, the factor applied to the cushion to derive the risky allocation
NAV	Net asset value — assets less liabilities, divided by units outstanding
Open-end fund	A fund that issues and redeems units continuously at NAV
Share class	A division of a fund with different fees, currency or distribution terms
Sharpe ratio	(Return − risk-free rate) ÷ standard deviation of return
SICAV	A European open-end investment company with variable capital; often an um
TER	Total expense ratio — annual costs as a percentage of NAV
Total return	Price change plus income, reinvested
Total return swap (TRS)	A swap exchanging the total return of a reference for a financing leg
Tracker	An index fund; passively managed to track a benchmark
Trailer fee	Paid by the asset manager to a distributor in lieu of ongoing advice
Upfront sales or subscription charge	A charge on subscription, reverted to the distributor
UCITS	Undertakings for the Collective Investment in Transferable Securities
Umbrella fund	One legal entity with several distinct sub-funds, traded individually
Underlying asset	What a direct-policy fund tracks, or what an indirect-policy fund's return is l
Underlying securities	The transferable securities constituting the underlying asset

Term	Meaning
Value at risk (VaR)	A statistical measure of risk over a period, in three variables: the potential los
Volatility	The dispersion of returns, measured by standard deviation or variance
Zero-coupon bond	Pays no interest during its tenor; sold at a discount and redeemed at face valu

UCITS was first adopted in **1986 (UCITS I)** and applies to funds marketed to retail investors. It brought costs down because providers no longer needed a new vehicle for each country. A UCITS-compliant fund, registered in its home state, can be **passported** to the public across the **30 countries of the European Economic Area**. UCITS funds are marketed globally and have succeeded in Asia and Latin America. In **July 2012** the European Commission published the draft **UCITS VI** directive.

9.1.2 Compared with Traditional Funds

Traditional active funds select assets to outperform a benchmark, using manager judgement on securities, sectors, timing and allocation. A structured fund instead fixes the payoff mechanism in advance — a formula, a floor or a participation rate — so the outcome is determined by the structure rather than by discretionary selection.

9.1.3 Compared with Trackers

Trackers are **passively managed** index funds tracking a chosen benchmark. A structured fund may use an index as its underlying, but its payoff is transformed by the formula or the derivative; it does not simply replicate the index. Index investing is a form of passive investing; a structured fund is neither wholly passive nor discretionary.

9.1.4 Components

A structured fund has a **principal component** and a **return component**:

- the principal component may be a bond, a zero-coupon bond, a deposit or a money-market instrument;
- the return component may be a call, a put, a range, a digital or a swap.

A typical structure might place **80%** of assets in a principal component designed to reach 100% at maturity, and **20%** in an S&P 500 call.

9.1.5 Structured Funds Using Derivatives

Total return swap funds. A TRS fund receives the total return of a reference asset or index and pays financing to a swap counterparty. The financing leg might be **SIBOR + 30bp**, reset over six months, for the five-year life. It provides efficient exposure but adds **counterparty, collateral, reset, liquidity and termination risk** — the fund's return depends on the swap counterparty performing.

Zero-plus structures. A zero-plus fund allocates to a zero-coupon component that is projected to reach 100 at maturity, and to a call option for the upside.

Worked example. Ten-year fund. **80%** is placed in a zero-coupon bond worth 80% today and projected to reach 100% in ten years — this is a projection, not a guarantee. **20%** buys a ten-year STI call. If the call costs **25%** of notional, the participation rate is:

$$\frac{20}{25} = \mathbf{80}\%$$

The floor is protected only if the zero-coupon component performs and the issuer pays.

9.1.6 CPPI

Constant proportion portfolio insurance allocates dynamically between a risky and a risk-free asset. It typically offers a guarantee of **80–100%** of principal; principal insurance bought from a bank costs roughly **1% p.a.**

9.1.6.1 How CPPI Works

$$\text{cushion} = \text{portfolio value} - \text{floor value}$$

$$\text{multiplier} = \frac{1}{\text{crash size}}$$

$$\text{risky allocation} = \text{multiplier} \times \text{cushion}$$

$$\text{risk-free allocation} = \text{portfolio value} - \text{risky allocation}$$

With a **15%** floor and a **20%** crash size, the multiplier is **5** and the risky allocation is

$$5 \times 15\% = \mathbf{75}\%$$

.

CPPI fund example. A USD 100,000 fund, five years, **100% guaranteed**, guarantee fee **1%**, management fee **1.5%**, index allocation between **0% and 120%**.

Path	Year 1	Year 2	Year 3	Year 4	Year 5	Outcome
Good	100	110	105	120	125	135
Weak	100	95	90	98	100	95 → **100 guaranteed**

On the weak path the fund falls to 90, the cushion approaches zero, the manager sells the risky assets and the fund finishes at the guaranteed 100 rather than at 95. That is the mechanism working — and also the reason the investor misses any recovery.

9.1.6.2 CPPI Risks Three assumptions underpin CPPI: the risky asset **appreciates**, has **low volatility**, and has **limited drawdown**. Each can fail.

- **Buy high, sell low.** The formula buys after rises and sells after falls, so a range-bound market erodes the fund.
- **Cash lock.** A sharp fall forces the fund entirely into the risk-free asset, after which it misses any recovery.
- **Gap risk.** A fall larger than the assumed crash size breaches the floor before the manager can de-risk.
- **High fees.** Guarantee and management fees reduce the cushion from the outset.

9.1.7 DPPI

Dynamic proportion portfolio insurance varies the multiplier rather than holding it constant. A common configuration uses a multiplier band of **3–5×**, reset to **4×**. It responds to market conditions but shares CPPI's path, fee, gap, liquidity and manager risks.

9.1.8 Fund Structures

9.1.8.1 Capital-Preservation Funds A capital-preservation fund guarantees a stated proportion of principal at maturity, typically by combining a zero-coupon bond with a call.

9.1.8.2 Formula Funds A **formula fund** allocates according to a pre-determined formula based on an anticipated view, with the final payout depending on the realised market outcome. The advantage is that the investor knows from the start how the investment will be allocated. Such funds can be set up to track indices.

9.1.8.3 Capitalised and Distribution Funds **Capitalised** funds — also called accumulating or reinvesting — automatically reinvest dividends into the fund. **Distributing** funds — also called income funds — pay dividends received to investors periodically.

9.1.8.4 Indirect Investment Policy Funds (Swap-Based) These give a return linked to the performance of an underlying without investing directly or fully in it. The fund may invest part or all of the net proceeds in one or more derivatives under the **Investment Restrictions** in the prospectus. The investor's return depends on the underlying's performance **and** on the performance of the derivative used to link the proceeds.

The fund may instead hold a **hedging asset** and exchange all or part of its performance or income for a performance linked to the underlying, using derivatives within the prospectus limits. The underlying may follow a **passive** strategy — typically a financial index or a rules-based strategy — or an **active** strategy in which the real or notional basket is actively managed within the Investment Restrictions.

9.1.9 Examples of Structured Funds

Most structured funds use equities as the underlying, with others on commodities or hybrids. A common pattern is a **fixed coupon for an initial period**, then variable coupons or payouts linked to the underlying. A high fixed coupon forces the **participation ratio lower**, because the fixed payments consume the available premium.

Formula fund on a commodity index. The ABC Fund is an agriculture-related open-end USD fund tracking an agricultural index with diversified sector exposure, giving access to **seven liquid commodities: corn, wheat, soybean, sugar, cotton, coffee and cocoa**. Each subcomponent's weight is fixed and **reset annually**. The fund requires a low initial investment and offers **daily liquidity at NAV**.

The index uses an **optimal yield** rolling mechanism to generate alpha. Periodic rolling of expiring futures at a fixed term, the usual approach, can be detrimental to return and can even trigger losses. Optimal yield instead **maximises roll return in backwardation and minimises roll loss in contango**.

DEF Fund. An SGD fund of **25 stocks** across infrastructure, utilities and real estate, paying a quarterly dividend, selling calls and buying puts as a stop-loss. It illustrates a covered-call overlay combined with a protective-put overlay inside a fund structure.

9.1.10 Risks of Structured Funds

Market, credit and counterparty (issuer, swap counterparty or custodian), liquidity, early termination, structure, correlation, currency and concentration risk all apply. A structured fund adds **governance and valuation risk** — NAV calculation, pricing policy and conflicts — that a direct portfolio does not carry in the same way.

9.1.11 Roles in a Structured Fund

9.1.11.1 Fund Manager The manager implements the mandate, manages the portfolio and the day-to-day trading, and produces semi-annual and annual accounts and reports. It also handles unit creation and redemption.

9.1.11.2 Fund Trustee The trustee holds the assets on trust and oversees the manager. The reporting clocks are strict: the **semi-annual report within two months** and the **annual report within three months**, with **MAS notified within three business days** of a breach.

9.1.11.3 Conflicts of Interest Conflicts arise where the manager, trustee, custodian, swap counterparty or distributor are affiliated. They are managed through **information barriers and separate reporting lines**, and disclosed.

9.1.12 Managing Underlying Risk Exposure

Controls are operational, financial and legal. Risk is measured and limited through stress testing, VaR and duration, and the manager's job is to ensure risk does not exceed the level at which the fund can absorb the losses of a probable worst outcome.

9.1.13 Fund Documentation

9.1.13.1 Fund Prospectus The prospectus must disclose, at minimum:

- **Basic information:** registration and expiry date; place of constitution; the trust deed and any supplemental deeds; semi-annual reports; disclaimers; whether the scheme is constituted outside Singapore.

- **The manager:** name, address and track record.
- **The representative**, and **the trustee**.
- **Other parties:** investment adviser, registrar and where the register may be inspected, and auditor.
- **Structure:** standalone or umbrella; details for a fund of funds, feeder fund or sub-managed fund.
- **Investment objectives**, focus and approach.
- **Facts of inclusion:** fees and charges, and a **fee table**.
- **Risks:** general and specific.
- **Subscription of units:** procedure; minimum subscription; initial purchase price and offer period; dealing deadline and pricing basis; a **numerical example** under the pricing system adopted; confirmation of purchase; minimum fund size; return of contributions; and classes of units.
- **Regular Savings Plan:** its details, and a statement that the investor may cease participation.
- **Realisation of units:** procedure; minimum holding and minimum realisation amounts; dealing deadline and pricing basis; a numerical example.
- **Performance:** past performance, comparison against a CIS, an index or other investments, the manager's record, future prospects, the benchmark, the **expense ratio** and the **turnover ratio**.
- **Soft dollars, conflicts, reports, specialised CIS warnings**, and the **complaints** procedure.

9.1.13.2 Product Highlights Sheet The PHS summarises the prospectus in a few pages: product summary, suitability, key features, key risks, fees and charges, definitions and contacts. It is a required companion, not a substitute.

9.1.13.3 Trust Deed The trust deed establishes the fund, the trustee's powers and duties, and the investor's rights. It is the constitutional document, and the prospectus must be read with it.

9.1.14 Investor Types

A **restricted scheme** may be offered only to accredited or relevant persons, or at **SGD 200,000** per transaction. A retail offer requires a **lodged and registered prospectus**. Institutional investors do not require an information memorandum.

9.1.15 After-Sales Information

9.1.15.1 Where to Get NAV The **administrative agent**; newspapers; and Bloomberg, Reuters or the issuer's website via the **ISIN**.

9.1.15.2 Performance Information

- **Semi-annual and annual statements:** the statement of net assets; the statement of changes in net assets; the **statement of investments**, listing the portfolio, with the same for each fund the company issues; and the **changes in the number of shares for each share class**, reconciling opening and closing units.
- **Investment manager report:** performance of the underlying and of each share class, AUM, annualised volatility for different share classes, and a performance outlook.
- **Factsheet:** launch date, manager information, key features, asset allocation, performance and applicable fees.
- **Monthly performance report:** the principal terms — issuer, launch date, management company, initial NAV, management fee per annum; an overview of the investment policy and portfolio methodology; and performance — **YTD, 1-month, 6-month, 1-year, 3-year and since-launch** returns, with volatility, Sharpe ratio and VaR.

9.2 Structured Exchange Traded Funds

9.2.1 What are Exchange Traded Funds?

An **ETF** is an **open-end investment fund** that aims to replicate the performance of a published market index. It is **listed and traded on a stock exchange**, and investors buy or sell ETF units to participate in the index. Indices may be by country, region or sector, and may be emerging or developed.

Investors use ETFs to gain exposure to all constituents, diversifying without purchasing each asset. Many ETFs now do more than track stock prices: some track volatility, some track only **daily** price movement, and some are **leveraged to 2–3×** an index. For some ETFs, performance depends more on the options and futures market than on the index itself.

9.2.2 The Global ETF Market

The global ETF market reached roughly **USD 7.7 trillion** in AUM in **2020**, with the **United States** accounting for about **70%**.

9.2.3 Types of ETFs

Equity; fixed income; cash; currency; commodity; index; alternative; leveraged; inverse; and protective ETFs.

9.2.4 Replication Methodologies

Approach	Method
Physical — full replication	Hold **all** assets underlying the index in the **exact proportion**. Returns after
Physical — representative sampling	Hold a **subset** of constituents, usually the more dominant, to reduce the numl
Synthetic — derivative embedded	Use derivatives and/or OTC transactions to replicate the index without holdi
Synthetic — swap based	Exchange the fund's return for the index return through a swap, with **collate**

Synthetic structures are subject to a **10% net counterparty exposure** limit, with collateral of at least **90% of NAV** held at a third-party custodian owned by the trustee. Swap-based funds may be:

- **unfunded** — the ETF buys the collateral itself;
- **funded** — the counterparty deposits the collateral;
- **combined** — a mix of both.

Collateral is rebalanced **daily** to keep it at or above 90% of NAV.

9.2.5 ETF Pricing and Valuation

- **Bid-ask spread** is the immediate cost of trading.
- **Creation and redemption** by authorised participants keeps the price near NAV; it can fail in stressed markets.
- **NAV premium or discount** arises when the traded price diverges from NAV.
- **TER** is expressed as a percentage or in basis points and **accrues daily**.
- **Tracking error** is the divergence of fund return from index return. Six causes:
  - **a.** transaction costs;
  - **b.** portfolio holdings differing from the index;
  - **c.** the replication method;
  - **d.** futures roll, which costs in contango and earns in backwardation;
  - **e.** cash drag; and
  - **f.** hedging costs.

Tracking error is generally **lower for synthetic** than physical replication, and **higher for emerging markets**.

9.2.6 ETFs versus ETNs

An ETN is unsecured debt, with issuer credit risk and no segregated assets. **Exchange-traded commodities (ETCs)** are commodity-backed ETNs. A **VIX ETN** rolls daily futures and comes in different tenors, so it diverges from the index over time and can be shut down.

9.2.7 ETFs versus Mutual Funds

An ETF has **two liquidity sources** — primary creation and redemption, and secondary exchange trading. Mutual funds deal at NAV once a day. There is no difference between physical and swap-based ETFs in **secondary** trading; the difference is in what the fund holds and who bears the counterparty risk.

9.2.8 ETF and Fund Governance in the US and Europe

The **first US ETF** launched in **1993**, tracking the S&P 500 physically. US funds are governed by the **Investment Company Act of 1940**, under **SEC** jurisdiction. In Europe, **UCITS III from 2001** enabled European synthetic ETFs, subject to a **10% NAV per single counterparty** limit, with some providers adopting voluntarily tighter limits.

9.2.9 Investor Types and Strategies

Investors range from retail to institutional. Common uses are core-satellite allocation, **portfolio-imbalance management** and **cash-position management** — using an ETF to gain immediate exposure while cash is being invested or during a transition.

9.2.10 Benefits of ETFs

- Intraday liquidity and exchange trading.
- Diversification and transparency of holdings.
- Long, short, leveraged and inverse exposure.
- Low cost and tax efficiency relative to many alternatives.
- Core-satellite building blocks, and swap-based structures that **minimise pre-fee tracking error**.

9.2.11 Risks of ETFs

- **Market and liquidity risk**, where liquidity is measured by the **underlying**, not by secondary volume.
- **Tracking error and NAV premium or discount.**
- **Counterparty and collateral risk** in synthetic structures.
- **Roll and cash-drag risk** in commodity and futures-based funds.
- **FX and time-zone risk** — an Asian-hours fund tracking a US index diverges from NAV outside overlapping hours.
- **Leverage and daily-reset risk**, where the daily multiple is not a long-run multiple.
- **Closure, merger and securities-lending risk.** VIX ETNs were shut down in 2018, with losses exceeding **70%** even after the VIX recovered.

9.2.12 ETFs After-Sales

Monitor NAV, tracking difference, corporate actions, distributions, securities lending and factsheets. Remember that an ETF's **liquidity is the underlying's liquidity**: a narrow secondary market does not mean the fund is illiquid, and a deep secondary market does not mean the underlying is.

9.3 Summary

Structured funds are CIS vehicles whose payoff depends on assets, formulae and sometimes derivatives or swaps. CPPI preserves a floor only subject to path, fee, gap and issuer risk. ETFs trade intraday but carry NAV, tracking, market, liquidity, FX and counterparty exposures. Governance, custody, disclosure and suitability are part of the product.

Exam cues

- **A structured fund is a CIS**, must **publish NAV**, and its capital preservation applies only at maturity and only if the issuer and strategy perform. It is not deposit insurance.
- **CPPI:** cushion = portfolio − floor; multiplier = 1/crash size; risky = multiplier × cushion. A 15% floor with a 20% crash gives multiplier **5** and **75%** risky.
- **CPPI fails** by buying high and selling low, by cash-locking at the floor, and by gap risk. It assumes an appreciating, low-volatility asset with limited drawdown.
- **DPPI** varies the multiplier, typically within a **3–5×** band reset to **4×**.
- **Zero-plus participation** = available premium ÷ call cost. 20% available against a 25% call gives **80%** participation.
- **Trustee clocks:** semi-annual report within **two months**, annual within **three**, MAS notified within **three business days** of a breach.
- **Prospectus** must include a fee table, a numerical pricing example, the RSP, realisation terms, performance with the expense and turnover ratios, soft dollars and conflicts.
- **Restricted scheme:** accredited or relevant persons, or **SGD 200,000** per transaction.

- **ETF replication:** physical full, physical representative sampling, synthetic derivative-embedded, synthetic swap-based (unfunded, funded, combined). Synthetic limits: **10% net counterparty exposure**, collateral at least **90% of NAV** rebalanced daily.
- **Six causes of tracking error:** transaction costs, holdings, replication method, futures roll, cash drag, hedging costs. Synthetic tracks **tighter** than physical; emerging markets track **wider**.
- **Trap:** an ETF's liquidity is the **underlying's** liquidity, and a leveraged or volatility ETF's daily target is not a long-run multiple.

Chapter 10 — Contracts for Differences (CFDs)

A CFD is a leveraged, cash-settled agreement on the price difference of an underlying the investor never owns. This chapter covers the provider models, the margin mechanics, the cost arithmetic and the strategies traders run.

Learning Objectives

The candidate should be able to:

- Define a CFD and distinguish it from ownership of the underlying.
- Distinguish the market-maker, direct-market-access and exchange-traded provider models.
- Explain initial margin, maintenance margin, margin calls and liquidation.
- Compute commission, GST and financing costs and the resulting ROI on margin.
- Explain stop-loss, stop-entry and guaranteed stops, and the gap risk they leave.
- Explain dividend capture, pairs trading and merger arbitrage.

10.1 History and Origin

CFDs emerged in **London in the early 1990s** from the OTC equity-swap market, originally for institutional hedging. They reached retail markets in the late 1990s and were **not available in the US retail market** in the same way. That history explains their OTC, leveraged and cash-settled character.

10.2 What are CFDs?

A CFD is an agreement to exchange in cash the difference between the opening and closing value of an underlying position — or the reverse for a loss. The investor does **not** own the share or asset, normally has **no voting rights**, and does not take physical delivery.

A long equity CFD may receive a dividend adjustment; a short position may be debited. Corporate-action adjustments follow the provider's terms.

Five contract terms define a CFD: the **underlying**; the **opening and closing price**; the **number of units**; the **direction**; and the **currency**. Alongside those sit margin, financing rate, expiry or rollover, and corporate-action treatment.

10.3 Characteristics and Features

Leverage; flexibility; a relatively low initial outlay; a transparent reference price; simplicity of long and short exposure; no fixed expiry in some provider models; global access; cash settlement; and corporate-action adjustments.

Countervailing risks: margin calls, forced liquidation, financing, commission and GST, rollover, spread, provider credit and market liquidity.

Singapore **charges no stamp duty** on CFDs.

10.4 Business Models

Model	Description
Market maker	The provider quotes the CFD and may internalise or hedge the flow. Price and spread
Direct market access (DMA)	Orders route through the provider into the underlying market. **DMA is predomina**
Exchange-traded CFD	Exchange and clearing structure can reduce bilateral counterparty risk, but the terms

10.5 Advantages over Other Instruments

A CFD trades like a share in many respects but differs in nine:

Feature	Shares	CFDs	Options
Ease of trading	Standard	Yes — one account, global access	Requires an options-approve
Profit from price movement	Up only, unless borrowed	**Both directions**	Both directions
Margin	Full consideration	**Percentage of notional**	Premium only
Short selling	Needs a stock borrow	**Direct**	Via puts or calls
Dividends	Received as owner	Adjustment only	None
Expiry	None	None in some models	**Fixed**
Stop-loss	Manual	Available and often required	Manual
Spread	Exchange spread	Provider spread	Bid-ask
Number of instruments	Listed universe	Very wide, provider dependent	Listed series

10.6 Mechanisms

10.6.1 Margin

Initial margin is a percentage of contract value. The investor controls a larger notional than the cash posted, so return on margin can be high while losses are based on the **full exposure**.

Example 1. At **10%** margin, 10,000 shares at **$2** give a contract value of $20,000 and an initial margin of $*$ $*$2,000** — **10× leverage**.

Example 2. Crude at **$105 × 100 barrels** is $10,500; at $*$ $*$202,100** — **5× leverage**.

The Risk Disclosure Statement should explain margin, liquidation and the client's liability for any shortfall.

10.6.2 Mark-to-Market

Positions are revalued as the underlying moves, continuously during the trading day, and the account's equity is compared with the maintenance requirement.

10.6.3 Margin Call

A margin call requires additional funds. It is triggered when equity falls below the **maintenance margin**. The provider may then:

- **partially liquidate**, selling enough positions to restore margin; or
- **fully liquidate**, closing everything;

and the investor remains **liable for the shortfall**, loss and expenses after liquidation.

10.6.4 Liquidation

Liquidation converts the position to cash without the investor's consent. It is the mechanism that turns a temporary adverse move into a realised loss — and why a margin buffer matters more than a view.

10.6.5 Orders

All orders are **day orders** unless stated otherwise, and use the **underlying's bid and ask**.

- **Limit order.** A limit buy at 10.00 $*$ $*$when the ask is $*$ $*$10.02** will not execute at 10.02; it queues at 10.00 or better.
- **Market-to-limit (MTL).** Executes at the market and leaves any unfilled balance at the prior execution price. A buy filling partially at **$10.02** stays at 10.02; it is cancellable and valid only during market hours.
- **Market order.** Matched to the **lowest ask** for a buy, or the **highest bid** for a sell.

10.6.5.2 Contingent Orders

- **Stop-loss.** Long 1,000 CFDs at $10.00 * *with a sell stop at * *\$9.50$ and a limit at **\$9.45**.
- **Stop entry.** Market at $20.80 * *; place 1,000 at * *\$21.00$ with a stop sell at $20.50 * *and a limit at * *\$20.46$. The limit must be within **10 ticks (\$20.40)** of the stop.
- **Guaranteed stop-loss.** A distinct paid service. Where offered, the provider fills at the stop price regardless of slippage; where not, a fast market can gap straight through the stop.

10.6.6 Financing, Commission and GST

- **Financing.** Long CFDs normally incur interest on the **full notional**; short CFDs may receive interest on deferred sale proceeds but can incur **stock-borrow costs**.
- **Commission.** Commonly a percentage of notional, subject to a minimum.
- **GST** applies to the commission.
- **Rollover** depends on the provider and the asset.

Worked example — long. Buy 10,000 at $2.00 * *, sell after 10 days at * *\$2.20$. Commission **0.4%**, GST on commission **8%**, financing **5% p.a.**

Component	Calculation	Amount
Buy commission + GST	$20{,}000 \times 0.4\% \times 1.08$	**\$86.40**
Sell commission + GST	$22{,}000 \times 0.4\% \times 1.08$	**\$95.04**
Financing, 10 days	$20{,}000 \times 5\% \times 10/365$	**\$27.78**
Total expense		**\$209.22**
Gross gain	$(2.20 - 2.00) \times 10{,}000$	\$2,000.00
Net profit		**\$1,790.78**

10.6.6.4 Rollover Charges A CFD with no fixed expiry is rolled periodically, and the provider charges or credits a rollover adjustment reflecting the financing and dividend differential between the contract periods.

10.6.7 Going Short Sell 10,000 at $3.30 * *, cover at * *\$3.10$, commission **0.4%**, GST **8%**, **20%** margin and **6%** financing over five days.

Scenario	Total expense	Net profit / loss	Capital posted	ROI on margin
Cover at **\$3.10**	\$303.98	**+\$1,696.02**	\$6,600	**+25.70%**
Cover at **\$3.45** (price rises)	\$319.10	**−\$1,819.10**	\$6,600	**−27.56%**

The second row is the point: a 4.5% adverse move in the underlying produced a **27.56%** loss on the margin.

10.6.8 Corporate Actions

- **Dividends.** Long equity positions may **receive** cash dividends; short positions **pay** them.
- **Splits and reverse splits** adjust quantity and price.
- **Scrip dividends, bonus issues and rights** may differ or require close-out, because the CFD cannot deliver the actual entitlement.

10.6.9 CFDs versus Futures

Feature	CFDs	Equity futures
Venue	Mostly **OTC**	**Exchange**
Expiry	Can be extended or rolled, subject to provider policy	**Fixed expiry**
Financing	Charged **explicitly**	**Embedded** in the price
Dividends	Passed as adjustments	Normally **no entitlement**
Counterparty	**Provider**	Clearing house
Margin	Yes	Yes

10.7 Global Trading Opportunities

Underlyings include Singapore **STI and SiMSCI**; global equity and sector indices (S&P 500, DJIA, FTSE 100, CAC 40, DAX, Nikkei 225, ASX 200, Hang Seng, China A50); currencies (AUD, CAD, EUR, CNH, JPY, CHF, GBP, USD); energy, industrial, precious metal and agricultural commodities; US, UK, Japan and Euro government bonds; ETFs such as SPDR Gold; and volatility and real-estate references. Availability is **provider-specific**.

10.8 CFD Trading Strategies

10.8.1 Dividend Capture

Buy before the ex-dividend date and close after, seeking to profit if the price does not fall by the full dividend. Costs, tax, timing and price behaviour can defeat the trade. Traders generally screen for shares with **strong earnings and high payout potential**.

10.8.2 Pairs Trading

Long an undervalued or stronger asset and short an overvalued or weaker related asset, aiming for mean reversion and market neutrality. Correlation can break and costs are **doubled**.

Industry-sector pairs compare companies in the same sector — CapitaLand against City Developments, for instance — leaving stock-specific and sector risk.

Statistical arbitrage uses historical correlation and quantitative signals; the relationship may not revert.

Merger arbitrage goes long the target and short the acquirer, or the reverse. **Deal risk** — a failed transaction — can hurt both legs.

Index pairs trade correlated indices such as the **S&P 500 against the Nikkei 225** when their relationship diverges. Index liquidity is usually deeper than single-stock liquidity.

10.9 Summary

CFDs are leveraged cash-settled contracts, not ownership. Their flexibility conceals financing, margin, liquidation, counterparty, liquidity, currency, corporate-action and short-sale risk. Know the provider model, the order mechanics, the total cost and the worst-case liability.

Exam cues

- **A CFD is not ownership** — no voting rights, no delivery, and no fixed expiry in some provider models. **DMA is the predominant model in Singapore.**
- **Margin examples:** 10% on $20,000 is **2,000 (10× leverage); 20% on $10,500 is **2,100 (5× leverage).
- **Cost arithmetic must include GST on commission and daily financing on full notional.** Long: $86.40 + $95.04 + 27.78 = **209.22**, net $1,790.78 **. *Short* : *expense* ***303.98**, net **$1,696.02**, ROI **25.70%**; at 3.45 *itis* **−1,819.10**, ROI **−27.56%**.
- **Long CFDs pay financing on the full notional**; shorts may receive interest but can incur stock-borrow cost. **GST applies to commission.**
- **Orders:** limit queues at the stated price or better; MTL leaves the unfilled balance at the prior execution price and is market-hours only; market fills at the best opposite quote.
- **A plain stop is not guaranteed.** A gap can fill well below the stop; only a **guaranteed stop**, a paid service, removes that risk.
- **CFD versus futures:** OTC vs exchange, rollable vs fixed expiry, explicit vs embedded financing, dividend adjustment vs no entitlement, provider vs clearing-house counterparty.
- **Pairs and merger arbitrage are not risk-free:** correlation can break, both legs can lose, and costs double.
- **Trap:** the smaller initial outlay does **not** cap the loss. Liquidation converts a temporary move into a realised one, and the investor remains liable for any shortfall.

Chapter 11 — Key Products and Investment Risks for Derivatives and Structured Products

This chapter is the risk register. It takes each product family — futures, options, warrants, CFDs — and names its specific risks, then layers the risks that structured products add, then sets out the controls a firm must operate.

Learning Objectives

The candidate should be able to:

- Explain the specific risks inherent in futures.
- Explain the specific risks inherent in options.
- Explain the specific risks inherent in warrants and in CFDs.
- Explain why structured products carry more complex, layered risks.
- Explain the trading controls and risk-management processes a financial organisation implements.

11.1 Investment Risks for Derivatives

11.1.1 Risks in Trading Futures

11.1.1.1 Basis Risk The basis is the difference between the futures price and the cash or spot price. Where the asset being hedged is identical to the asset underlying the contract, the basis is **zero at maturity**. Where the contract and the asset are liquidated **before** maturity, the hedger faces **basis risk**, because futures and cash prices do not move in perfect lockstep; gains and losses on the contract and on the asset need not exactly offset.

Basis risk arises because:

- **a.** the asset to be hedged may not be exactly the same as the asset underlying the contract;
- **b.** the hedger may be **uncertain about the date** of the purchase or sale;
- **c.** the hedge may require the contract to be **closed out well before** expiry; and
- **d.** the size and unit of measurement of the contract may not correspond to the hedged position, so it is **under- or over-hedged**.

11.1.1.2 Liquidity Risk Although futures trade on exchanges, liquidity is not assured. Unless the investor is willing to take delivery, the only way to close a position is to **offset** it with another contract.

Liquidity can dry up **near maturity**, and can decline in a rapidly rising or falling market. If the price has moved by the maximum daily limit and no counterparties will transact, there are no trades — a **"lock limit" market** — and the investor cannot offset.

Two indicators gauge liquidity:

- **a.** trading **volume**; and
- **b. open interest** — the number of contracts not yet closed by an offsetting trade and not yet delivered.

11.1.1.3 Leverage Risk Futures use margin financing, so a relatively small price move produces a significant profit or loss. The **percentage** loss on the futures position is greater than the percentage move in the underlying.

The investor posts **initial margin**. If the market moves against the position, or the broker raises margin levels, additional funds are needed **at short notice**. Failure to comply may lead to liquidation.

11.1.2 Risks in Trading Options

Holders can lose the entire **premium**, especially for out-of-the-money options near expiry. European options may be difficult to monetise before expiry, and OTC exercise provisions may be unusual.

Naked writers face unlimited loss on a short call and a loss down to zero on a short put, and American-style writers **do not control exercise timing**.

Both sides face volatility, complex-strategy, trading-halt, foreign-market timing and currency risk. OTC options add **counterparty risk**.

11.1.3 Equity Index Options

Index options may reference illiquid or niche indices, and OTC index options add counterparty risk. Settlement is usually cash, but the index's composition and the contract multiplier must be checked.

11.1.4 Bond Options

Bond options inherit the bond's **interest-rate**, **credit-spread**, **liquidity**, **embedded-option** and **sovereign** risks.

In a **flight to safety**, Treasury yields fall while credit spreads widen most. The "bond vigilantes" of the **2010 Eurozone crisis** — Greece, Spain and Portugal — are the standard illustration: government yields rose as investors demanded a premium for credit risk, so bond prices fell.

Bond options also carry **embedded call and put** features and **change-of-control or event risk**.

11.1.5 Currency Options

Currency options involve two currencies, huge notionals and **round-the-clock** price movement, so monitoring is continuous. Physical delivery and OTC counterparty risk add to the exposure.

11.1.6 Warrants

Company warrants can expire **worthless**, carry **no voting or dividend rights**, and **dilute** shareholders if exercised. Structured warrants add **issuer credit risk**, underlying market risk, leverage, time decay, volatility, liquidity and currency exposure.

11.1.7 Risks in Trading CFDs

Leverage, market volatility, counterparty, liquidity, currency, financing, commission and GST, rollover, corporate actions and **provider-model risk** all apply. Foreign markets may have weaker protections and exchange controls. Market-maker pricing may be less transparent than DMA.

11.2 Key Risks of Structured Products

Structured products combine risks that a headline yield does not reveal.

11.2.1 Market Risk

Macro rates, inflation, employment, recession, earnings and political change all move the underlying and the payoff.

The crucial framing is that a **short option sells protection**:

- selling a **call** sells protection against a rise;
- selling a **put** sells protection against a fall;
- a **short receiver or payer swaption** sells protection against a rate move.

Four payoff profiles illustrate it: the short call, the short put, the short receive-fixed (call) swaption, and the short commodity call. In each, the investor receives premium and takes the loss if the protection is called upon.

11.2.2 CPPI Risk

CPPI **buys more risky assets as they rise and sells as they fall**, and a sharp fall can lock the fund at its floor, after which it misses any recovery.

11.2.3 Credit and Counterparty Risks

The **issuer**, the **SPV**, a **swap counterparty**, the **reference obligor** or the **collateral** may fail. An SPV may be **not guaranteed by its parent**, so the claim is limited to the SPV's own assets and the documents' terms.

11.2.4 Types of Credit Default

The ISDA credit events are **bankruptcy**, **failure to pay**, **restructuring**, **obligation acceleration**, **obligation default** and **repudiation/moratorium**. The **2014 Credit Derivatives Definitions** reformed these definitions and the settlement mechanics.

In a CLN the structure runs: the investor buys notes issued by an **SPV**; the SPV uses the proceeds to buy **AAA-rated securities**; the investor receives enhanced interest at **LIBOR + x bp**; the investor is **selling credit protection to the SPV**; and the SPV in turn sells protection to a **bank** through a CDS, receiving a protection payment that subsidises the enhanced interest.

If the reference entity defaults, the SPV — protection seller to the bank — **liquidates the AAA securities and pays the bank**. The investor, protection seller to the SPV, pays the contingent payment and recovers the remainder, **par minus losses**.

11.2.5 Liquidity Risks and Secondary Market Trading

Structured products are usually designed for investors willing to hold to maturity. Being customised to a specific risk-reward profile, they have a **limited secondary market**, so an early exit is hard.

A **lock-up** prevents withdrawal until a stated date, from a few months to a few years. A further reason for illiquidity is that the embedded derivatives — exotic options, CDS — may not be actively traded.

Liquidating before maturity may cost part or a **substantial part** of principal.

The issuer or its affiliates usually provide a secondary market even where not obliged. Investors should assess the issuer's quality, commitment and **track record in providing liquidity**, while recognising that the price ultimately depends on prevailing market conditions.

11.2.6 Early Termination Risk

Unwinding the bond and the derivative before maturity incurs **discount and break costs**. Full principal preservation may exist **only at maturity**.

11.2.7 Structure Risk

Combinations vary, and the loss may be **unlimited** or the principal fully exposed despite a familiar wrapper.

A **receive-fixed** swaption has its loss capped at the fixed rate; a **pay-fixed** swaption's loss is **unlimited**. Negative interest rates are a scenario the structure may not have contemplated.

11.2.8 Reinvestment Risk

The risk that coupons or principal must be reinvested **below the yield to maturity**. If rates fall, proceeds can only be reinvested at the prevailing lower rates, so the realised return is below the YTM.

It is greatest for bonds with **large coupons**, **higher payment frequency**, **long maturities** and those **selling at a premium**. **Zero-coupon bonds have none**; **callable bonds have the most**, because the issuer chooses when to return the money.

11.2.9 Volatility Risk

Applies to bonds and instruments with **embedded options** — callable, putable and convertible bonds. Their prices respond to volatility as well as to rates. A call option rises in value when rate volatility rises, which **reduces** the value of the callable bond because the holder has sold that call component to the issuer.

11.2.10 Correlation Risk

The likelihood that an event affecting one underlying directly affects another. A structured product is exposed whenever its underlyings are correlated — baskets, worst-of structures and credit pools, where banking or sector relationships make simultaneous deterioration likely.

11.2.11 FX Risk

The underlying and the note may be in different currencies, so the investor bears the currency move between them as well as the underlying's performance.

11.2.12 Legal Risk

The investor owns a **note or a claim**, not necessarily the underlying assets. Creditor priority and recourse depend on the documents, and an SPV's position in a group structure matters.

11.2.13 Transactional Risk

Pricing and contract commencement, and settlement timing **across currencies**, can create exposure between the trade and the settlement.

11.2.14 Mis-selling Risk

The product may not meet the client's objective. The PHS, prospectus or private-placement documents, and the advice given, must explain the **real payoff** — not the headline.

11.3 Trading Risk Control and Management

The **front office**, **middle office** and **back office** manage market, credit and settlement risk.

11.3.1 Futures Controls

Limit	Purpose
Open-contract limit	Smaller for volatile or illiquid products, and restricted relative to **gross open interest**. Separat
Maximum-loss limit	A weekly or monthly trader or desk loss triggers squaring-off, suspension of trading and escalatio
Maturity limit	Restricts illiquid deferred contracts.
Stress-test limit	Measures the portfolio under adverse scenarios.

Fixed-income measures support these:

- **1 basis point = 0.01%.**
- **DV01 / PVBP** is the dollar price change per basis point.
- **Duration** is a weighted cash-flow time measure; **modified duration** measures price sensitivity:

$$\frac{\Delta P}{P} \approx -\text{modified duration} \times \Delta YTM$$

A modified duration of **2** and a **100bp** rise implies about a **2%** fall. A bond priced at $980 ** with PVBP of ** 1$ falls to **$975** on a 5bp rise.

VaR combines volatility and correlation into a loss threshold. A **1-day 95% VaR of SGD 1m** means a **5% probability** that the loss exceeds SGD 1m over one day, under the model's assumptions — about **one day in twenty**. VaR has three components: confidence, horizon and the loss estimate. It is neither a guarantee nor a substitute for stress testing.

11.3.2 Options Controls

Greek	Limit measures
Delta	Directional, or currency, exposure
Gamma	Change in delta for a spot move, often per one standard deviation, with tenored limits
Vega	Premium change per **1%** volatility
Theta	Time decay; often limited jointly with gamma and vega for short-option books
Rho	Rate sensitivity; can be reduced by a swap

Gamma is especially volatile **near the money and near expiry**. Maturity concentration, total book size, premium payment and intraday limits complement the Greek limits. Limiting **premium paid** prevents options being used as a substitute for an unsecured loan.

11.3.3 Credit and Settlement Controls

OTC counterparty limits depend on credit rating, settlement amount and tenor. A **Credit Support Annex (CSA)** specifies collateral.

Clearing-broker risk arises because margin is held with the broker; exchange and CCP risk remains despite default resources.

The **exchange** itself is a risk to be assessed, not assumed away.

11.3.4 Settlement Risk

Back-office controls require timely, accurate confirmations stating the **option style**; the **expiry date and time**; the **premium payment date**; the **settlement date**; the **strike**; and **both currency amounts**.

11.4 Summary

Risk management must match the product: basis, liquidity and leverage for futures; premium, exercise, volatility and Greek risk for options; issuer and gearing for warrants; margin, financing and provider risk for CFDs; and layered market, credit, liquidity, legal, structure, correlation, FX and reinvestment risk for structured products.

Exam cues

- **Futures basis risk has four causes** — a different asset, an uncertain date, an early close-out, and a size mismatch causing under- or over-hedging. Basis is zero at maturity only if the assets are identical.
- **"Lock limit"** means the price has hit its daily limit and no counterparty will trade, so the position cannot be offset. **Volume and open interest** are the liquidity indicators.
- **Option losses:** a holder's maximum is the **premium**; a naked short call is **unlimited**; a short put loses down to zero.
- **Structured products sell protection:** a short call sells upside protection, a short put sells downside protection, a short swaption sells rate protection.
- **Reinvestment risk ranking: zero-coupon has none**, then low coupons, then high coupons, then **callable bonds have the most**. Large coupons, frequent payments, long maturities and premium prices all increase it.
- **Volatility and callables:** rising rate volatility raises the call's value and therefore **lowers** the callable bond's value.
- **Duration and PVBP:** $\Delta P/P \quad -D \times \Delta YTM$; D = 2 with a 100bp rise is −2%; $980 with PVBP $1anda5bpriseis * *975**$.
- **VaR:** 1-day 95% VaR of SGD 1m is a **5%** chance of losing more than SGD 1m in a day, roughly one day in twenty — not a maximum loss.
- **Controls:** open-contract, maximum-loss, maturity and stress-test limits; Greek limits for delta, gamma, vega, theta and rho; CSA collateral; and a confirmation checklist covering style, expiry date and time, premium date, settlement date, strike and both currency amounts.
- **Trap: receive-fixed** swaption loss is capped at the fixed rate, but **pay-fixed** loss is unlimited. And a 1-day VaR says nothing about what happens beyond the horizon or outside the model.

Case Studies

These cases are decision exercises, not product endorsements. A representative should establish suitability, knowledge, risk willingness, financial capacity, external exposure, internal limits, buffers, leverage, stop/offset discipline, profit targets and current market information before recommending a trade.

12.1 Enhanced yield with currency investments — Mrs Goh

Mrs Goh holds USD, GBP and SGD. She expects a short-term USD rebound but longer-term USD weakness and accepts conversion into SGD/GBP. A USD/SGD DCI invests USD1m for three months at 11% p.a., strike 1.3800, spot 1.3569.

- If expiry spot is 1.3300, no conversion at strike; USD proceeds = 1,000,000 + 1,000,000 × 11% × 3/12 = USD1,027,500. Converted at 1.33 = SGD1,366,575.
- If spot is **1.3800**, proceeds remain USD1,027,500.

- If spot is **1.4100** and conversion applies, USD1,027,500 is converted at 1.3800 = SGD1,417,950.

From the bank's perspective, fund the deposit at 1.3919% and sell the option at 0.0250, giving about **USD1,028,566.74** after three months — annualised around **11.43–11.44%**, above the 11% client return. DCI yield is option premium plus deposit interest and conversion risk.

The GBP/USD DCI example at 10% p.a., strike 1.3000 shows that if spot is 1.2800, the conversion calculation yields about USD1,009,230.76; if spot is 1.2600, proceeds about USD993,461.54 imply annualised return around −2.62%. A high headline yield does not prevent capital loss in the base currency.

12.2 Short-term downside volatility — Brian

Brian expects a two-month equity correction, higher volatility and longer-term recovery. Before trading, verify risk willingness/understanding, capacity and external positions, experience, internal limits, buffers, leverage, stop/offset plans, profit taking and market information. Limits, margin buffers and education help; maximising leverage does not.

Possible tools:

- long index/futures puts: limited premium loss, direct downside insurance, time decay and expiry risk;
- put warrants: similar limited-premium downside exposure, time decay/issuer risk;
- short index futures: direct daily-settled exposure, margin and potentially large losses if the market rises;
- short or long inverse/leveraged DLCs: daily reset and amplified risk;
- inverse ETFs/ETNs: exchange access but tracking, cost, roll and leverage risk;
- long volatility futures ETF: benefits from a volatility spike but suffers from leverage and contango/roll decay; long-term holding can trend toward zero.

Timing is difficult; short products are costly to maintain and should be closed/rolled with discipline.

12.3 Portfolio protection — William

William holds OCBC, DBS and CapitaLand worth SGD92.72m and wants protection beyond a 7% fall. Compare put-warrant strikes, breakeven, expiry and stock coverage. In the 7% scenario, DBS/OCBC put warrants B and D are in-the-money; CapitaLand E is selected because it is the only available CapitaLand protection and is less OTM than alternatives. A and D offer more time protection; E covers CapitaLand.

Higher volatility raises warrant prices because the strike is more likely to be reached. OTC alternatives are lending the shares for a 4–5% fee, buying tailored protective puts 3–4% below spot, or shorting MSCI Singapore futures. The futures hedge is imperfect because the three stocks represent only about 31% of the index. Futures have immediate daily MTM obligations; a long put has a right, not an obligation; a share loan earns a fee.

12.4 ELN yield and discounted blue-chip purchase — Mr Sunjoyo

OCBC spot = SGD13.30; ELN purchase = SGD975,000; notional = SGD1m; strike = SGD12.50; 80,000 shares; three months; annualised headline yield = 10.66%.

- At expiry SGD12.40, shares are worth SGD992,000; return over three months = 1.7435%; annualised compound return 7.16%.
- At SGD12.00, shares are worth SGD960,000; return = −1.5385% over three months; annualised −6.29%.
- Breakeven = 975,000/80,000 = SGD12.1875 SGD12.19.

The client is effectively short a put: below the strike, he owns the falling stock and can lose up to almost the entire investment if the stock goes to zero.

12.5 Range accrual note — Robert

AUD/USD range = 0.7600–0.8200, maturity three months, maximum yield 6%, minimum 1%, MMR 2.75%. If inside for two months and outside for one, annualised payout is

$$6\% \times \tfrac{2}{3} + 1\% \times \tfrac{1}{3} = 4.33\%$$

(the answer text displays approximately 4.32% due to rounding).

To make the digital spread cheaper or raise maximum yield: narrow the range, skew it so spot is nearer a barrier, extend maturity, or lower the minimum yield. Widening the range makes inside outcomes more likely and the option more expensive; shortening maturity can increase cost. A double-no-touch version observes the path: touching either barrier terminates the enhanced payout and leaves only the minimum return.

12.6 Equity accumulator — Harry

ABC spot SGD1.10; strike SGD1.00; knock-out SGD1.30; one year/250 business days; 10,000 shares per observation; monthly settlement. An accumulator combines long call and short put economics and can be funded or margin/unfunded. Below strike under a 1×2 arrangement, the client may have to buy 2×; above strike but below the knock-out, 1×; at/above the barrier, the note terminates.

- If the barrier is touched on day 13, days 1–12 accumulate: 120,000 shares.
- Twenty days in month 1 plus 12 days in month 2 before day-13 knock-out = 320,000 shares; cost = SGD320,000.
- If marked at SGD1.30, value = SGD416,000 and gain = SGD96,000.

Worst case: price zero for 250 days, 2.5m shares purchased at SGD1 = **SGD2.5m loss**. There is no capital preservation; the bank may margin-call/liquidate; gains are capped by the knock-out while losses continue below strike; early-break costs can be large.

12.7 Gold, silver and USD volatility — Brandon

Rising US rates may strengthen USD and weaken gold/silver, while political/equity uncertainty may support safe-haven gold. Gold's widening premium to silver and different USD sensitivities create relative-value ideas. If the direction is uncertain but a large move is expected, buy a gold straddle: long call and long put at the same strike. A strangle uses different OTM strikes and is cheaper but needs a larger move. Combine a bull call spread with a bear put spread, or a bull call with a bear call, to express two-sided volatility; do not use two bearish or two bullish structures if the view is genuinely two-sided.

12.8 Yield-curve movement — Tracy

Tracy expects the US yield curve to flatten/invert: short rates rise faster than long rates. A constant maturity swap (CMS) resets a floating leg against a long-term constant-maturity rate; unlike a vanilla IRS, the reference maturity/duration remains constant. It can express steepening/flattening views. To benefit from short rates rising faster, swap the long end for the short end — pay the long-end rate and receive the short-end rate — subject to the exact swap convention and market quotation.

Exam cues

The case studies test the same method, not the products. For each, work the numbers and identify the embedded option and the investor's obligation.

- **DCI (Goh):** yield = option premium + deposit interest; conversion risk is the real exposure. A high headline yield does not prevent a base-currency loss.
- **Downside volatility (Brian):** puts / put warrants cap loss at premium but decay; short futures and inverse/leveraged products add margin, daily reset and compounding risk.
- **Portfolio protection (William):** compare put-warrant strikes, breakeven and coverage; a futures hedge is imperfect when the basket is only part of the index; a long put is a right, a futures position is an obligation.
- **ELN (Sunjoyo):** effectively short a put — breakeven = purchase / shares; below strike the investor owns the falling stock and can lose almost everything.
- **Range accrual (Robert):** payout = max × (days inside / total) + min × (days outside / total); a double-no-touch observes the path and a barrier touch leaves only the minimum.
- **Accumulator (Harry):** 1× above strike to the knock-out, 2× below strike; gains capped by the knock-out while losses continue — worst case is multiples of the notional.
- **Two-sided volatility (Brandon):** a straddle/strangle for a large move; combine a bull call spread with a bear put spread — never two structures on the same side.
- **CMS (Tracy):** pay the long end and receive the short end to profit from short rates rising faster than long rates.

Appendix A — CPPI strategy worked examples

Parameters: initial \$100; crash size 20%; multiplier

$$1/20\% = 5$$

; risky asset S&P index fund; five-year tenor.

- Day 1 floor 85%: cushion 15; risky = 5×15 = \$75; risk-free = \$25.
- New floor 86% with total still \$100: cushion 14; risky = \$70; risk-free = \$30.
- Portfolio at \$86 after risky asset falls 20%: risky falls \$70→\$56; plus \$30 risk-free = \$86; liquidate all risky into risk-free, so later equity appreciation is missed.
- Portfolio at \$96: risky value \$66, required risky allocation

$$5 \times (96 - 86)\% = \$50$$

  ; sell \$16 risky and hold \$46 risk-free.
- Portfolio at \$110: required allocation

$$5 \times (110 - 86)\% = \$120$$

  ; if capped, put all \$110 in risky; if leverage permitted, borrow \$10 and hold \$120 risky.
- If leveraged risky value falls 20% from \$120 to \$96, repay \$10 and move remaining \$86 to risk-free. The crash has again locked the floor.

The reallocation repeats periodically. CPPI assumes an appreciating, relatively low-volatility asset with limited drawdown; a range-bound or gap-down market can cause buy-high/sell-low behaviour.

Appendix B — MAS Product Highlights Sheet practice note

An excerpt of **MAS Practice Note SFA PN-02** (issued 8 June 2018), referencing SFA 13-G13. Key points:

- the note guides issuers and professional advisers in preparing a PHS under the SFA framework and is read with the relevant offers regulations;
- the PHS is submitted to MAS at prospectus lodgement;
- it is made available with the prospectus, including online if the prospectus is online;
- the prescribed format, tabular structure, headings/subheadings and yellow Pantone 109C right-edge feature are minimum presentation requirements;
- additional headings and material information should be added where needed so investors understand the product;
- a PHS should plainly state product summary, suitability, key features, risks, fees/charges, definitions and contacts, and prominently disclose possible loss of all principal where applicable.

Appendix C — Futures contract specifications

These specifications were updated as at **April 2022**; always use current SGX specifications.

Contract	Main terms to remember
SGX EUR/CNH FX futures (`EC`)	EUR100,000; tick CNH10; minimum fluctuation CNH0.0001; cash-settled in CNH; last
SGX 10-year full-size JGB (`JG`)	JPY100m notional, 10-year 6% coupon; quarterly March/June/September/December
SGX Nikkei 225 (`NK`)	JPY500 × index; 6 serial + 32 quarterly months; outright tick 5 points/JPY2,500, str
SGX FTSE Taiwan (`TWN`)	USD40 × index; 2 serial + 12 quarterly months; 0.25-point/US\$10 standard tick; 10%
SGX MSCI Singapore (`SGP`)	SGD100 × index; 2 serial + 4 quarterly months; 0.05 point/SGD5 tick; ±10%/±15%
SGX STI (`ST`)	SGD10 × index; 2 serial + 4 quarterly months; 1 point/SGD10 tick; +10%/+15% coo

Exam habit: identify contract size, tick value, trading hours/session, price limits, last trading day, settlement basis, final settlement reference and position-accountability threshold before calculating.

Appendix D — Formulae sheet (consolidated)

Futures

$$F_{0,t} = S_0(1 + C_{0,t})$$

$$P = 100(1 - R)$$

for short-rate futures.

$$\left(1 + \frac{d_1 r_1}{360}\right)\left(1 + \frac{d_2 r_2}{360}\right) = \left(1 + \frac{DR}{360}\right)$$

for long-rate futures.

$$F = S \times \frac{1 + R_c(n/360)}{1 + R_b(n/360)}$$

for interest-rate parity.

$$F = S\left[1 + r(t/360)\right] - D$$

for equity-index futures.

$$\text{target hedge rate} = \text{futures rate} + \text{target-rate basis}$$

$$V_t = S_t + (F - F_t) = F + (S_t - F_t)$$

$$\text{bond hedge ratio} = \frac{\text{PVBP(hedged security)}}{\text{PVBP(CTD)}} \times \text{conversion factor}$$

$$h = \frac{\Delta S}{\Delta F}$$

; contracts for a delta-neutral loan

$$= \frac{-h \times \text{loan value}}{\text{contract size}}$$

$$N = \frac{VP}{F \times T} \times \beta$$

for equity futures hedging.

Options

$$\text{option price} = \text{intrinsic value} + \text{time value}$$

$$\text{call IV} = \max(0, S - X)$$

;

$$\text{put IV} = \max(0, X - S)$$

$$C + \text{PV}(X) = P + S$$

$$\text{delta} = \frac{\text{change in option price}}{\text{change in underlying}}$$

;

$$\text{gamma} = \frac{\text{change in delta}}{\text{change in underlying}}$$

Warrants and CBBCs

$$\text{gearing} = \frac{\text{share price}}{\text{warrant price} \times n}$$

;

$$\text{effective gearing} = \text{delta} \times \text{gearing}$$

$$\text{call IV} = \max[0, (S - X)/n]$$

;

$$\text{put IV} = \max[0, (X - S)/n]$$

$$\text{call conversion price} = X + nWP$$

;

$$\text{put conversion price} = X - nWP$$

$$\text{call cash settlement} = (S - X)/n$$

;

$$\text{put cash settlement} = (X - S)/n$$

$$\text{bull CBBC} = \frac{(\text{underlying} - \text{strike}) + \text{financial cost}}{\text{conversion ratio}}$$

$$\text{bear CBBC} = \frac{(\text{strike} - \text{underlying}) + \text{financial cost}}{\text{conversion ratio}}$$

R-CBBC residual: bull

$$(\text{settlement} - \text{strike})/n$$

; bear

$$(\text{strike} - \text{settlement})/n$$

; N-CBBC residual is zero at MCE.

Structured products

$$\text{CPPI cushion} = \text{portfolio} - \text{floor}$$

;

$$\text{multiplier} = 1/\text{crash size}$$

;

$$\text{risky} = M \times \text{cushion}$$

;

$$\text{risk-free} = \text{portfolio} - \text{risky}$$

$$\text{RAN payout} = P_1 \frac{n}{N} + P_2 \frac{N - n}{N}$$

$$\text{inverse floater coupon} = \max\left[X\% - \text{leverage} \times \text{floating index},\ \text{minimum coupon}\right]$$

$$\text{participation rate} = \frac{\text{discount sum}}{\text{call premium}}$$

under the structured-product convention.

Appendix E — Review Questions

Candidates should note that the **sole purpose of the Review Questions is to familiarise candidates with the scope and general nature of the examinations, and the format of the examination questions**. They are **not intended as preparatory study material**, and **do not cover all the material tested**. The correct answer for each question is in **bold** in the source booklet and is reproduced as **Answer** below.

Chapter 2 — Futures

1. There are certain risks involved in hedging in the forward or futures market. Which of the following is TRUE?

- **a. Futures contracts have lower counterparty risk, while forward contracts carry the credit risk of the counterparty.**
- **b.** Futures contracts have basis risk, whereas forward contracts do not.
- **c.** Forward contracts have the risk of the exchange.
- **d.** Forward contracts have interim partial settlements.

Answer: a.

2. Advantages of futures contracts over OTC instruments include:

- **a.** More transparency in pricing.
- **b.** Market liquidity.
- **c.** Reduced credit risk.
- **d.** Increased credit risk.

Options: (a) and (c) · (b) and (c) · **(a), (b) and (c)** · (a), (b) and (d)

Answer: (a), (b) and (c).

3. What is the economic importance and impact of the futures market?

- **a.** Futures are not standardised contracts and therefore not readily available.
- **b.** The futures market is both a less active and less important source of vital market information.
- **c. Participants can use contracts to leverage on long or short positions.**
- **d.** Participants cannot use contracts to leverage on long positions.

Answer: c.

4. Spot GBP/USD is trading at **1.2422**. Given annualised US interest rates at **1.799%** and annualised British interest rates at **0.70444%**, what should be the theoretical price for a 1-year forward contract?

- **a. 1.2557.**
- **b.** 1.2487.
- **c.** 3.1743.
- **d.** 0.9989.

Answer: a. — By interest rate parity,

$$F = S \times \frac{1 + r_{USD}}{1 + r_{GBP}} = 1.2422 \times \frac{1.01799}{1.0070444} \approx 1.2557$$

.

5. What are the factors affecting basis in the market of a futures contract?

- **a.** Cost / return of carry.
- **b.** Non-difference in coupons between fixed income instruments.
- **c.** Market rates versus administrated rates.
- **d.** Yield curve changes.

Options: (a) and (d) · (a), (c) and (d) · (a), (b) and (c) · (b) and (c)

Answer: (a), (c) and (d).

Chapter 3 — Futures Strategies

6. Using bond futures to hedge a bond portfolio is more complicated because, in the case of US T-bond futures, certain options are embedded for the sellers — including delivery timing and choice of deliverable bonds. As a result, we would expect futures pricing to be _____ than that implied by cash-and-carry arbitrage.

- **a. Lower.**
- **b.** Higher.
- **c.** Same.
- **d.** Not enough information provided.

Answer: a.

7. Which are the main investment and trading strategies used in the futures market?

- **a.** Outright trades.
- **b.** Hedging.
- **c.** Spread trades.
- **d.** Bid trades.

Options: **(a), (b) and (c)** · (b), (c) and (d) · (a), (c) and (d) · (a), (b), (c) and (d)

Answer: (a), (b) and (c).

8. A local company wishes to take a **6-month USD 50 million** loan; the prevailing 6-month USD lending rate is **2.15%**. The September Eurodollar futures contract is trading at **98.15**; at the time of entering the hedge it was **98.65**. The Eurodollar futures contract size is **$1,000,000**, each representing a **3-month LIBOR period (0.25 years)**. Assuming the loan rate is correlated one-to-one with the Eurodollar rate and the company intends to fully hedge using September futures, calculate the number of contracts needed.

- **a.** 25
- **b.** 50
- **c. 100**
- **d.** 150

Answer: c. — Contracts

$$= \frac{\text{loan}}{\text{contract size}} \times \frac{\text{exposure period}}{\text{contract period}} = \frac{50{,}000{,}000}{1{,}000{,}000} \times \frac{0.5}{0.25} = 100$$

.

9. Calculate the hedge ratio required to hedge the interest rate on a **6-month loan for USD 100 million**. Assume the loan rate is correlated one-to-one with the Eurodollar rate of the nearest 90-day contract, and that a **stack hedge** is employed. Contract size **$1,000,000**, each representing a **3-month LIBOR period (0.25 years)**.

- **a.** 200
- **b.** 20
- **c. 2**
- **d.** 0.2

Answer: c. — In a **stack** hedge the whole exposure is covered by contracts of the nearest expiry, so the contract count is

$$\frac{100{,}000{,}000}{1{,}000{,}000} \times \frac{0.5}{0.25} = 200$$

contracts — but expressed against the **notional of the stack** the hedge ratio is **2**, the ratio of exposure tenor to contract tenor.

10. A player in the financial markets who sells contracts to manage his cash market exposure while holding the underlying stock is a/an:

- **a.** Arbitrageur.
- **b. Hedger.**
- **c.** Market maker.
- **d.** Speculator.

Answer: b.

Chapter 4 — Options

11. There are positive relationships between the value of a call option and _____.

- **a.** Interest rates.
- **b.** Time to expiration of the call option.
- **c.** Dividend rate of the underlying stock.
- **d.** Volatility of the underlying stock.

Options: (a) and (c) · (a) and (b) · (a), (c) and (d) · **(a), (b) and (d)**

Answer: (a), (b) and (d). — A higher **dividend rate reduces** call value, so (c) is excluded.

12. The exercise price of a call option is $50 * *. The price of the call is * *\mathbf{6}$, and it expires in 3 months. The current share price is **$54**. What is the intrinsic value?

- **a.** $50.
- **b. $4.**
- **c.** $54.
- **d.** $6.

Answer: b. — Intrinsic value

$$= \max(S - K, 0) = 54 - 50 = 4$$

. The remaining $2 * * of the * *\mathbf{6}$ premium is time value.

13. Which of the following statements is TRUE about Theta and Vega?

- **a.** Theta measures the change in asset price with respect to time, while Vega measures the volatility of the option's portfolio.
- **b.** Vega measures the change in asset price with respect to time, while Theta measures the volatility of the option's portfolio.
- **c.** Vega is positive for long calls and short puts.
- **d. Vega is positive for a long volatility investment position with call or put options.**

Answer: d.

14. Which of the following statements about zero-cost options are TRUE?

- **a.** Since the option is zero cost, it is possible to buy as much as possible to take advantage of its zero-cost structure without having to worry about the risk associated with it.
- **b.** Zero-cost structures give unlimited profits and limited losses.
- **c.** A zero-cost collar is defined as buying a protective put and selling an out-of-the-money covered call.
- **d.** A zero-cost (costless collar) option is sold with a strike price where the premium received is equal to the premium paid for the put purchased.

Options: (b) and (c) · (a) and (d) · (a) and (c) · **(c) and (d)**

Answer: (c) and (d).

15. A trader buys a Nikkei 225 futures contract at **10,000**, a put struck at **9,500** (premium **200**) and sells a call struck at **10,500** (premium **190**). What is the synthetic position created?

- **a. Bull spread.**
- **b.** Bear spread.
- **c.** Ratio put spread.
- **d.** Short call.

Answer: a.

Chapter 5 — Structured Warrants and Daily Leveraged Certificates

16. Which of the following will increase the value of both a call option and a put option?

- **a.** Increase in the exercise price.
- **b. Increase in volatility.**
- **c.** Increase in interest rates.
- **d.** Decrease in time to maturity.

Answer: b.

17. The warrant premium is affected by:

- **a.** The length of time before expiration date.
- **b.** The size of the cash dividends paid on the underlying share.
- **c.** The intrinsic value of the warrant.
- **d.** The credibility of the individual buyer.

Options: (a), (b) and (d) · **(a), (b) and (c)** · (b), (c) and (d) · (a), (b), (c) and (d)

Answer: (a), (b) and (c).

18. A call warrant with an exercise price of $1.00 * *istradingat * *\mathbf{1.20}$. One warrant converts to one share. What is its conversion price?

- **a.** $1.00.
- **b.** $1.20.
- **c. $2.20.**
- **d.** $0.20.

Answer: c. — Conversion price

$$= \text{exercise price} + \text{warrant price} = 1.00 + 1.20 = 2.20$$

for a **1:1** conversion ratio.

19. The warrant is trading at $0.40 * *, theexercisepriceis * *\mathbf{0.80}$, and the share price is **$2.40**. What is its gearing? (Assume a company-issued warrant with a conversion ratio of 1.)

- **a.** 2X.
- **b. 6X.**
- **c.** 3X.
- **d.** 5X.

Answer: b. — Gearing

$$= \frac{\text{share price}}{\text{warrant price}} = \frac{2.40}{0.40} = 6$$

X (with a 1:1 ratio).

20. Company A's warrant is trading at $0.35 * *, itsexercisepriceis * *\mathbf{0.50}$, and the share price is **$0.65**. What is its intrinsic value?

- **a.** $0.80.
- **b. $0.15.**
- **c.** $0.30.
- **d.** $1.00.

Answer: b. — Intrinsic value

$$= \max(S - K, 0) = 0.65 - 0.50 = 0.15$$

.

21. The market price of Company B's common share is $5 * *andtheexercisepriceofthewarrantis * *\mathbf{3.50}$. **Two warrants** convert into **one** common share. The warrant price is currently **$2**. What is the warrant premium?

- **a.** 25%.
- **b.** 30%.
- **c.** 50%.
- **d.** 100%.

Answer: 50%. — Because **two warrants** are needed per share, the warrant outlay per share is

$$2 \times 2 = 4$$

. Premium

$$= \frac{\text{warrant outlay} + \text{exercise price} - \text{share price}}{\text{share price}} = \frac{4 + 3.50 - 5}{5} = \frac{2.50}{5} = 50\%$$

. The examinable point is that a **non-unit conversion ratio must be applied to the warrant price** before computing the premium.

Chapter 6 — Barrier Options, Binary Options and Callable Contracts

22. For a single "knock-out", up-and-out call barrier option, the barrier level is:

- **a.** Below the current spot price of the asset.
- **b.** Equal to the current spot price of the asset.
- **c.** Equal to the strike price.
- **d. Above the current spot price of the asset.**

Answer: d. — An **up-and-out** call is knocked out when the price rises to the barrier, so the barrier must sit **above** spot.

23. Which of the following statements are CORRECT?

- **a.** For a Category-N callable bull/bear contract (CBBC), there is no residual value when a mandatory call event occurs.
- **b.** For a Category-R CBBC, there is always a positive residual value when a mandatory call event occurs.
- **c.** The Category-N CBBC call price is always equal to the strike price.
- **d.** The Category-R CBBC call price is always different from the strike price.

Options: **(a), (c) and (d)** · (b), (c) and (d) · (a), (b) and (c) · (a), (b) and (d)

Answer: (a), (c) and (d). — A Category-N CBBC has **no residual value** on a mandatory call event, and its call price **equals** the strike; a Category-R CBBC has a call price **different** from the strike so that a residual value remains.

24. In a double "knock-out" barrier option, the barrier levels _____.

- **a.** For both barriers are higher than the strike price for a call option.
- **b.** For both barriers are lower than the strike price for a call option.
- **c.** Will vary, as there will be two strike prices with one barrier level for each strike price.
- **d. Will vary, as there will be one barrier level above the strike price and another barrier level below the strike price.**

Answer: d.

25. All the following are common features of callable bull/bear contracts (CBBC) and structured warrants **except one**. Which characteristic is TRUE for CBBC products but **NOT** for structured warrants?

- **a. Call price specified.**
- **b.** Fixed maturity period.
- **c.** No margin requirement.
- **d.** Traded on an exchange.

Answer: a.

26. What are the DISADVANTAGES of barrier options?

- **a.** Low risk of loss of premium if barrier conditions are breached.
- **b.** High risk of loss of premium if barrier conditions are breached.
- **c.** Investor is exposed to counterparty risk on the issuer, which arises when dealing in OTC products.
- **d.** Barrier options are not over-the-counter (OTC) products for which investors do not pay an option premium.

Options: (a) and (c) · **(b) and (c)** · (c) and (d) · (b) and (d)

Answer: (b) and (c).

Chapter 7 — Structured Deposits and Other Structured Products

27. What should investment professionals consider when helping clients invest in structured products?

- **a.** Risk appetite.
- **b.** Investment experience and knowledge.

- **c.** Investment horizon.
- **d.** Liquidity needs.

Options: (a), (b) and (c) · (a), (c) and (d) · (b), (c) and (d) · **(a), (b), (c) and (d)**

Answer: (a), (b), (c) and (d).

28. What would be the purposes and uses of OTC structured products?

- **a.** Structured products can be customised to meet needs that cannot be fulfilled by standard traditional investments.
- **b.** Structured products provide optimisation of yield and returns.
- **c.** Structured products are an alternative rebalancing investment to the direct purchase of the underlying asset.
- **d.** Structured products allow clients to gain leveraged exposure that can potentially augment gains using a smaller investment outlay.

Options: (a), (b) and (c) · **(a), (b), (c) and (d)** · (b), (c) and (d) · (a), (c) and (d)

Answer: (a), (b), (c) and (d).

29. Which of the following are structured products?

- **a.** Accumulators.
- **b.** Callable range accrual notes.
- **c.** Step-up digital options.
- **d.** Equity-linked notes.

Options: (a), (c) and (d) · (b), (c) and (d) · **(a), (b), (c) and (d)** · (a), (b) and (d)

Answer: (a), (b), (c) and (d).

30. Which of the following instruments can be used as the principal component of a structured note?

- **a.** AAA-rated bond.
- **b.** Common shares.
- **c.** Range accrual deposit.
- **d.** Zero-coupon or corporate bonds.

Options: **(a), (c) and (d)** · (b), (c) and (d) · (a) and (d) · (b) and (c)

Answer: (a), (c) and (d). — The principal component is a **debt or deposit** instrument; **common shares** are not.

31. A structured product offered a minimum **100% return of capital** at the end of the investment term, plus **80% of any rise in the FTSE100 index**. If the index rose by **40%** over the period, the investor would receive the initial capital in full plus an additional return of **32%**. What is the participation rate?

- **a.** 32%.
- **b.** 104%.
- **c. 80%.**
- **d.** 100%.

Answer: c. — Participation rate

$$= \frac{32\%}{40\%} = 80\%$$

.

Chapter 8 — Structured Notes

32. The underlying instruments of a structured note could include products related to movements of _____.

- **a.** Interest rates.
- **b.** Equities.
- **c.** Indices.
- **d.** Maturity dates.

Options: **(a), (b) and (c)** · (b), (c) and (d) · (a), (c) and (d) · (a), (b), (c) and (d)

Answer: (a), (b) and (c).

33. A range accrual note is a structured note where a client receives a higher return if:

- **a. The reference index is within a certain range.**
- **b.** The reference index is beyond a certain range.
- **c.** The coupon rate is within a certain range.
- **d.** The coupon rate is beyond a certain range.

Answer: a.

34. Which structured note would typically be structured with a principal preservation feature?

- **a. Range accrual note.**
- **b.** Dual currency investment.
- **c.** Inverse floater note.
- **d.** Exchange-traded note.

Answer: a.

35. Which of the following statements describe a structured note?

- **a.** A structured note is a debt instrument whose return characteristics are linked to the performance of other underlying instruments.
- **b.** A structured note is governed under the Securities and Futures Act.
- **c.** A structured note has return characteristics that are not linked to the performance of other underlying instruments.
- **d.** A structured note is governed by the Guidelines on Outsourcing.

Options: (a) and (d) · (b) and (d) · (a), (b) and (c) · **(a) and (b)**

Answer: (a) and (b).

36. Investors buy structured notes mainly because they want yield enhancement and market access. Which of the following factors should an investor consider before investing?

- **a.** Own liquidity needs.
- **b.** Terms and conditions of the structured note.
- **c.** Fees and charges of the structured note.
- **d.** Risk and return.

Options: (a), (c) and (d) · (a), (b) and (c) · (b), (c) and (d) · **(a), (b), (c) and (d)**

Answer: (a), (b), (c) and (d).

Chapter 9 — Structured Funds

37. Which of the following statements are TRUE about structured funds?

- **a.** A combination of equity and fixed income securities and/or derivatives, which achieve specific risk-return profiles or cost savings.
- **b.** Structured funds range from simple to highly complex funds and cover a wide range of risk exposures.
- **c.** Structured funds offer some capital preservation and are **not** designed to earn a market-linked return.
- **d.** Structured funds are **not** governed by the Code on Collective Investment Schemes under the Securities and Futures Act.

Options: **(a) and (b)** · (b), (c) and (d) · (a), (b) and (d) · (a), (b), (c) and (d)

Answer: (a) and (b). — Statements (c) and (d) are both false: structured funds **are** designed to earn a market-linked return and **are** governed by the Code.

38. A structured fund combines both _____ and _____ to provide investors with a degree of both capital preservation and capital appreciation.

- **a. Fixed income products; derivatives.**
- **b.** Fixed income products; equity products.
- **c.** Alternative assets; derivatives.
- **d.** Commodities; equity products.

Answer: a.

39. Which of the following parameters can be used to judge the performance of any fund?

- **a.** Sharpe ratio.
- **b.** Total expense ratio (TER).
- **c.** Underlying asset.
- **d.** Share class.

Options: **(a) and (b)** · (a), (c) and (d) · (a), (b) and (c) · (a), (b), (c) and (d)

Answer: (a) and (b).

40. What is the ADVANTAGE of investing in a structured fund using the constant proportion portfolio insurance (CPPI) strategy?

- **a. It has a capital preservation feature.**
- **b.** In a ranging bond market, the manager would be able to buy low and sell high.
- **c.** It has lower fees compared with a direct investment in the underlying asset.
- **d.** It may not have a 100% exposure to the underlying asset on day one.

Answer: a.

Chapter 10 — Contracts for Differences (CFDs)

41. If an investor has a negative view on a stock and believes the stock price will fall, which products will allow him to profit from this view?

- **a.** Callable bull/bear contracts or CFDs only.
- **b.** CFDs or options only.
- **c.** Options or callable bull/bear contracts only.
- **d. CFDs, options or callable bull/bear contracts.**

Answer: d.

42. Which statement regarding CFDs and equity futures is **INCORRECT**?

- **a.** CFDs have counterparty risk but equity futures do not.
- **b.** CFDs can be extended by the investor for as long as the investor wishes, but equity futures have a fixed maturity date.
- **c. CFDs have implicit financing cost as part of their price, whereas the financing cost for equity futures is explicitly computed.**
- **d.** CFDs are entitled to dividends and equity futures have no dividend entitlements.

Answer: c.

43. What are the terms of a CFD?

- **a.** Underlying asset.
- **b.** Price of the underlying asset.
- **c.** Number of units of the underlying asset specified in the contract.
- **d.** Volatility of the currency.

Options: (a), (b) and (d) · (a) and (d) · (b), (c) and (d) · **(a), (b) and (c)**

Answer: (a), (b) and (c).

44. An investor using a CFD trading strategy involving dividend capture should look for companies that _____.

- **a.** Are start-ups at an early growth stage.
- **b.** Do stock splits and issue scrip dividends regularly.
- **c. Have strong earnings and a high payout ratio.**
- **d.** Have price performance strongly driven by macroeconomic factors.

Answer: c.

45. Which of the following statements about CFDs and futures contracts is TRUE?

- **a. CFDs are mostly traded OTC whereas equity futures are traded on exchanges.**

- **b.** Both CFDs and equity futures have counterparty risk.
- **c.** Both CFDs and equity futures are entitled to dividends.
- **d.** CFDs have margin and leverage whereas equity futures have no margin and leverage.

Answer: a.

Chapter 11 — Key Products and Investment Risks for Derivatives and Structured Products

46. Trading in a CFD on an international stock outside the investor's home country has:

- **a.** No financing cost.
- **b.** No leverage risk.
- **c.** Market risk.
- **d.** Credit risk.

Options: (a) and (c) · (b) and (d) · **(c) and (d)** · (a) and (b)

Answer: (c) and (d).

47. Which of the following types of credit default are classified under the International Swaps and Derivatives Association?

- **a.** Restructuring.
- **b.** Failure to pay.
- **c.** Obligation acceleration.
- **d.** Obligation deceleration.

Options: **(a), (b) and (c)** · (b) and (c) · (a) and (d) · (a), (b) and (d)

Answer: (a), (b) and (c).

48. Which of the following factors contribute to basis risk?

- **a.** The asset whose price is to be hedged may not be exactly the same as the asset underlying the futures contract.
- **b.** The hedger may be uncertain about the exact date when the asset will be bought.
- **c.** The hedge may require the futures contract to be closed out well before its expiration date.
- **d.** The size and unit of measurement of futures contracts may not correspond with the size of the underlying hedged position.

Options: (a), (b) and (d) · (a), (c) and (d) · (b), (c) and (d) · **(a), (b), (c) and (d)**

Answer: (a), (b), (c) and (d).

49. What are the key risks involved in investing in structured products?

- **a.** Market risk.
- **b.** Liquidity risk.
- **c.** Reinvestment risk.
- **d.** Currency risk.

Options: (a) and (b) · **(a), (b), (c) and (d)** · (b) and (c) · (a) and (c)

Answer: (a), (b), (c) and (d).

Answer key at a glance

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	a	14	(c), (d)	27	all four	40	a
2	(a), (b), (c)	15	a	28	all four	41	d
3	c	16	b	29	all four	42	c
4	1.2557	17	(a), (b), (c)	30	(a), (c), (d)	43	(a), (b), (c)
5	(a), (c), (d)	18	$2.20	31	80%	44	c
6	a	19	6X	32	(a), (b), (c)	45	a
7	(a), (b), (c)	20	$0.15	33	a	46	(c), (d)
8	100	21	50%	34	a	47	(a), (b), (c)

Q	Ans	Q	Ans	Q	Ans	Q	Ans
9	2	22	d	35	(a), (b)	48	all four
10	b	23	(a), (c), (d)	36	all four	49	all four
11	(a), (b), (d)	24	d	37	(a), (b)		
12	$4	25	a	38	a		
13	d	26	(b), (c)	39	(a), (b)		

What these questions test

The set is a **format and scope orientation**, not a syllabus. Read against the chapters, it reveals where the examination puts its weight.

Calculation discipline. Six questions require a number, and each uses a single formula from Appendix D: **interest rate parity** (Q4), the **contract-count hedge** (Q8, Q9), **intrinsic value** (Q12, Q20), **conversion price** (Q18), **gearing** (Q19), **warrant premium with a non-unit conversion ratio** (Q21) and **participation rate** (Q31). Q21 is the trap — a **2:1** conversion ratio doubles the effective conversion cost before the premium is computed.

Sign and direction conventions. Q11 excludes the dividend rate because a higher dividend **reduces** call value; Q22 requires the barrier of an **up-and-out** call to sit **above** spot; Q16 identifies **volatility** as the one factor that raises both call and put value.

Product boundaries. Q1, Q25, Q42 and Q45 turn on distinguishing products that look alike: futures versus forwards (counterparty risk and interim settlement), CBBC versus structured warrants (the **call price**), CFDs versus equity futures (OTC versus exchange, implicit versus explicit financing, dividend entitlement), and the set of instruments that let an investor profit from a falling price.

Structural facts worth memorising. Q8 and Q9 both use the same hedge formula —

$$\frac{\text{notional}}{\text{contract size}} \times \frac{\text{exposure tenor}}{\text{contract tenor}}$$

— applied to a **strip** and a **stack** respectively. Q26 separates the **disadvantages** of barrier options from their advantages: the premium-loss risk on breach and **issuer counterparty risk** on OTC products.

The expanded question bank in `notes/cmsip/CMSIP_QUESTIONS.md` covers these themes and adds application questions beyond the orientation set.

Appendix F — Essential readings

Recommended: Bodie, Kane & Marcus, *Investments*; Hicks, *Managing Currency Risk Using Foreign Exchange Options*; Walmsley, *Foreign Exchange and Money Markets Guide*; Fabozzi & Leibowitz, *Fixed Income Analysis*; the CME Group *Risk Management Handbook*; Chance, *Analysis of Derivatives*; Hull, *Options, Futures & Other Derivatives*; Kolb & Overdahl, *Financial Derivatives: Pricing and Risk Management* and *Futures, Options, and Swaps*; Wilmott, *Derivatives*; Blumke, *How To Invest in Structured Products*; Lancaster, Schultz & Fabozzi, *Structured Products and Related Credit Derivatives*; Swiss Structured Products Association; Fabozzi, Davis & Choudhry, *Introduction to Structured Finance*; and current MAS, MoneySense and SGX materials.

Appendix G — Glossary and cross-reference

General finance language used throughout the chapters. The most important terms are defined in context above; use this compact cross-reference when revising:

- **Market basics:** absolute/relative return, alpha, benchmark, beta, bid/ask, blue chip, asset class, asset allocation, diversification, position, portfolio, risk appetite, risk tolerance, systematic/specific risk, spot market/rate, primary/secondary market, supply/demand and price discovery.
- **Accounting/economics:** accounts payable/receivable, balance sheet, cash-flow statement, current assets/liabilities, equity, book value, net income/profit margin, ROA/ROE, EBITDA, GDP, inflation/deflation, recession, fiscal/monetary policy, business cycle, balance of payments, current/capital account, exports/imports, trade balance, goodwill, amortisation, depreciation, fixed/variable costs, WACC, DCF/NPV, EPS, P/E/P/B/P/S/P/CF, enterprise value and EV/S.

- **Derivatives:** derivative, forward, futures, options, premium, strike/exercise, call, put, American/European, ITM/ATM/OTM/moneyness, intrinsic/time value, delta, gamma, theta, vega, rho, compound/exotic/plain-vanilla option, options on futures, warrant, structured warrant, conversion ratio, gearing, effective gearing, CBBC, DLC, CFD, swap, swaption, IRS, currency/commodity/equity swap, CDS, cap, floor, collar, basis, basis point, carry, mark-to-market, initial/maintenance/additional margin, notional, delivery, settlement, contract specification, hedge/hedge ratio, stack/strip, spread, calendar/horizontal, vertical, diagonal, butterfly, condor, ratio and seagull.
- **Structured products/funds:** structured product, structured deposit, structured note, ELN, CLN, BLN, RAN, CDRAN/VMRAN, inverse floater, zero-plus, accumulator/decumulator, discount/capped certificate, participation rate, basket/worst-of, barrier, knock-in/knock-out, one-touch/no-touch, CPPI, cushion/multiplier, DPPI, TRS, zero-coupon bond, ETN, ETF, NAV, AUM, TER/expense ratio, Sharpe/Treynor/information ratio, tracking error, UCITS, CIS, SICAV, umbrella, open/closed-ended fund, custodian, trustee, prospectus, PHS, trust deed, accredited/institutional/retail investor.
- **Risk/control:** market, basis, leverage, liquidity, credit/counterparty, settlement, operational, legal, transaction, structure, correlation, volatility, currency, financing, reinvestment, early termination, mis-selling, sovereign, political, model, gap and roll risk; VaR, stress test, open-contract/maturity/maximum-loss/Greek/book/intraday limits, CSA, front/middle/back office, DMM and authorised participant.
- **Corporate/market terms:** dividend/stock dividend, bonus/rights issue, share split/reverse split/buyback, conversion value, straight value, convertible bond, current yield/YTM, credit spread/rating, default/credit event, repo, T-bill/T-bond, MBS/ABS/CDO, dark pool, ECN, dealer/market maker, brokerage, proprietary trading, hedge fund, private equity, venture/growth/buyout/distressed funds, REIT, real estate, alternative investment, index/price-weighted/capitalisation-weighted, index fund, passive manager, technical analysis, EMH, APT, CAPM, efficient frontier, GIPS and information ratio.

The less product-specific entries — annuity, appraisal, appreciation/depreciation, assets, banks, board, foundations, fraud, insurance, legal/regulatory terms, policies/procedures/processes, reserve currency/requirement, retained earnings, shareholders/voting rights, tariffs, underwriters, trade bills, trust deed and similar general terms — are ordinary definitions supporting the product chapters. When a question uses one, translate it back to the underlying cash flow, risk or legal relationship.